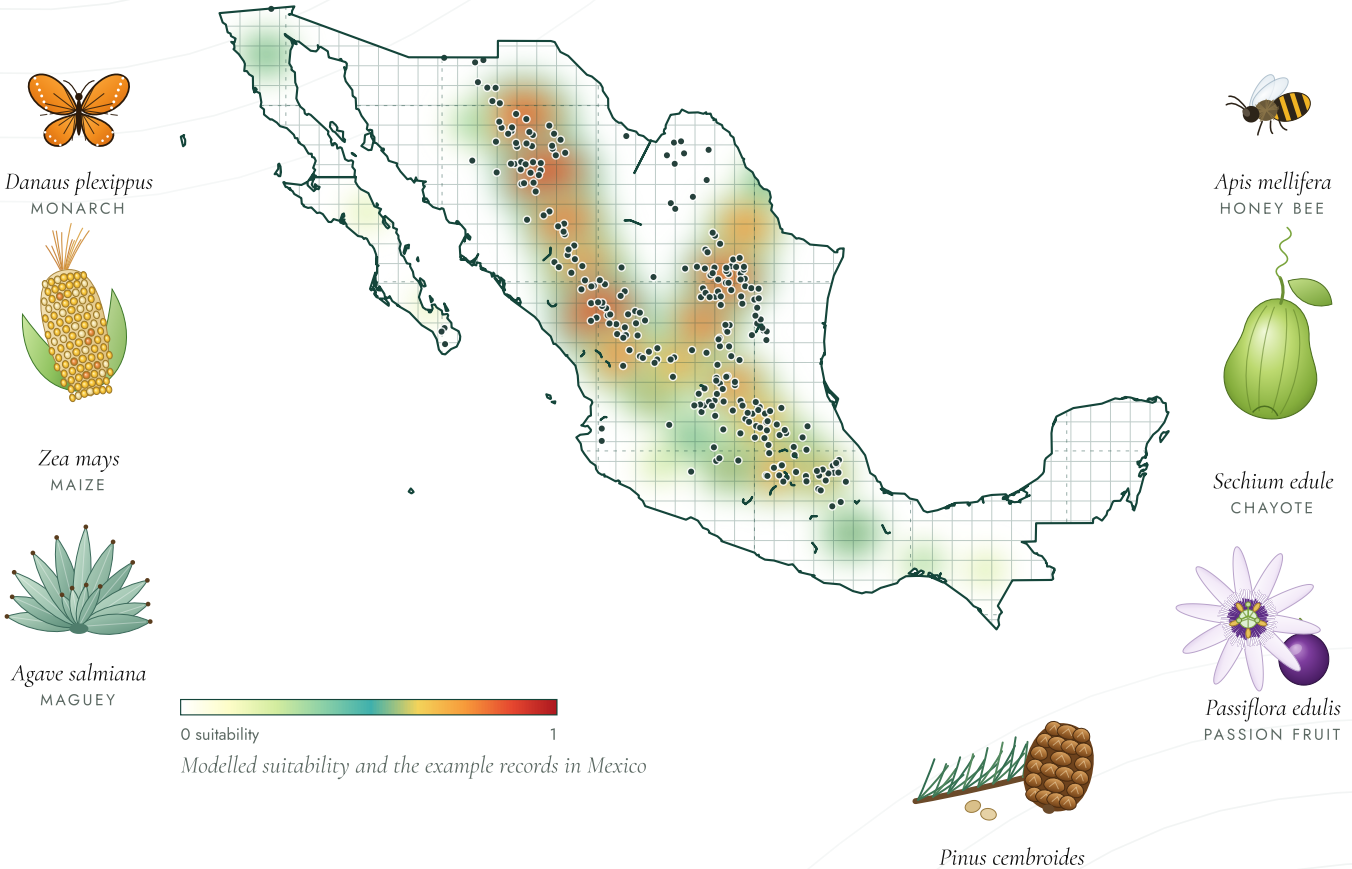
BioModelling Pro

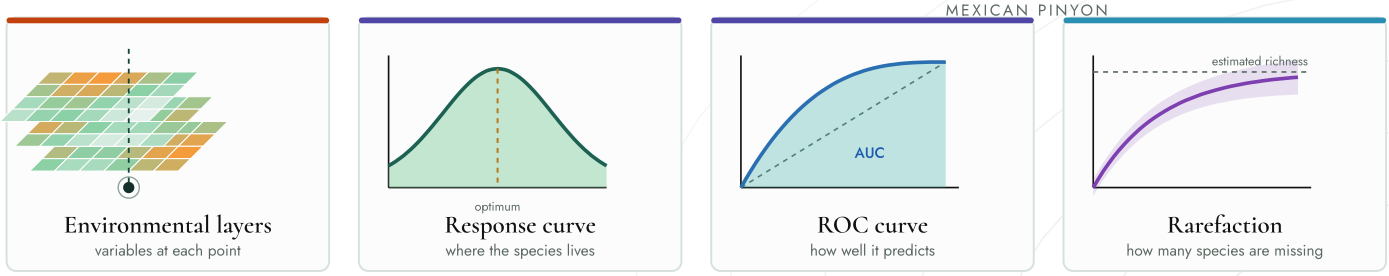
version 1.1.1

Biogeography and species distribution modelling

Biogeografía y modelado de la distribución de especies



Modelled suitability and the example records in Mexico



Luis Ángel Barrera-Guzmán · Gabriela Ramírez-Ojeda · Jorge Cadena-Iñiguez
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2026 · LICENSED UNDER GPL-3.0 OR LATER

Ten steps, from records to maps

BioModelling Pro

User's Manual · Biogeography and species distribution modelling

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SOFTWARE	BioModelling Pro, version 1.1.1 (September 2026), the one this manual describes
MANUAL	First English edition, 2026, translated from the first Spanish edition.
SOFTWARE LICENCE	GNU General Public License, version 3 or later (GPL-3.0-or-later).
MANUAL COPYRIGHT	© 2026 the authors. All rights reserved.
INTERFACE LANGUAGE	Spanish or English, with the ES · EN buttons on the top bar (Spanish by default). Button and menu names are given as they appear **in English**, e.g. Run models →.

HOW TO CITE THE SOFTWARE

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HOW TO CITE THIS MANUAL

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That DOI is the **concept DOI**: it always resolves to the latest version, and each version also has its own. The citation is also given in the How to cite section of the app, together with its BibTeX entry.

> **YOUR DATA NEVER LEAVE YOUR COMPUTER**
>
> BioModelling Pro runs inside the browser: the files you open are read on your own machine and **none of your data are uploaded to any server**. An internet connection is needed to query GBIF, to draw the base map and to download the computing engine the first time. The only query that sends coordinates is the soil-type lookup in Block 5, which is optional.

The illustrations, diagrams and screenshots in this manual are original. The example data are described in section I.7.

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How to read this manual

The manual follows the same order as the app: ten numbered blocks, laid out from left to right on the top bar. Each chapter explains one block and carries its number: chapter 3 is Block 3, and its sections, figures and tables are numbered 3.1, 3.2… Each chapter opens with the essential theory, goes on to what the app computes and how to read it, and closes with a step-by-step example using the data that already come with the program. The introduction comes before the blocks and is numbered separately: I.1, I.2…

One colour per block

The block colour appears on the opening page of its chapter, in the section numbers, on the tab at the edge of the page and in the contents, and it matches the strip on the cover. That way you can tell at a glance which part of the app you are in.

Block	Name in the app	What it does
1	1 GBIF	Searches GBIF for the species and fetches its records with coordinates, or loads your own from a file
2	2 Filter	Filters by taxon, country, year, basis of record, geographic bounding box and Mexican state or region
3	3 Clean	Nine coordinate-quality rules with a before-and-after report
4	4 Map	Records map and map studio: base layers, legend, scale bar, compass rose and export
5	5 Variables	Extracts at each point the 19 bioclimatic variables, elevation, the Köppen-Geiger climate and soil
6	6 Correlation	Correlation matrix, VIF and a selection of weakly collinear variables that keeps the ones you lock
7	7 Statistics	Descriptive statistics, histograms, box plots, normality and transformations, with fully editable figures
8	8 ML and ecology	PCA, clustering, correspondence analysis, range size, niche breadth and diversity
9	9 SDM modelling	Ten algorithms and an ensemble, validation, thresholds, maps and projection to future scenarios
10	10 Agroclimate	Agroclimatic indices and crop suitability on the same climate, now and in the future

Types of box

NOTE

Clarifications and practical tips that save time.

IMPORTANT

Things that can change your conclusions if overlooked.

THEORY IN BRIEF

The idea behind an analysis, without unnecessary formulas.

EXAMPLE

A case worked step by step with the data bundled with the program.

DECISION RULE

If the result falls in this case…	**then** this is the recommended reading.

These are guides to interpretation, not laws: what you know about your species and your region always comes first.

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KEY FACT

Help cards built into the app, one per index or statistic, each with its reading scale, the most common mistake and the reference it comes from.

Typographic conventions

Written like this	Means
Prepare model data →	A button, checkbox or menu in the app, with its text exactly as it appears on screen in English.
Map studio › Export › Clip to the country	A path: open the first item, then the second, and so on.
`decimalLatitude`, `.tif`	A column name, file name or value, written exactly as shown.
BIO1, BIO12 · AUC, TSS	Variables and indices; their meaning is explained the first time they appear and collected in the glossary.
Pinus cembroides	Scientific names are in italics, as in any biological text.
● ◐ —	Available · partly available or with conditions · not applicable.

IF YOU ARE COMING FROM ANOTHER PROGRAM

Anyone who has modelled niches before will recognise the route: cleaned records, environmental layers, variable selection, fitting, validation and projection. What changes is that here everything happens in a single window, with nothing to install, and every result comes with its reading card alongside. The methods go by the names used in the literature and are cited by their original papers.

In

INTRODUCTION · BEFORE THE BLOCKS

Introduction to BioModelling Pro

What the program is, which biogeographical questions it answers, which data it accepts and how to work through it from start to finish. It includes the five ideas of niche modelling on which the rest of the manual rests.

I.1 What is BioModelling Pro?

I.2 Questions it answers

I.3 Five key ideas of niche modelling

I.4 Data it accepts

I.5 What you need and how to open it

I.6 A tour of the interface

I.7 The bundled example: the Mexican pinyon

I.8 Recommended workflow

I.9 How to read the numbers

In

I.1 What is BioModelling Pro?

BioModelling Pro is a platform for carrying out a complete species distribution study, **from records to maps**: it downloads presence records, cleans them, extracts climate, elevation and soil at each point, explores them with statistics and multivariate methods, fits and validates distribution models, projects the result onto future climate scenarios and assesses the agroclimatic suitability of crops over the same territory.

Everything happens inside the browser. There is nothing to install, no programming is needed and your data never leave your computer. The same route works for a wild species, for a crop wild relative or for a crop: the question changes, the workflow does not.

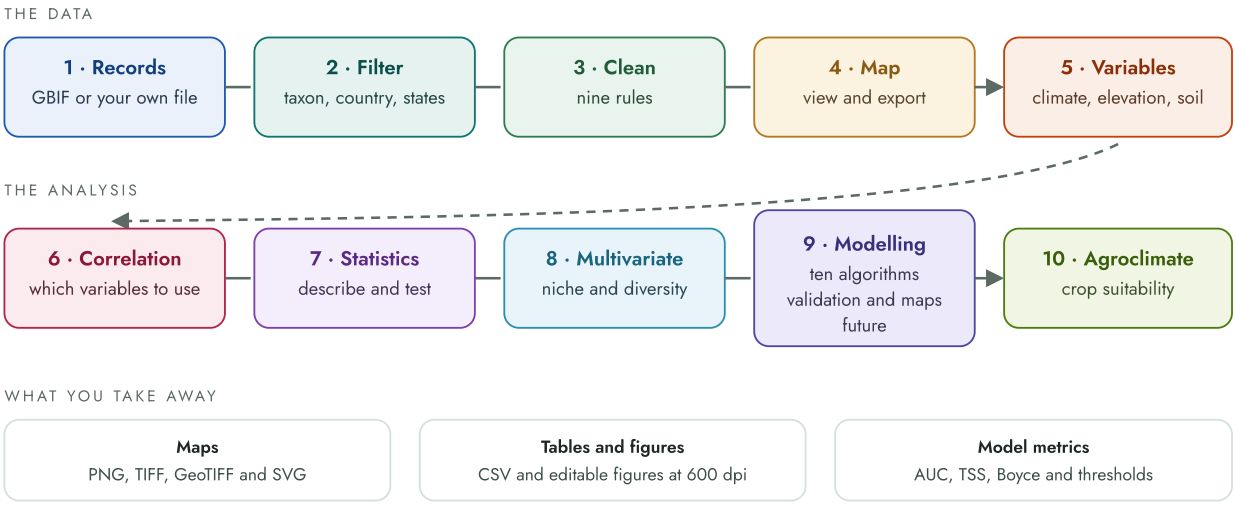


Figure I.1. The ten blocks. The first five prepare the data; the next five analyse them. Each block leaves the input of the next one ready, and all of them draw on the same cleaned data set.

I.2 Questions it answers

These are real questions from theses, papers and projects that the program answers with the data you already have or can download in a few minutes:

- **Where does my species live today, and where *could* it live?** The suitability map of Block 9 points to the areas with the climate the species occupies, even where nobody has collected it yet.
- **Which climate variables explain its distribution?** Permutation importance and the response curves of Block 9 tell you, once Block 6 has discarded the variables that duplicate each other.
- **How well does my model predict?** Spatial block validation with AUC, TSS and the Boyce index, and thresholds to turn suitability into a presence–absence map.
- **What will happen under climate change?** Projection to several scenarios, periods and climate models, with the area gained and lost, the shift of the centre of the distribution, and refugia.

- **How threatened is it?** The extent of occurrence (EOO) and area of occupancy (AOO) of Block 8, with the Red List thresholds.
- **Do the subtaxa occupy different environments?** Niche overlap, ordinations and clustering in Block 8.
- **Which crop suits this region, today and thirty years from now?** Block 10 computes growing degree-days, frosts, chill hours, water balance and suitability for each crop, with its planting calendar and its limiting factor.

> **WHAT A DISTRIBUTION MODEL IS NOT**
>
> A suitability map is **not** a map of confirmed presence, nor a census, nor a prediction of abundance. It shows where the environment resembles that of the places where the species has been recorded. An area coming out as suitable does not mean the species is there: it may never have reached it, it may have been eliminated, or it may be excluded by a competitor, by the soil or by land use. That distinction, spelled out in full, is the subject of section I.3.

I.3 Five key ideas of niche modelling

Five ideas are enough to understand everything the app does. They come back in every chapter.

1. The niche is a space of variables; the distribution is a geographical space

Hutchinson (1957) defined the niche as the set of conditions under which a population persists: a volume in a space of *n* variables (temperature, rainfall, elevation…). That volume cannot be seen on a map, but every point on the map has its place inside it. Modelling means moving back and forth between the two spaces: the niche is learnt in environmental space and drawn in geographical space.

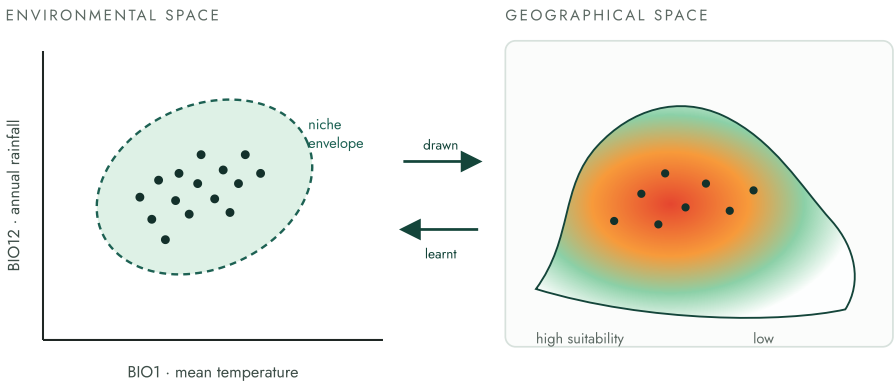


Figure I.2. The two spaces. The model learns in environmental space —where the records fall within the range of temperature and rainfall— and projects that result onto the map. The same points appear on both sides.

2. Fundamental niche, realised niche and what the model can see

The **fundamental** niche is everything physiology allows; the **realised** niche is the part the species actually occupies, already trimmed by competitors, predators and history. Presence records estimate something close to the realised niche, and only within the environments that exist in the study region. That is why the area chosen for modelling changes the result: it is a decision for the researcher, not a technical detail.

3. The BAM diagram: not everything suitable is occupied

Soberón and Peterson (2005) summed it up in three sets: **B**, what biotic interactions allow; **A**, what the abiotic environment allows; and **M**, what the species has been able to reach through its mobility. The observed distribution lies in the intersection. The app models mainly the **A** part, which is why there are suitable areas with no presence.

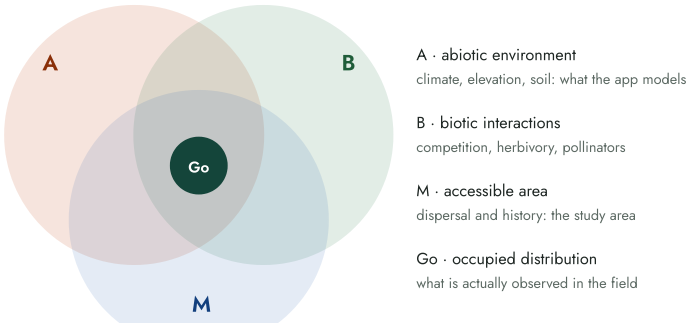


Figure I.3. The BAM diagram. The app estimates mainly **A**; the study area you choose in Block 9 is your hypothesis about **M**, and **B** almost always stays outside the model. It is worth saying so in the paper.

4. Records carry biases, and bias looks like signal

Collections pile up along roads, near cities and in protected areas. A naive model learns that sampling pattern instead of the niche. That is why Block 3 removes duplicates, country centroids and impossible coordinates, Block 9 can keep a single presence per cell, and validation is done by **spatial blocks** rather than at random: with nearby points, random validation hands out excellent results that do not hold at a new site.

5. Extrapolating is not interpolating

When the model is projected to another time or another region, combinations of climate appear that it never saw. There the prediction is an extension of the formula, not an estimate backed by data. The MESS map of Block 9 marks those areas, and the manual insists on showing them whenever a future projection is published.

I.4 Data it accepts

What	Format	Where it goes in and what to keep in mind
Presence records	GBIF download	Block 1. The scientific name is enough: the app resolves the taxon and fetches the records with coordinates, with a progress bar, an adjustable cap and a country filter —Mexico, Central America and South America— that is applied within the query itself.
Your own records	`.csv`, `.txt`, `.xlsx` or pasted	Block 1. Three columns are enough: species, longitude and latitude. It accepts decimal commas, degrees-minutes-seconds and files saved on Windows; you can add them to the GBIF records or use them on their own.
Climate layers	`.tif` (GeoTIFF)	Block 5. The 19 bioclimatic variables and elevation from WorldClim 2.1, at whatever resolution you download. The app reads only the window it needs, so it does not load the whole file into memory.
Köppen-Geiger climate	`.tif`	Block 5. Categorical: the app includes the table of class codes and names.
Soil type	online query	Block 5, optional. It queries a public service point by point; it is the only query that sends coordinates.
Future scenarios	`.tif`	Block 9. One 19-band file or separate layers per variable, for each climate model, pathway and period.
Monthly layers	`.tif`	Block 10. Minimum temperature, maximum temperature and precipitation month by month, which is what the agroclimatic indices require.
Your own regions	`.geojson`	Block 2. Municipalities, watersheds, protected areas: any polygon layer can be used to filter and to clip the maps.

i YOU DO NOT NEED TO DOWNLOAD ANYTHING TO GET STARTED

The program comes with a complete example: 2,400 GBIF records and climate layers clipped to the whole of Mexico. With them you can work through the ten blocks, try every button and reproduce every figure in this manual before turning to your own data.

I.5 What you need and how to open it

1. **An up-to-date web browser.** Any of the widely used ones, on Windows, macOS or Linux. Nothing else is installed.

2. **Double-click `index.html` to open it.** That is enough for the example records and everything done with them. The climate layers are downloaded once (section I.4) and loaded with the first two buttons of Block 5, which also work when the app is opened by double-clicking. You can also use the local server that comes with the program if you prefer to open it from another computer on the same network.

In

3. **An internet connection** for three things: querying GBIF, drawing the base map and downloading the computing engine the first time you enter a block that uses it. Everything else —reading your files, the calculations and the figures— works offline.
4. **Choose the language and theme** on the top bar: ES or EN, and light or dark. The app opens in Spanish; once you switch to English the choice is remembered, and the figures redraw themselves.

> **THE FIRST TIME YOU ENTER A COMPUTING BLOCK**
>
> Blocks 7, 8, 9 and 10 use a Python engine that runs **inside** the browser. The first time, it takes from a few seconds to a couple of minutes to download, depending on your connection, and a progress notice appears. After that it stays in the browser's memory and subsequent analyses are immediate.

1.6 A tour of the interface

The window always has the same three parts:

- **The top bar**, with the name of the program, the ES · EN buttons, the theme switch and the **block route**: ⌂ Home and the ten numbered steps. You can jump from one to another at any time; whatever you have already computed is kept.
- **The home page** (⌂ Home), with the summary of the workflow, the cards for the ten steps, the methods gallery, the theory, the references, how to cite and a **virtual species lab**: an invented landscape with a niche you define, presences sampled with or without bias, and three algorithms that try to recover it. It is the best place to understand what a model does before using real data.
- **The block panel**, always with the same structure: at the top a box saying what the step is for and what you decide in it, then the controls, then the results in cards, tables and figures, and finally a collapsible **interpretation guide**. Next to each number there is a help button that opens its card.

> **THE HELP CARDS**
>
> Each card tells you four things in this order: **what that number measures** in plain words, **how to read it**, **on what scale** —with a bar of coloured bands that marks where your value falls— and **the most common mistake** when using it, and it closes with the reference it comes from. When the scale is a reading convention rather than a statistical law, the card says so; when no accepted scale exists, it says that too instead of inventing one.

I.7 The bundled example: the Mexican pinyon

All the examples in this manual use the same data set, which comes with the program: the records of the **Mexican pinyon**, *Pinus cembroides* Zucc., a pine with edible seeds from the dry mountain ranges of northern and central Mexico. It is a good case study because its distribution is wide but fragmented, it is tied to a well-defined semi-arid montane climate, and it has described subspecies and varieties.

In the example data set	What it contains
Records	2,400 downloaded from GBIF on 21 September 2026, out of the 5,329 reported by the query
Countries	Mexico 1,719 · United States 670 · New Zealand 8 · France 2 · Austria 1
Names	7 forms: the species (2,281) and six subspecies and varieties (*orizabensis*, *cembroides*, *lagunae*)
Basis of record	human observation 1,343 · preserved specimen 967 · other 90
Years	from 1982 to 2026
Layers	the 19 bioclimatic variables, elevation and the Köppen-Geiger climate at 10 arc-minutes (≈ 18 km), clipped to lon −122° to −86° and lat 13° to 40°: the whole of Mexico and the south-western United States

▸ THE ELEVEN RECORDS WORTH A LESSON

Eleven of those 2,400 records are in New Zealand, France and Austria: they are **cultivated** trees in botanic gardens and arboreta, not natural populations. GBIF delivers them without telling them apart, and if they slip through, the model tries to explain the climate of Europe and Oceania with the niche of a Mexican mountain range. Chapters 2 and 3 show how to detect them and what to do with them. They are not a flaw in the example: they are what turns up in any real download.

I.8 Recommended workflow

The order of the blocks is the order in which it pays to work, but it is not a straitjacket: you can go back, change a filter and recompute. This is the short route for a typical study:

1. **Fetch the records** (Block 1) and check how many there are and where they come from.
2. **Filter** (Block 2) by taxon and by country or region. This is where the study area is decided, and that is an ecological decision, not one of convenience.
3. **Clean** (Block 3) and look at the before-and-after table: each rule tells you how many records it removes and why.
4. **Look at the map** (Block 4). A glance reveals the errors that no rule catches: points in the sea, in the wrong country or in a single botanic garden.

In

5. **Extract the environment** (Block 5) at each point and check how many were left without data.
6. **Choose the variables** (Block 6). Keep those that add distinct information and those that make sense for your species, even if they are correlated.
7. **Describe them** (Block 7): that way you will know the range of temperature and rainfall in which your species lives before modelling anything.
8. **Explore the structure** (Block 8): ordination, groups, range size and niche breadth.
9. **Model and validate** (Block 9). Start with two or three algorithms and block validation; leave the ensemble and the future for when the base model convinces you.
10. **If your question is agricultural**, go on to Block 10 with the monthly layers.

> **SAVE AS YOU GO**
>
> Every block lets you download what it produces: the CSV of the cleaned records, the table of points by variables, the metrics, the maps and the figures. Downloading the cleaned CSV at the end of Block 3 costs one click and saves you repeating the GBIF download if you close the browser.

1.9 How to read the numbers: scales and thresholds

This manual, like the app, distinguishes three kinds of numerical reference, and it is worth not confusing them when writing up the results:

Kind	Examples	How to treat it
Reading convention	|r| > 0.7 is high collinearity; AUC > 0.8 is a good model; VIF > 10 is a problem	Useful habits, not laws: cite them as such and give the exact value alongside.
Published threshold	EOO < 20,000 km² and AOO < 2,000 km² from the Red Lists; MESS < 0 marks extrapolation	They come from a formal criterion; cite them with their source and with the caveat that a threshold on its own does not assign a category.
No accepted scale	kurtosis, Dunn index, accumulated degree-days, observed richness	Compare them with one another —between regions, between models, between years—, never against a magic number.

Whenever the manual gives a band of values, it will say which of the three kinds it belongs to. And whenever a number depends on the sample size, the study area or the resolution of the layers —which is almost always—, it will also say what it depends on.

✓ THREE QUESTIONS BEFORE PUBLISHING A MAP

What area did I model?	Say so in the text and show it on the map: the study area is a hypothesis about where the species has been able to reach.
How did I validate?	Spatial blocks, with AUC, TSS and Boyce, and the number of presences that went in.
Where did I extrapolate?	Accompany every future projection with its MESS map or, at the very least, with the percentage of the area under extrapolation.

The three answers fit in two lines of the methods section and head off half of the reviewers' comments.

1

BLOCK 1 · THE STARTING POINT

Presence records: GBIF and your own data

Where the points to be modelled come from. How a scientific name is resolved, how GBIF records are downloaded without losing control of what arrives, and how a table of your own from the field or the herbarium is brought in.

1.1 What a presence record is

1.2 GBIF in two minutes

1.3 Resolving the scientific name

1.4 Fetching the records and filtering by country

1.5 What arrives and what it looks like

1.6 Your own records

1.7 How many records do you need?

1.8 Common problems

1.1 What a presence record is

A **presence record** is the statement that an organism of a certain species was in a certain place at a certain time, made by someone who saw or collected it. Everything this program does rests on that sentence, so it is worth taking it apart.

A record that is useful for modelling needs three things: a **name** we can trust, **coordinates** in decimal degrees and some **idea of their precision**. Everything else —who collected it, which herbarium holds it, what year it was— helps with filtering and with giving credit, but does not enter the model.

> **PRESENCE IS NOT THE OPPOSITE OF ABSENCE**
>
> The lack of records in an area can mean three very different things: that the species is not there, that it is there but nobody has gone to look for it, or that it was looked for and not found. GBIF data almost never distinguish between them. That is why the methods in this program are **presence-only**: they learn from the contrast between the sites where the species was recorded and a sample of the environment available in the region, not between true presences and absences. It is the difference between saying "here the climate resembles that of the places where it lives" and "it lives here".

Field	What it is and why it matters
`decimalLatitude` `decimalLongitude`	The coordinate in decimal degrees, negative west of Greenwich and south of the equator. The whole of Mexico lies between −118° and −86° longitude and between 14° and 33° latitude: in a national study, any value outside that range is suspect.
`coordinateUncertaintyInMeters`	The radius within which the true point lies. An old herbarium record georeferenced from "10 km north of Durango" may carry 10,000 m; a phone observation carries 5 or 10 m. If the uncertainty is larger than the cell size of your layers, the point may fall in the wrong climate.
`basisOfRecord`	The kind of evidence it comes from: preserved specimen, human observation, living specimen (botanic gardens), fossil material... The living specimen is the most troublesome: these are plants cultivated outside their natural range.
`taxonRank, taxonKey`	The rank of the name (species, subspecies, variety) and its numerical key in GBIF, which is what really identifies the taxon, regardless of how the name is spelt.
`year, month`	The date. It is used to discard very old records, whose site may no longer exist, and to study change over time.
`issues`	The flags GBIF raises on its own: rounded coordinates, country inferred from the coordinate, coordinate probably inverted... They are kept because they help with decisions in Block 3.

1.2 GBIF in two minutes

GBIF —the Global Biodiversity Information Facility— does not collect data: it **brings them together**. Herbaria, museums, citizen-science projects and national monitoring schemes publish their catalogues there, and GBIF serves them in a common format with a unified taxonomy. This has two consequences worth keeping in mind at all times:

- **The quality is that of the publisher, not of GBIF.** A transcription error in a herbarium reaches your download intact. That is why Block 3 exists.
- **Coverage is uneven.** Some countries, groups and regions have thousands of records and others are almost empty. The density of points says as much about collecting effort as about the species.

> **CITE THE DATA, NOT JUST THE SOFTWARE**
>
> Every publication that uses GBIF records must cite the download. The app queries the online service, which does **not** issue a permanent identifier; so, at the time of downloading, write down the **date**, the **name searched**, the **taxon key** and the **filters** you used —including the countries, if you filtered by them—: with that, anyone can repeat the query. If your journal requires a DOI for the download, repeat the same query from the GBIF portal and cite the identifier it gives you.

1.3 Resolving the scientific name

Before downloading anything, we need to agree on which species we are talking about. Type the name in the box and press `Look up name`: the app asks GBIF which name is accepted and under which key it is registered.

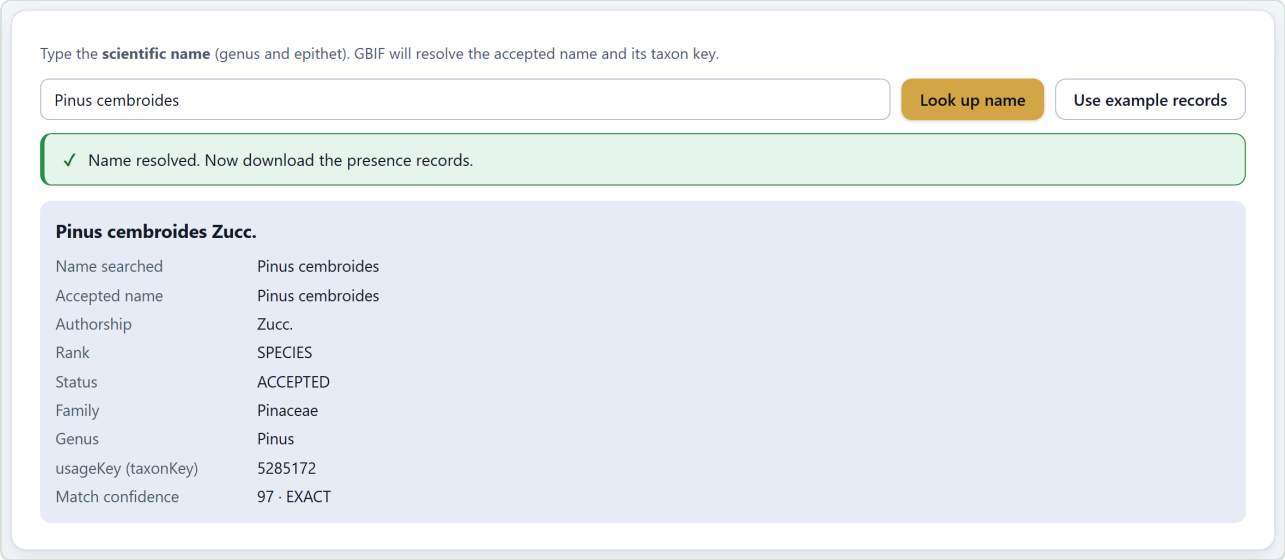


Figure 1.1. The resolved name. The key `5285172` is what is sent to GBIF; the typed name only served to get there. The `Use example records` button, on the right, loads the 2,400 records that come with the program without going online.

What it says	How to read it
Accepted name	If it differs from what you typed, you typed a synonym. That is not an error: GBIF takes you to the valid name, and records published under the synonym are included.
Rank	`SPECIES`, `SUBSPECIES`, `VARIETY`… If you search for a subspecies, only its records will arrive; if you search for the species, you can include or exclude infraspecific taxa when downloading.

What it says	How to read it
Status	`ACCEPTED` is what you want to see. `SYNONYM` or `DOUBTFUL` call for a taxonomic check before going on.
Match confidence	How sure GBIF is that it understood the name, with the type of match (`EXACT`, `FUZZY`…). A fuzzy match with low confidence is usually a misspelt name.

IF THE APP CANNOT FIND THE NAME

Check the spelling and make sure it is the full scientific name (genus and epithet). Common names do not work: "Mexican pinyon" is not a scientific name and GBIF will not resolve it. If the name is recent and not yet in the GBIF taxonomy, use the name of the species it used to belong to and then filter by the written name in Block 2.

1.4 Fetching the records

Once the name is resolved, the download card controls *what* is requested, *from where* and *how much*. All three decisions travel inside the query to GBIF: whatever is excluded here is not downloaded, and the total the app announces is already filtered.

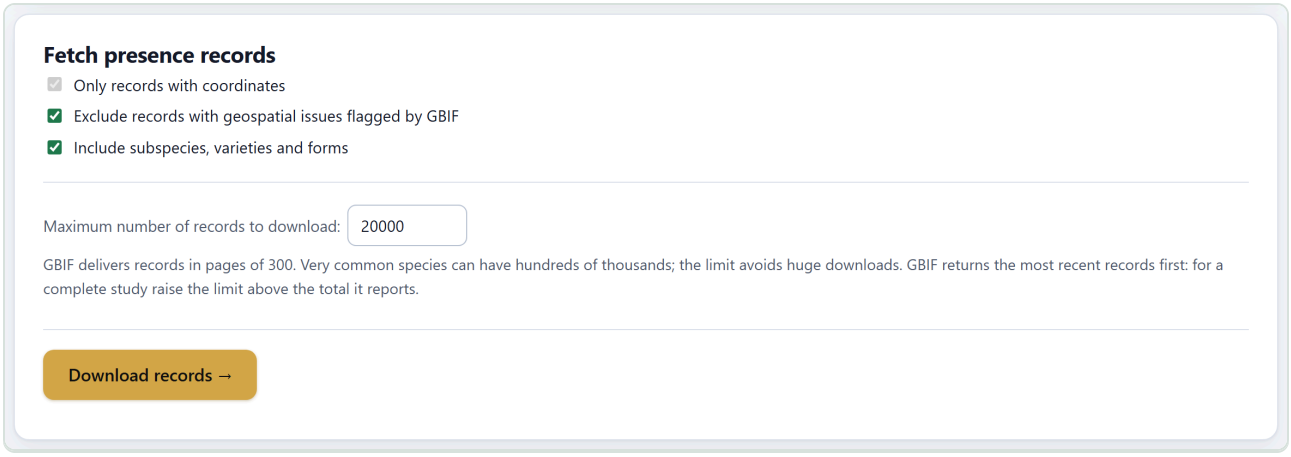


Figure 1.2. The three checkboxes and the cap. The download proceeds page by page, with a progress bar, and can be stopped.

Option	What it does and when to change it
Only records with coordinates	Discards those without a point. It is always on: a record without coordinates is of no use for modelling.
Exclude records with geospatial issues flagged by GBIF	Asks GBIF to exclude the records it has itself flagged with serious coordinate issues. It is a cheap first filter, but it does **not replace** Block 3.
Include subspecies, varieties and forms	Also fetches subspecies and varieties. Tick it if you want to model the whole species or compare its subtaxa; untick it if your question concerns only the species in the strict sense.

Option	What it does and when to change it
Countries	Requests only the records from the ticked countries (next subsection). It is the filter that shrinks the download the most.
Maximum number of records to download	The cap on the download. It saves you from waiting longer than necessary while you explore.

The country filter

Below the checkboxes is the **country** filter, which does not trim what has been downloaded: it asks GBIF, from the outset, for **only** the records from the countries you tick. The difference is large when the species is common: maize has 153,301 records with coordinates worldwide and **51,288** in Mexico. The count GBIF announces is already filtered, so you know from the start how many you are going to fetch.

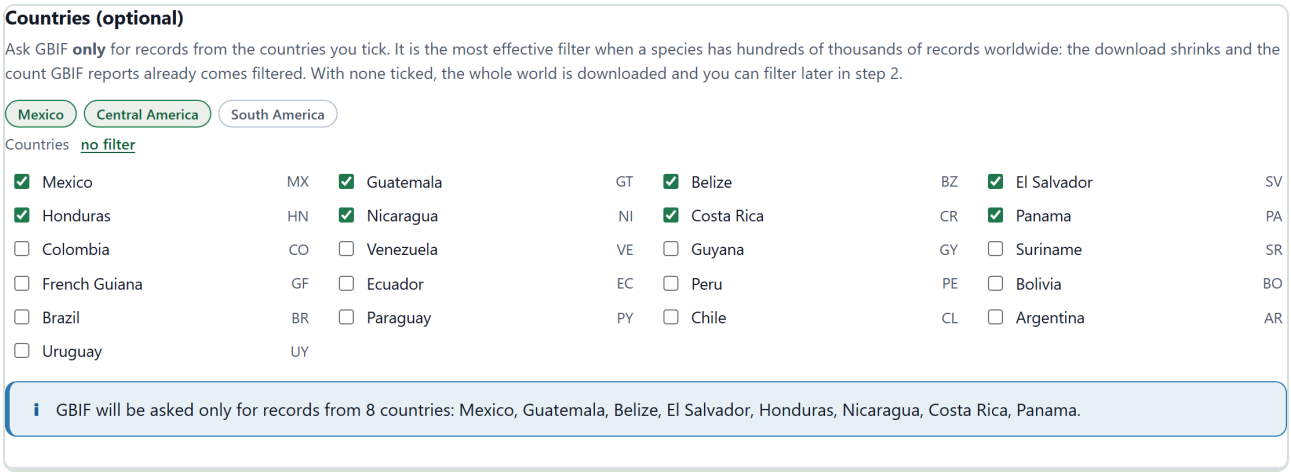


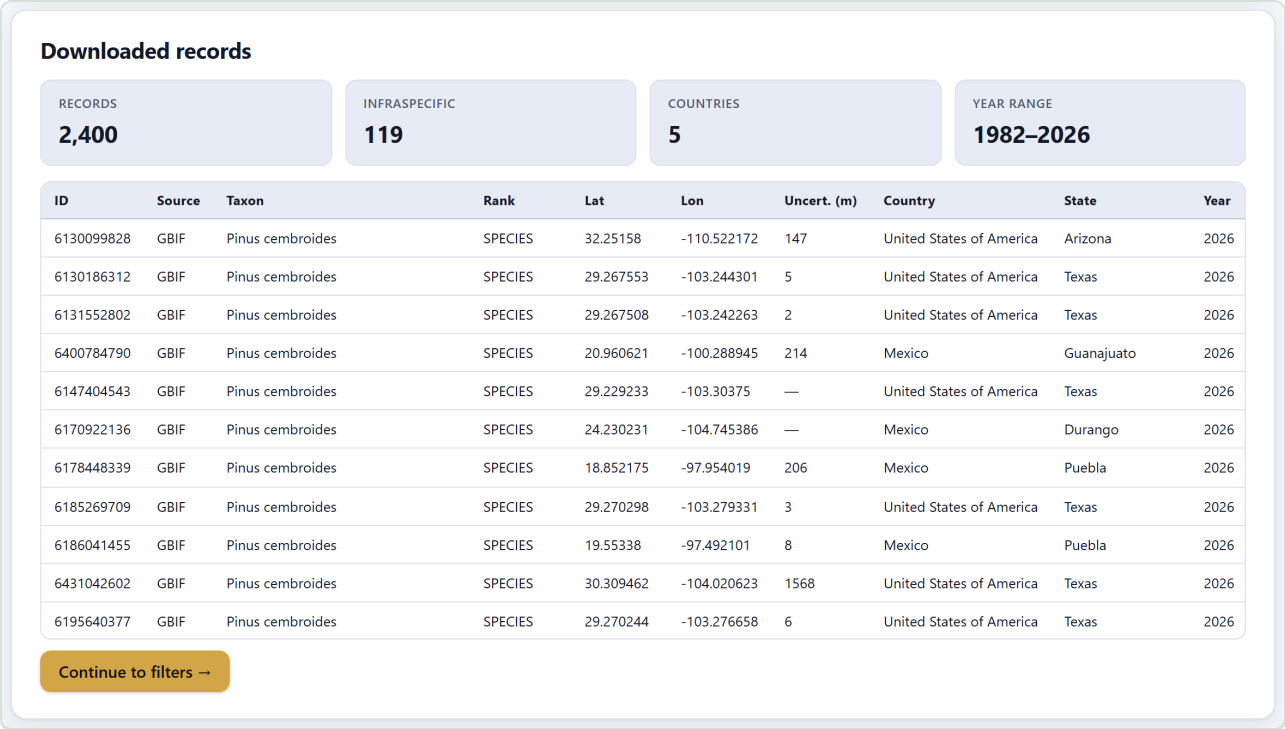
Figure 1.3. The twenty-one countries of Mexico, Central America and South America, with one shortcut per region. The buttons work as toggles, and no filter unticks them all: with none ticked, records are downloaded from the whole world, as usual.

> **WHEN *NOT* TO FILTER BY COUNTRY**
>
> When the distribution crosses a border to a country outside the list. The Mexican pinyon of the example also grows in the south-western United States: filtering to Mexico would leave out 670 records and truncate the real niche, not just the map. The rule is the one in chapter 2: the study area is decided by biology, not by convenience.

> **THE CAP DOES NOT TAKE A RANDOM SAMPLE**
>
> GBIF delivers the **most recent records first**. If your species has 20,000 records and you set a cap of 2,000, you will not get 2,000 representative records: you will get the 2,000 newest, which are biased towards citizen science and towards areas with phones and signal. For a serious study, raise the cap above the total the search reports and let everything come down.

1.5 What arrives and what it looks like

Once the download has finished, the last card summarises what there is and shows the first rows. This is the moment for the first check: how many records, from which countries, from which years.



ID	Source	Taxon	Rank	Lat	Lon	Uncert. (m)	Country	State	Year
6130099828	GBIF	Pinus cembroides	SPECIES	32.25158	-110.522172	147	United States of America	Arizona	2026
6130186312	GBIF	Pinus cembroides	SPECIES	29.267553	-103.244301	5	United States of America	Texas	2026
6131552802	GBIF	Pinus cembroides	SPECIES	29.267508	-103.242263	2	United States of America	Texas	2026
6400784790	GBIF	Pinus cembroides	SPECIES	20.960621	-100.288945	214	Mexico	Guanajuato	2026
6147404543	GBIF	Pinus cembroides	SPECIES	29.229233	-103.30375	—	United States of America	Texas	2026
6170922136	GBIF	Pinus cembroides	SPECIES	24.230231	-104.745386	—	Mexico	Durango	2026
6178448339	GBIF	Pinus cembroides	SPECIES	18.852175	-97.954019	206	Mexico	Puebla	2026
6185269709	GBIF	Pinus cembroides	SPECIES	29.270298	-103.279331	3	United States of America	Texas	2026
6186041455	GBIF	Pinus cembroides	SPECIES	19.55338	-97.492101	8	Mexico	Puebla	2026
6431042602	GBIF	Pinus cembroides	SPECIES	30.309462	-104.020623	1568	United States of America	Texas	2026
6195640377	GBIF	Pinus cembroides	SPECIES	29.270244	-103.276658	6	United States of America	Texas	2026

Figure 1.4. The example just after loading. Scroll the table sideways to see the remaining fields. The four figures at the top are the first sniff test: five countries for a Mexican pine already tells you that something needs checking.

> **THE EXAMPLE IN NUMBERS**
>
> The 2,400 Mexican pinyon records that come with the program are distributed as follows: **Mexico 1,719**, **United States 670**, **New Zealand 8**, **France 2** and **Austria 1**. There are seven named forms —the species, with 2,281 records, and six subspecies and varieties— and the dates run from 1982 to 2026. By kind of evidence: 1,343 human observations, 967 preserved specimens and 90 of other kinds.
>
> The eleven European and New Zealand records are trees cultivated in botanic gardens and arboreta. They are legitimate as records —those trees exist— but they say nothing about where the species *can live* on its own. They will be removed in Block 2 or Block 3.

1.6 Your own records

Not everything is in GBIF: data from your own sampling, from a thesis not yet published or from a local herbarium almost never are. The second card of the block accepts such data and treats them just like GBIF records.

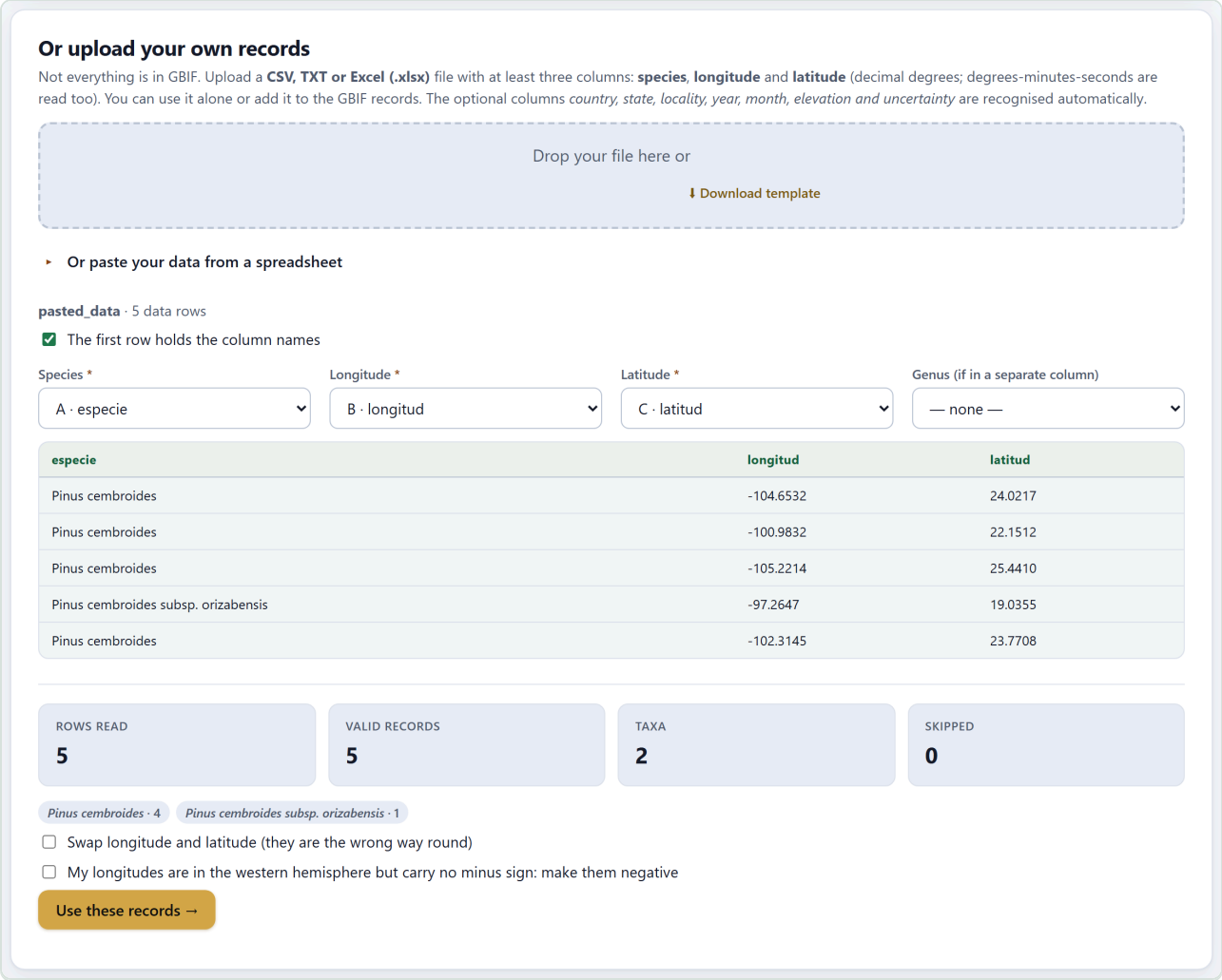


Figure 1.5. Five rows pasted from a spreadsheet. The app recognised the columns `especie`, `longitud` and `latitud` by itself; had it got them wrong, the three menus would let you correct the assignment.

1. **Prepare the table.** Three columns are enough: species, longitude and latitude. The headers can be in English or in Spanish. If you want something to start from, Download template gives you a file with the columns already in place.

2. **Upload or paste it.** Drag a `.csv`, `.txt` or `.xlsx` file onto the card, or copy the cells from your spreadsheet and paste them into the text box.

3. **Check the column assignment.** The app suggests which column is which; correct it if necessary. The optional columns —country, state, locality, year, month, elevation and uncertainty— are recognised automatically and put to use in the filters.

4. **Look at the four tiles** for rows read, valid records, taxa and skipped, and at the list of taxa with their counts. If "skipped" is not zero, some row had empty, text or out-of-range coordinates.

5 **Press** Use these records → and decide whether they replace the GBIF records or are added to them.

If your data look like this…	The app does this
Degrees, minutes and seconds 24° 01′ 18″ N, 104° 39′ 12″ W	Converts them to decimal degrees, with the sign that corresponds to the hemisphere.
Decimal comma –104,6532	Reads it as the decimal separator.
Longitude and latitude swapped	Tick Swap longitude and latitude (they are the wrong way round) and they are all corrected at once.
Longitudes without the minus sign	The checkbox below makes them negative, which is the usual case in the Americas.
File saved on Windows with garbled accented letters	Recognises the old encoding and repairs the accents.

1.7 How many records do you need?

The question always comes up, and the honest answer is: **fewer than people think, if they are well spread**. What a model needs is not a large number of points but points that cover the range of conditions in which the species lives. Fifty records spread across the whole mountain range teach more than five hundred crowded into two ravines.

✓ ORDERS OF MAGNITUDE

Fewer than 15	Do not model with flexible algorithms. A simple envelope model and a description of the environment are more honest.
From 15 to 30	Modelling is possible, with few variables (three or four), simple models and careful validation. Studies that have measured the effect of sample size find that this is where usable results begin.
More than 50	Comfortable ground for the ten algorithms and the ensemble of Block 9.
Many thousands	The problem is no longer the number but **bias**: thinning to one presence per cell usually improves the model more than adding data.

These are orders of magnitude from the literature, not official thresholds: the real minimum depends on how narrow the niche is and on how much environmental variation there is in your region.

1.8 Common problems

What you see	What happened and what to do
"Could not query GBIF"	No internet connection, or the service is not responding. The app works just the same with the bundled example or with your own files; try the download again later.
The download stops halfway "the connection to GBIF was lost"	In downloads of tens of thousands of records the connection may drop while a page is being read. The app retries up to six times with increasing waits and, if it still fails, offers a button to **use the records that have already arrived** instead of losing them. You can carry on working with that set or try the full download again later.
Far fewer records arrive than you expected	Check the cap and the checkboxes: Only records with coordinates discards many old herbarium records that were never georeferenced.
Countries appear where the species does not live	They are almost always cultivated specimens or coordinate errors. Do not delete them by hand: Block 2 filters them by country and Block 3 detects them by rules, so there is a record of how many there were.
All the records are from the last five years	The cap was too low and GBIF delivered only the most recent ones. Raise it.
Your file loads, but "valid records" is much lower than "rows read"	Wrongly assigned columns, coordinates containing text ("s/d", "—") or empty cells. Correct the assignment or clean the sheet and load it again.
The points end up in the Indian Ocean	Longitude and latitude swapped, or longitudes without a sign. The two checkboxes at the bottom of the card fix it.

BEFORE MOVING ON TO BLOCK 2

Download the raw CSV if you are going to work over several days: that way you do not depend on GBIF delivering exactly the same records next time. The downloaded records are also listed in the final report, with their date and query.

2

BLOCK 2 · WHAT GOES IN AND WHAT STAYS OUT

Filters: taxon, country, years, bounding box and states

This is where the study area is decided, together with the records that represent it. It is the block with the greatest influence on the final model, and the one that usually gets the least attention.

2.1 Filtering means deciding the study area

2.2 The four facets

2.3 Filtering by state or region

2.4 Geographic clip

2.5 The result

2.6 How to choose the study area

2.7 Common problems

2.1 Filtering means deciding the study area

A filter looks like a matter of convenience —removing whatever gets in the way— but in niche modelling it is a substantive decision. The records you keep define which environments count as "occupied", and the territory they span defines where the "available" environment they are compared against comes from. Changing the study area changes the model, even when the algorithms and the variables stay the same.

THE ACCESSIBLE AREA

In the introduction we saw the BAM diagram: **M** is the part of the world the species has been able to reach during its history. A model should learn by comparing the occupied sites against that M, not against a whole continent. If the area is too large, the model faces a trivial task —telling the Mexican sierras apart from the Amazon rainforest— and its quality measures come out inflated. If it is too small, the model never sees the conditions the species avoids and learns nothing. A reasonable M is usually the biogeographic region where the species lives, plus a margin.

2.2 The four facets

The first card shows four groups of checkboxes with the number of records in each category. Changes take effect at once: every click recalculates the number on the result card.

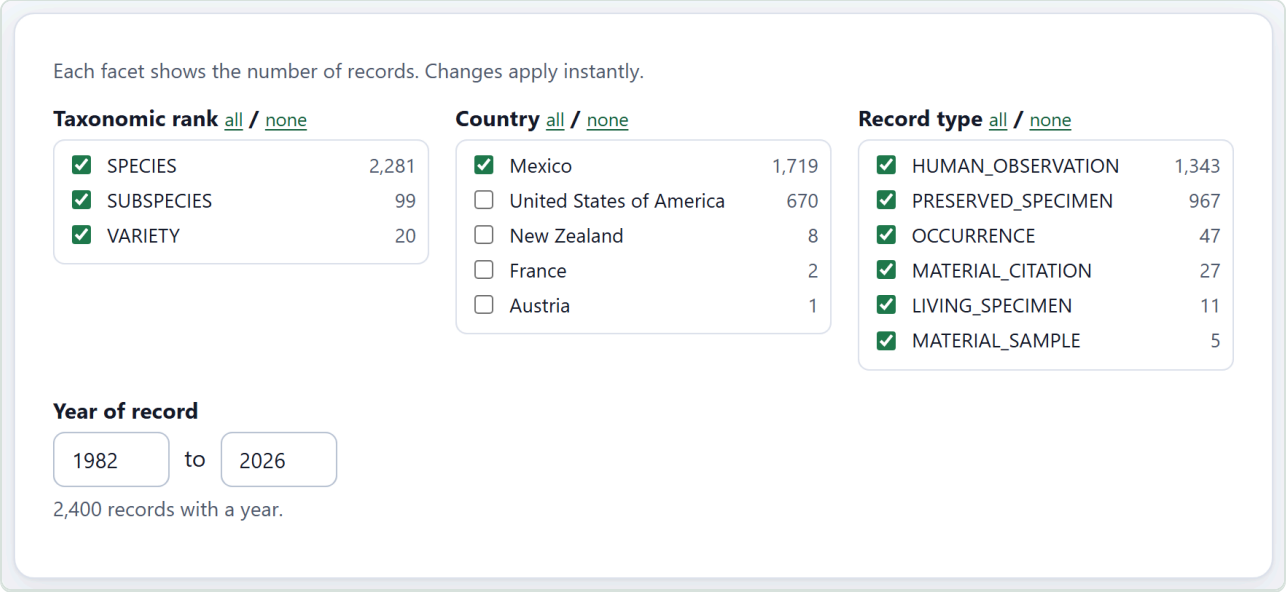


Figure 2.1. The facets of the example, with Mexico as the only country ticked. Each row carries its count, which is worked out on the **downloaded records**, not on those that survive the other filters: that way you always know how much of each thing there is in the original set.

Facet	What it is really for
Taxonomic rank	Separating the species from its subspecies and varieties. If you are going to model the whole species, keep them all; if your question is about one infraspecific taxon, keep only that one —and check that it has enough records (section 1.7). It is also useful for comparisons: model each subspecies separately and see whether they occupy different environments.
Country	The quickest way to remove specimens cultivated on other continents and to confine the study to one country. If you already knew which countries you would work with, it is better to filter them in the download itself (section 1.4): here you filter what has arrived, there you avoid bringing it in. Beware: if your species crosses a border —as the Mexican pinyon does between Mexico and the United States— removing the neighbouring country truncates the real niche, not just the map.
Record type	`LIVING_SPECIMEN` are living plants in collections: botanic gardens and arboreta. They almost always have to go. `PRESERVED_SPECIMEN` are herbarium specimens, whose identification can be checked but whose georeferencing is sometimes coarse; `HUMAN_OBSERVATION` are observations, with a precise coordinate but variable identification.
Year	Discarding old records whose site may have changed its land use, or keeping a period that matches the climate of your layers. If you use WorldClim 2.1, the climate represents 1970–2000: mixing it with records from 1900 is comparing different things.

2.3 Filtering by state or region

Many studies are regional rather than national: a mountain range, a state, a watershed. The program comes with the **32 states of Mexico** and their boundaries built in, and it can assign each record the state it belongs to.

Where each record's state comes from

GBIF supplies a text field with the state, but many records arrive without it and others carry it misspelt. That is why there are three modes:

Mode	What it does
from the coordinates when the text is missing (recommended)	Uses the record's text when there is one, normalising spellings and abbreviations ("Edo. de México", "Mexico", "Michoacan" → the same state), and when there is none it decides from the coordinate. It is the mode that makes use of the most records.
always from the coordinates	Ignores the text and relies only on the point-in-polygon test. Useful when you suspect the text field is unreliable.
from the record text only	Keeps whatever the publisher wrote. It is for auditing the data, not for modelling.

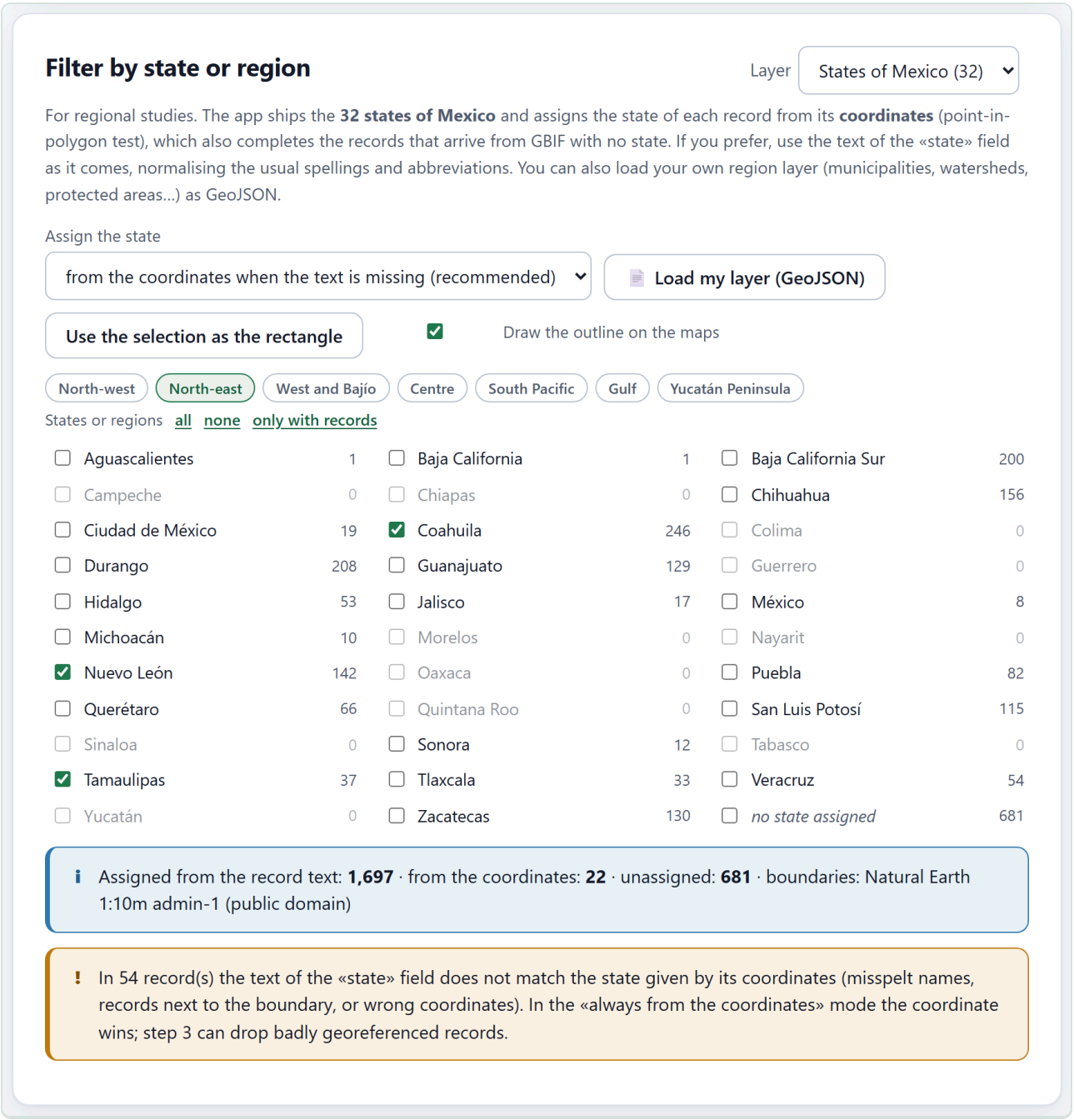


Figure 2.2. The state filter with the North-east group switched on: Coahuila, Nuevo León and Tamaulipas. States with no records are greyed out, and the last row gathers the records that could not be assigned to any state.

THE EXAMPLE, STATE BY STATE

With the 2,400 records of the Mexican pinyon: **1,697** got their state from the record text, **22** got it from their coordinates —Mexican records that GBIF delivered with no state— and **681** remained unassigned, which are exactly the 670 from the United States and the 11 from Europe and New Zealand: there is not a single Mexican record without a state.

The states with the most records are Coahuila (246), Durango (208), Baja California Sur (200), Chihuahua (156), Nuevo León (142) and Zacatecas (130): the Sierra Madre Occidental, the Sierra Madre Oriental and the Sierra de La Laguna, which is where the Mexican pinyon lives.

WHEN THE TEXT AND THE COORDINATE DISAGREE

In the example there are **54 records** whose written state is not the state their coordinate falls in. There are three causes: misspelt names or municipalities taken for states, records right on the boundary between two states —where an error of 200 m changes the answer— and plainly wrong coordinates. The app warns you and does not decide for you: in the recommended mode the text wins, in the coordinate mode the coordinate wins, and badly georeferenced records are dealt with in Block 3.

Shortcuts and options

- **The group buttons** — North-west, North-east, West and Bajío, Centre, South Pacific, Gulf, Yucatán Peninsula — tick all the states of that region at once. They work as a toggle: if the group was already complete, a second click removes it.
- all · none · only with records are the list shortcuts. The last one leaves ticked only the states that have at least one record, which is a convenient starting point.
- Use the selection as the rectangle turns the chosen states into the rectangle of the geographic clip in section 2.4, and that same bounding box is later offered as the map extent in Block 9.
- Draw the outline on the maps draws the boundary of your selection on the maps of Blocks 4 and 9, so that it is clear which region is being discussed.
- Load my layer (GeoJSON) replaces the states with your own polygons: municipalities, watersheds, protected areas, management units, ejidos. The app takes the name of each polygon from its properties and builds the list just as it does with the states.

WHERE THE BOUNDARIES COME FROM

The 32 states come from Natural Earth at a scale of 1:10 million, which is in the public domain, simplified to keep them light. They are accurate enough to classify records and to draw outlines on a page-sized map, but they are **not** an official boundary: do not use them to measure areas or to settle a boundary dispute.

2.4 Geographic clip

When the region you are interested in does not match any state —an elevation belt, a rectangular study zone, the area of a project— there is the clip by coordinates.

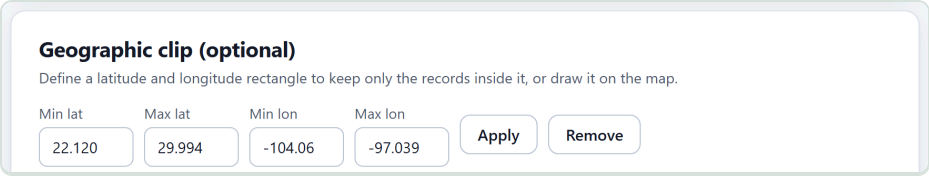


Figure 2.3. The four numbers of the bounding box, already filled in with the envelope of the three North-east states. Below them, the card has a mini map on which the rectangle can be drawn with the mouse: press Draw rectangle (it then reads Click 2 corners...) and click two opposite corners; the rectangle is applied straight away and its numbers fill the four boxes.

The bounding box and the state filter can be used together: the box clips and the state selection refines. If you are going to model, the box should be somewhat larger than the known distribution, so that the model also sees conditions the species does *not* occupy.

2.5 The result

The last card states how many records survived and shows the table with the **State** column already filled in. This is the set that goes on to Block 3.

Filtering result **425 records**

ID	Source	Taxon	Rank	Lat	Lon	Uncert. (m)	Country	State	Year	Type
6196011593	GBIF	Pinus cembroides	SPECIES	25.191397	-101.401108	5	Mexico	Coahuila	2026	HUMAN_OBSERVATI
6196047919	GBIF	Pinus cembroides	SPECIES	25.191263	-101.400153	6	Mexico	Coahuila	2026	HUMAN_OBSERVATI
6413853565	GBIF	Pinus cembroides	SPECIES	25.450819	-100.740012	67	Mexico	Coahuila	2026	HUMAN_OBSERVATI
5890202863	GBIF	Pinus cembroides	SPECIES	25.229244	-100.825924	—	Mexico	Coahuila	2025	HUMAN_OBSERVATI
5827384530	GBIF	Pinus cembroides	SPECIES	25.234973	-100.435197	35	Mexico	Coahuila	2025	HUMAN_OBSERVATI
5291976067	GBIF	Pinus cembroides	SPECIES	25.19555	-101.484428	3	Mexico	Coahuila	2025	HUMAN_OBSERVATI
5828058488	GBIF	Pinus cembroides	SPECIES	25.360299	-100.831743	720	Mexico	Coahuila	2025	HUMAN_OBSERVATI
5237583716	GBIF	Pinus cembroides	SPECIES	25.312735	-100.903189	1838	Mexico	Coahuila	2025	HUMAN_OBSERVATI
5291913515	GBIF	Pinus cembroides	SPECIES	25.247402	-100.486832	—	Mexico	Coahuila	2025	HUMAN_OBSERVATI
5292649018	GBIF	Pinus cembroides	SPECIES	25.271055	-100.557372	—	Mexico	Coahuila	2025	HUMAN_OBSERVATI
5828245197	GBIF	Pinus cembroides	SPECIES	24.245777	-99.909063	—	Mexico	Nuevo León	2025	HUMAN_OBSERVATI

Continue to cleaning →

Figure 2.4. The example narrowed down to a regional study: from 2,400 records to 425. The **State** column appears even where GBIF did not supply it, because the coordinate assigned it.

425

THE PATH OF THE EXAMPLE

2,400 records downloaded → **1,719** when keeping only Mexico → **425** when keeping Coahuila, Nuevo León and Tamaulipas. Each step is a decision that should be written down in the methods section.

2.6 How to choose the study area

There is no single rule, but there is a sequence of questions that works:

1. **What is the question?** A national study of a species calls for the whole country; an assessment of a management unit calls for that unit and its surroundings; a comparison between subspecies calls for the area of both.
2. **How far has the species been able to spread?** That is the biological criterion: the biogeographic region it occupies, not the rectangle that fits on the screen. Barriers —a desert, a sea, a mountain range— make good limits.
3. **Is the contrast represented?** The area must contain sites with conditions similar to those the species occupies and also sites that are clearly different. An area that contains only its habitat prevents the model from learning what it avoids.
4. **Can I justify it in one sentence?** If the answer is "because it looked good", go back to point 2.

2.7 Common problems

What you see	What happened and what to do
The count does not drop when you tick a box	The facet counts refer to the downloaded set. If you had already filtered by another facet, the number on the result card does drop, but the one next to the box does not change.
Many records with "no state assigned"	Normal if your download includes other countries: no Mexican state applies to them. If they are Mexican, check the assignment mode: with `from the record text only`, every record GBIF delivered without that field is left out.
A state with records you did not expect	It may be a little-known real population, a cultivated specimen or a wrong coordinate. Look at it on the Block 4 map before deciding.
The bounding box leaves out records you wanted	Remember that longitude is negative in the Americas: a box from −104 to −97 does not include −110. The `Use the selection as the rectangle` button saves you from typing the numbers by hand.

What you see	What happened and what to do
Your GeoJSON loads but the list is empty	The polygons carry no recognisable name in their properties, or the file is in a projection other than latitude–longitude. Save it in geographic coordinates (EPSG:4326).
Fewer than fifteen records remain	The area is too small for modelling. Enlarge it, or lower the goal to describing the environment of the points (Blocks 5 and 7) without fitting a model.

3

BLOCK 3 · DATA HYGIENE

Coordinate cleaning

Nine rules that look for the typical errors of a biodiversity database, each with its count before it is applied, and a before-and-after report. None of them removes anything unless you decide so.

3.1 Why clean

3.2 The nine rules

3.3 The three parameters

3.4 Informative rules

3.5 The before-and-after report

3.6 What cleaning does not do

3.7 What to download

3.8 Common problems

3.1 Why clean

Biodiversity databases accumulate two centuries of work done with very different means: handwritten labels, localities written from memory, georeferencing done afterwards from a text, hurried digitisation. The resulting errors are not random: they recur in the same forms, and that is why they can be searched for with rules.

What is at stake is not how the map looks. A record bearing the coordinates of the centre of the country brings into the model a climate the species never occupied; fifty duplicate records of the same tree tell the algorithm that this climate is fifty times more important than the rest; a specimen grown in a European garden widens the niche until it becomes unrecognisable.

> **DUPLICATES CARRY WEIGHT**
>
> A presence-only model learns from the contrast between where there *are* records and what the environment is like in general. If a site appears fifty times —because fifty people photographed the same tree next to the viewpoint— the model takes that environment to be fifty times more common than it is. Removing duplicates is not cosmetic tidying: it stops sampling effort from passing itself off as ecological preference.

3.2 The nine rules

Each rule is switched on or off separately and shows **how many records it would flag** before you apply anything. That live count is the useful part: seeing that a rule flags 400 records forces you to understand why before you delete them.

Rule	What it looks for and when to switch it off
Missing or out-of-range coordinates	Latitude or longitude empty, containing text, or impossible (|lat| > 90, |lon| > 180). There is no reason to switch it off.
Coordinate (0, 0)	The "null island" in the Gulf of Guinea, which appears when a program writes zeros instead of leaving the field empty.
Latitude ≈ longitude	Both values almost identical and far from zero: the pattern left by copying one cell over the other. Since it can genuinely occur along a diagonal band of the planet, the app treats it as informative.
GBIF geospatial issues	GBIF's own flags: country not matching the coordinate, presumably swapped coordinate, and others. Informative, because some flags are harmless.
Exact coordinate duplicates	Records with the same latitude and longitude, down to the last decimal. The first is kept. It is usually the rule that removes the most records.
Duplicates after rounding	Records that coincide after rounding to N decimals: the same tree recorded with different precisions. With 4 decimals, "the same site" means about 11 m.
Low precision	Declared uncertainty above the threshold, or coordinates with two decimals or fewer — which already amounts to more than a kilometre. Informative, because its threshold depends on the resolution of your layers.
Country centroid	Points within about 10 km of the geographic centre of the stated country: the classic mark of a record georeferenced "at country level".
Spatial outliers	Records whose distance to the centre of the point cloud exceeds k times the interquartile range, **computed per taxon**. It detects whatever lies far from where everything else is.

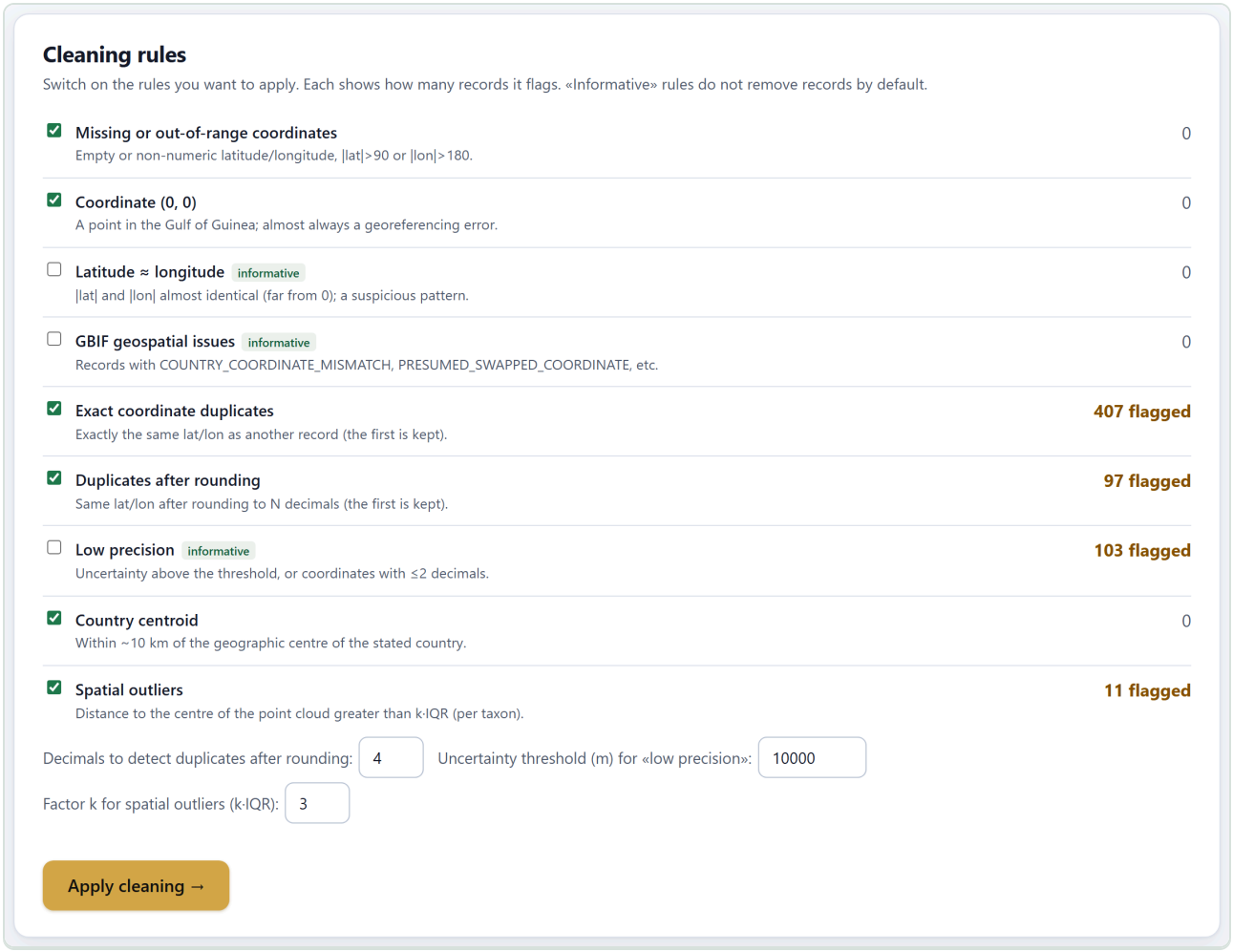


Figure 3.1. The nine rules with the counts for the example. The three marked as informative come switched off: they point out suspicious records without removing them.

3.3 The three parameters

Parameter	What it changes
Decimals to detect duplicates after rounding default 4	Defines how close counts as "the same site": 4 decimals ≈ 11 m, 3 ≈ 110 m, 2 ≈ 1.1 km. With 1 km layers it makes sense to go down to 2; with 18 km layers, like those of the example, the thinning by cell in Block 9 does that job better.
Uncertainty threshold default 10,000 m	The declared uncertainty above which a record is considered low precision. A sensible rule: the threshold should not be larger than the cell size of your layers.
Factor k for outliers default 3	How many interquartile ranges of distance are tolerated. Lowering it to 1.5 flags many more records —and starts flagging real populations—; raising it to 5 lets almost everything through.

3.4 Informative rules: flagging without deleting

Three rules come switched off on purpose. It is not that they matter less: it is that the decision depends on your study. A record with 5 km of uncertainty is useless for a 1 km model and perfectly usable for an 18 km one. The app tells you how many there are and you decide.

> **! NEVER DELETE WITHOUT LOOKING**
>
> A spatial outlier may be an error… or the most interesting disjunct population in your study. Before applying the rule, go to the Block 4 map and look at where the flagged points are. If they are real populations, switch the rule off and remove by hand the records that really are errors. The manual insists: the rules propose, the researcher disposes.

3.5 The before-and-after report

When you press `Apply cleaning →` the report appears: how many records entered, how many remained, how many left and through which rule.

▸ THE EXAMPLE, RULE BY RULE

With the default rules, the 2,400 records of the Mexican pinyon come down to **1,885** (78.5% kept). The 515 that go are: **407** exact coordinate duplicates, **97** duplicates after rounding to four decimals and **11** spatial outliers. The other rules flagged nothing: there are no impossible coordinates, no (0, 0) and no country centroids.

The informative rules also pointed out **103** low-precision records, which were kept. With 18 km cells they do no harm; with 1 km layers it would be worth reviewing them.

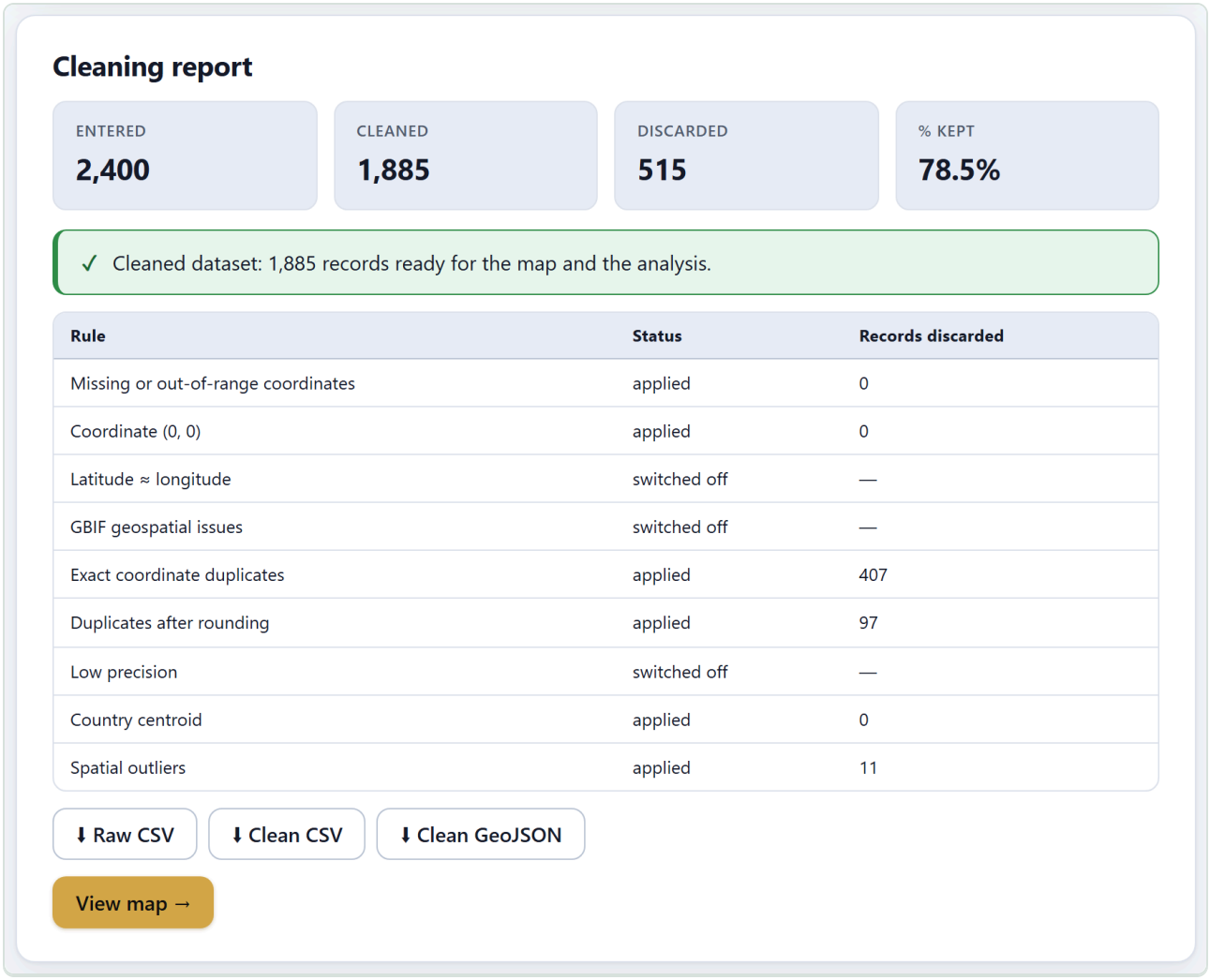


Figure 3.2. The report for the example. The table is the one to copy into the methods section: it says exactly what was applied and what each rule cost.

THE ELEVEN OUTLIERS, ONE BY ONE

It is worth seeing who the 11 records removed by the outlier rule are, because they explain what it is for:

- **Nine are trees cultivated outside the Americas**: six in New Zealand, two in France and one in Austria. The rule found them without knowing anything about countries or cultivation: they are simply very far from the rest of the cloud.
- **One is at (23.000, –102.000)**, whole degrees, in the middle of Zacatecas. It is a record of *Pinus cembroides* subsp. *orizabensis*, whose other 65 records are in Puebla (38), Veracruz (19), Tlaxcala (5), San Luis Potosí (3) and Querétaro (1). A coordinate in round degrees, hundreds of kilometres from its population, is a placeholder that someone never corrected.
- **The last one** is another *orizabensis*, on the boundary between Guanajuato and Querétaro, also outside the range of the subtaxon.

The fact that the rule works **per taxon** is what makes the second finding possible: (23, –102) lies within the range of the species, but very far from that of its subspecies.

3.6 What cleaning does not do

- **It does not check identifications.** If someone confused two pines, the record has a perfect coordinate and a wrong name, and no geographic rule will detect it. That can only be fixed by examining specimens or by trusting sources.
- **It does not correct sampling bias.** Removing exact duplicates helps, but the build-up of records along roads is still there. Thinning to one presence per cell is done in Block 9, and is explained there.
- **It does not judge age.** If you are interested in the recent climate, filter by year in Block 2.
- **It does not decide for you.** Everything it does is recorded in the report table, and the raw set remains available for download.

3.7 What to download

File	What it is for
↓ Raw CSV	Everything downloaded, neither filtered nor cleaned. It is your backup: keep it together with the date of the query.
↓ Clean CSV	The set the following blocks will work with. It is what goes into the supplementary material of a paper.
↓ Clean GeoJSON	The same points as a geographic layer, to open them in a mapping program.

3.8 Common problems

What you see	What happened and what to do
More than half of the records go	They are almost always exact duplicates, and that is almost always fine. Check the report table: if most of them left as duplicates, your set had many observations of the same site.
The outlier rule flags populations you know to be real	Raise the factor k, or switch the rule off and remove by hand whatever really is an error. With species of disjunct distribution —those of isolated mountain ranges— it is always worth reviewing it.
All the counts are zero	Your data were already clean, or you are cleaning a small set of your own. It is not an error.
Fewer records remain than you need	Go back to Block 2 and enlarge the area or the period before lowering the demands of the cleaning.
I want to know which ones left	Download the raw and the clean CSV and compare them by record identifier: the difference is the discarded records.

FOR THE METHODS SECTION

One sentence is enough, and with these data it writes itself: "Of the 2,400 records downloaded from GBIF, 407 exact coordinate duplicates, 97 duplicates at four decimals and 11 spatial outliers (k = 3, per taxon) were removed, giving a final set of 1,885 records".

4

BLOCK 4 · SEEING IS BELIEVING

The records map and the map studio

The first map serves two quite different purposes: checking the data by eye, and producing the figure that will end up in the paper. The map studio does both, and exports at whatever resolution and in whatever format the journal asks for.

4.1 The map as a quality check

4.2 What the map shows

4.3 The map studio

4.4 Exporting for publication

4.5 The preview and the clip

4.6 What a publishable map needs

4.7 Common problems

4.1 The map as a quality check

No automatic rule sees what a trained eye sees in thirty seconds of looking at a map. Before extracting a single environmental variable, it pays to look at the points and ask yourself five questions.

1. **Are there points in the sea?** A coordinate in the ocean a few kilometres off the coast is almost always a poorly georeferenced coastal record; far from the coast, it is a sign error or a transcription error.
2. **Does the distribution have the shape you expect?** The pinyon pine should trace the two Sierras Madre. If a uniform cloud appears over the high plateau, something is wrong.
3. **Are there suspicious alignments?** Points lined up along a road are roadside records: legitimate, but they show where people collect, not where the species lives.
4. **Is there a lone point very far away?** A botanic garden, a cultivated specimen or, now and then, the most interesting find of the whole study.
5. **Are the subtaxa where they should be?** Colouring by taxon reveals at a glance a subspecies in the wrong place.

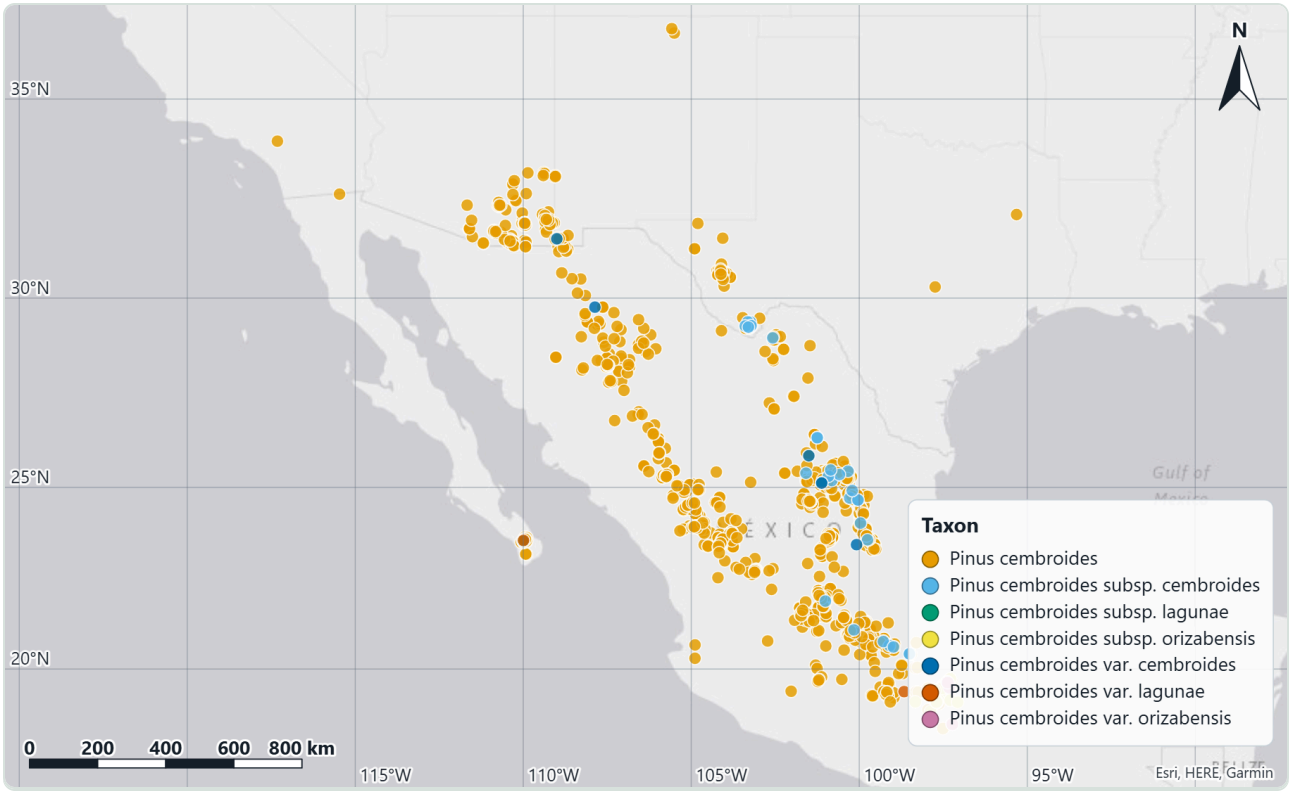


Figure 4.1. The 1,885 cleaned records of the example, exported with the map studio. The two Sierras Madre stand out, as do the Sierra de La Laguna in Baja California Sur and the population in central Mexico; the colours separate the species from its subspecies and varieties. This image came straight out of the app, untouched.

4.2 What the map shows

The map plots the records that survived Block 3, coloured by the criterion you choose: by taxon, by record type or in a single colour. On top of it you can switch on the cartographic elements (legend, scale bar, north arrow, grid) and the outline of the region you chose in Block 2.

THE BASE MAP COMES FROM THE INTERNET

The background tiles are downloaded from an online service. Without a connection, the map shows the points over a plain background, which is still perfectly usable, and the scale bar and the coordinates keep working. If a base map does not respond, the app switches to another one by itself.

4.3 The map studio

Below the map sits the **map studio**, folded away until you need it. Its seven tabs cover everything that separates a map on screen from a publishable figure.

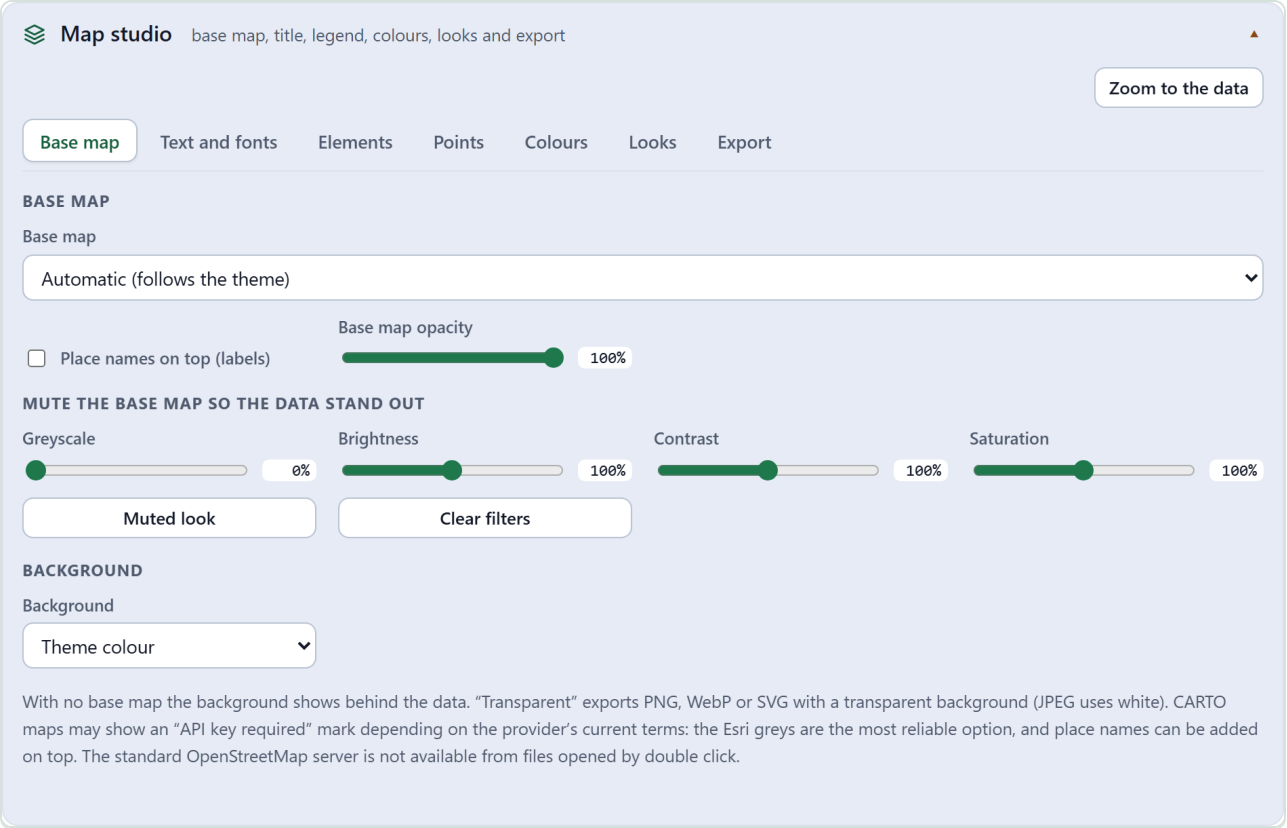


Figure 4.2. The studio open on Base map. Fourteen different backgrounds, and four controls to *tone it down*: in a scientific map the background should help readers find their way, not compete with the data.

Tab	What it controls
Base map	The background: greys, streets, topographic, terrain, imagery or none; its opacity and how far it is toned down in greyscale, brightness, contrast and saturation. The Muted look button makes the background discreet with a single click.
Text and fonts	Title, subtitle and credit, and the typeface of **each element separately**: the title, the legend labels, the scale bar numbers, the grid coordinates and the north arrow letter, each with its own size, bold, italic and colour.

Tab	What it controls
Elements	What is drawn: legend (inside or outside the map, in one or several columns), scale bar, north arrow, coordinate grid and frame.
Points	Shape, size, outline and transparency of the symbols, and the heat map for when the points pile up.
Colours	The palette for the taxa or the groups, and the colour ramps for the continuous maps of Block 9.
Looks	Ready-made presets, and the option of saving your own look as a file to reuse it in another figure or another study.
Export	Framing, size, resolution, format, clip and preview: section 4.4.

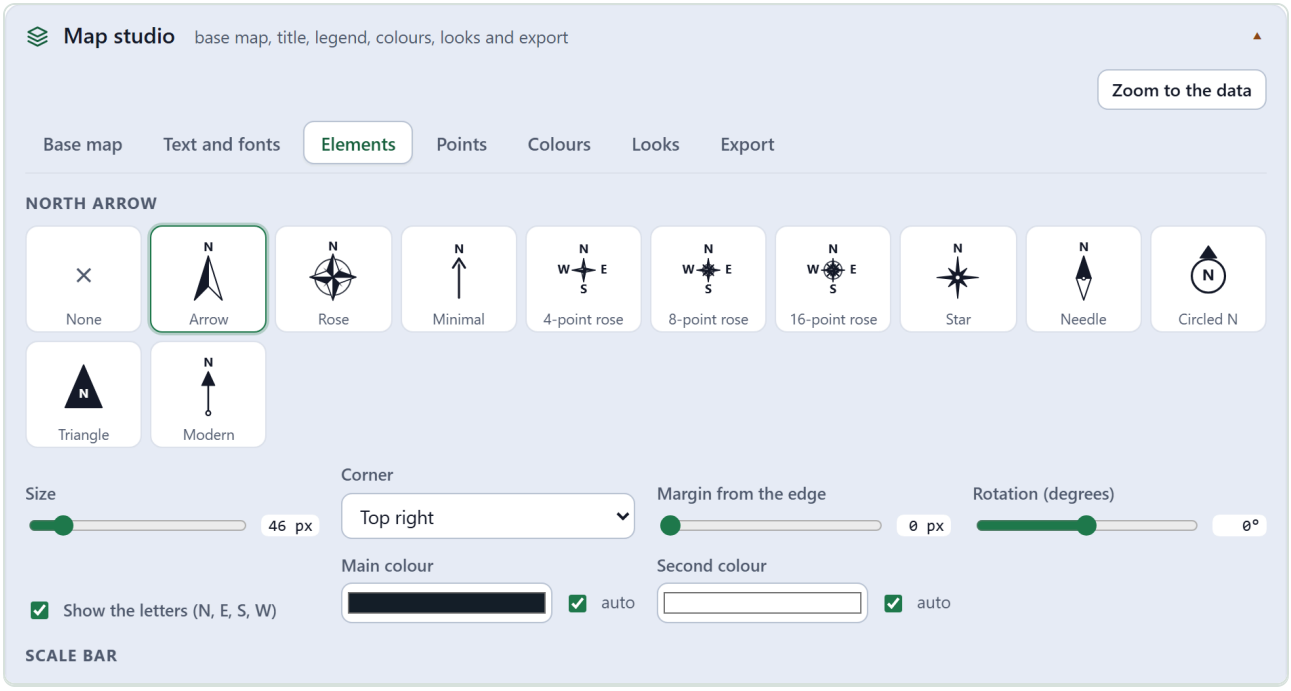


Figure 4.3. The Elements tab: each cartographic piece is switched on, placed and sized separately. The north arrow and the scale bar come in several styles.

4.4 Exporting for publication

The Export tab answers the three questions any journal asks: **what is shown**, **at what size** and **in what format**.

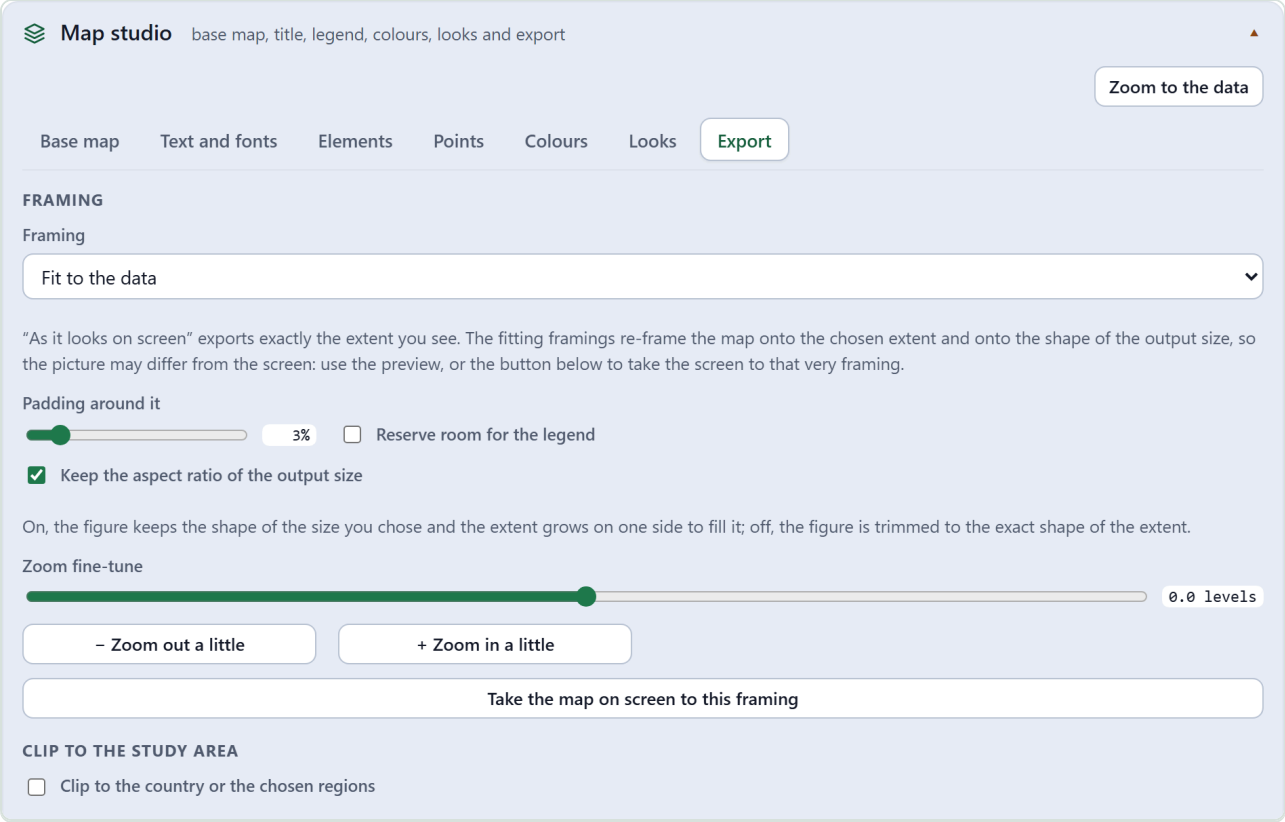


Figure 4.4. The framing decides which part of the world goes into the figure. `As it looks on screen` exports exactly what you see; the other framings recompose the map over the chosen extent.

Control	Options and what they are for
Framing	`As it looks on screen`, `Fit to the data`, `Fit to the chosen states or regions`, `The whole of Mexico` and `Manual extent` with its four coordinates. The padding and the zoom fine-tune refine the result without touching the map on screen.
Size	As shown, A4 landscape or portrait, single or double journal column (85 and 180 mm), a 16:9 slide, square, or a custom size typed in millimetres.
Resolution	A multiplier for the on-screen size, or 150, 300 and 600 dpi when the size is in millimetres. Most journals ask for 300 dpi for halftone figures and 600 for line art.
Format	**PNG** (lossless, supports a transparent background), **JPEG**, **WebP**, **TIFF**, georeferenced **GeoTIFF** and **SVG** (vector, with the data and the text, but without the base map).

WHICH FORMAT TO CHOOSE FOR WHAT

PNG for a document or a presentation. **TIFF** when the journal requires it or when you are going to keep editing the map in a design program. **GeoTIFF** if you want to open the image in a mapping program and have it land in its place on the globe: it carries the coordinates and the reference system inside. **SVG** when you want to change the text or the colours later without losing sharpness; since it does not include the base map, it is mainly useful for figures of points on a plain background.

4.5 The preview and the clip

Two features spare you the back-and-forth of exporting, looking at the file, correcting and exporting again.

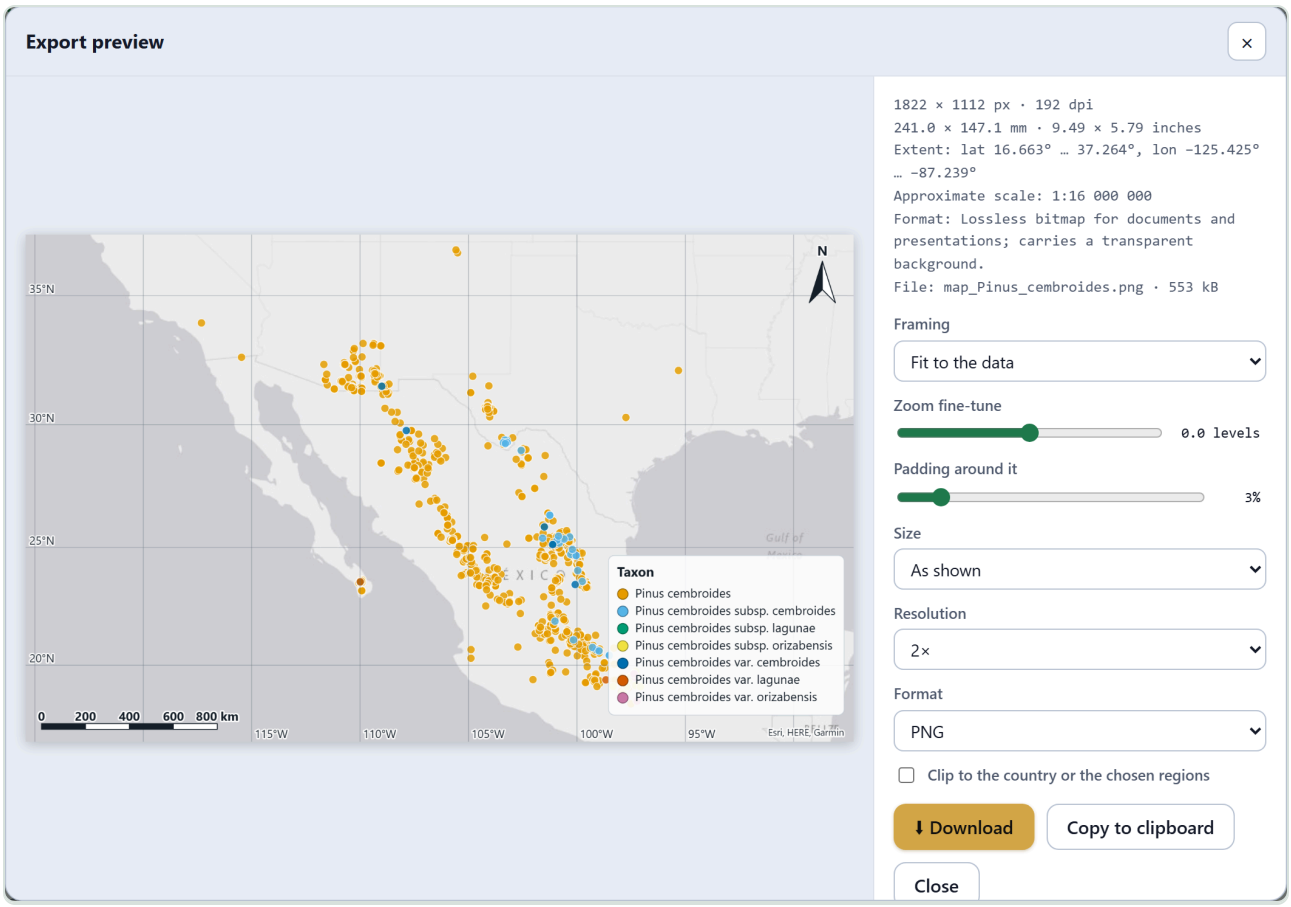


Figure 4.5. The preview shows **the image that will be saved**, not an approximation, together with its size in pixels and millimetres, its resolution, its geographic extent and its approximate scale. The controls on the right change the image without closing the window.

The **clip to the study area** leaves out of the image everything that is not the country or the regions chosen in Block 2: the background tiles, the suitability surfaces and the points outside disappear, leaving only the silhouette of the territory, with its outline drawn and the outside transparent or in the colour you choose. It is the way to deliver a map of Mexico (or of a single state) without the neighbouring country peeking in.

1:16M

KEY FACT

The preview works out the **approximate scale** of the figure from its width in millimetres and the extent it covers. It is the number a map's figure caption asks for, and the one almost nobody remembers to calculate.

4.6 What a publishable map needs

Element	Why
Scale bar	Without it no distance can be judged. A bar is better than a "1:16,000,000" text: the bar remains valid if the figure is reduced.
North arrow	Essential when the map is not oriented with north at the top, and good practice always.
Coordinates	The grid with its degrees lets readers locate any point and compare with other maps.
Legend	What each colour and each symbol means, with the scientific names in italics.
Base map credit	Tile providers require attribution; the app writes it in the corner by itself and it is exported with the image. The figure caption should also give the source of the records, how many there are and their date.

4.7 Common problems

What you see	What happened and what to do
The map shows the whole world	It was drawn before its space had a size. Press Zoom to the data, in the corner of the studio.
The background is grey, has no detail or carries a watermark	There is no connection, the provider did not respond, or it adds its mark under its terms of use. Switch base map (the greys are the most reliable) or use None (background colour): for a figure of points, a plain background is usually enough.
The image does not match the screen	The framing is not As it looks on screen. Open the preview, or use Take the map on screen to this framing.
The lettering comes out tiny in the figure	When exporting to a paper size, the text is sized with the figure. Increase the font size of each element in Text and fonts and check the preview again.
The points pile up and nothing can be made out	Reduce the symbol size, increase its transparency, or switch to a heat map in the Points tab.

5

BLOCK 5 · THE ENVIRONMENT OF EACH POINT

Environmental variables: climate, elevation, Köppen and soil

Here the points stop being coordinates and become a table of conditions: nineteen bioclimatic variables, elevation, the climate type and, if you ask for it, the soil. That table is the raw material for everything that follows.

5.1 What the bioclimatic variables are

5.2 Three ways to load the layers

5.3 How the value at a point is extracted

5.4 Köppen-Geiger and soil type

5.5 The result

5.6 The thematic map

5.7 Extremes give records away

5.8 Which resolution to use

5.9 Common problems

5.1 What the bioclimatic variables are

The nineteen bioclimatic variables are not raw data: they are **summaries** calculated from the monthly series of temperature and rainfall, designed to make biological sense. They do not tell you how much it rained on a given day, but how much rain falls in the driest quarter, or how far the temperature drops in the coldest month: the things that limit a plant.

Code	What it measures	Unit
BIO1	Annual mean temperature	°C
BIO2	Mean diurnal temperature range	°C
BIO3	Isothermality (BIO2 / BIO7 × 100)	%
BIO4	Temperature seasonality (standard deviation × 100)	°C×100
BIO5	Max temperature of warmest month	°C
BIO6	Min temperature of coldest month	°C
BIO7	Temperature annual range (BIO5 – BIO6)	°C
BIO8	Mean temperature of wettest quarter	°C
BIO9	Mean temperature of driest quarter	°C
BIO10	Mean temperature of warmest quarter	°C
BIO11	Mean temperature of coldest quarter	°C
BIO12	Annual precipitation	mm
BIO13	Precipitation of wettest month	mm
BIO14	Precipitation of driest month	mm
BIO15	Precipitation seasonality (coefficient of variation)	%
BIO16	Precipitation of wettest quarter	mm
BIO17	Precipitation of driest quarter	mm
BIO18	Precipitation of warmest quarter	mm
BIO19	Precipitation of coldest quarter	mm
ELEV	Elevation above sea level	m

WHY THEY REPEAT ONE ANOTHER

All nineteen come from the same twelve monthly temperatures and the same twelve monthly rainfall totals. BIO1 and BIO11, or BIO12 and BIO16, measure almost the same thing over most of the world. Putting them all into a model adds no new information and does cause collinearity: that is why Block 6 exists, to decide which ones to keep.

5.2 Three ways to load the layers

The app neither downloads layers nor carries them inside: the WorldClim layers **cannot be redistributed without permission**, so the program ships with none of them. You download them once (`datos/LEE-ME.md`) and the app **reads them from your disk**. Köppen-Geiger, on the other hand, is included.

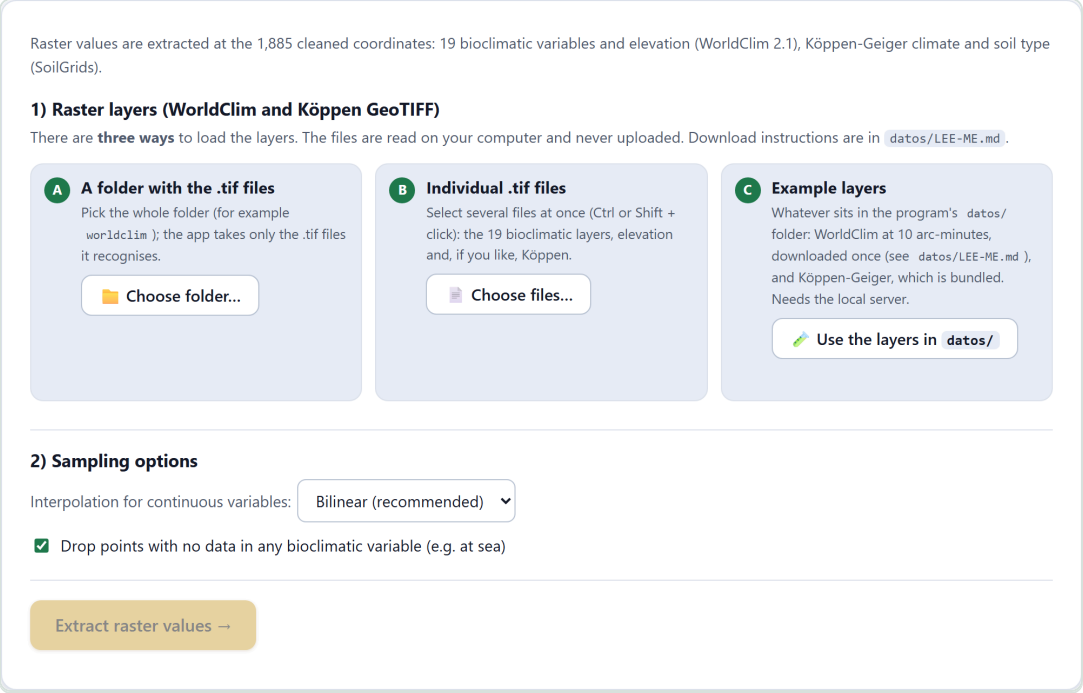


Figure 5.1. The three ways to load the layers, each with its own button and its own status note, so there is never any doubt about which one was used.

Way	When it suits you
Choose folder...	You have the WorldClim folder unzipped. You pick the whole folder and the app takes from it only the files it recognises. It is the most convenient option for a real study.
Choose files...	Your layers are scattered or you want to load only some of them. Select several at once with Ctrl or Shift.
Use the layers in `datos/`	Reads the program's `datos/` folder, which is the best place to put the layers when you download them. It needs the local server.

WHICH NAMES IT RECOGNISES

The app identifies the layers by their file name, following the WorldClim pattern: `wc2.1_res_bio_1.tif` … `bio_19.tif` and `elev.tif`. If your files have different names, simply rename them to that pattern. Appendix A sets out the full rules.

5.3 How the value at a point is extracted

For each record and each layer, the app converts the coordinate into a row and a column of the grid and reads the value. Two details matter:

- **It reads only the window it needs.** A 30 arc-second file has billions of cells; the app opens the file, works out where your points fall and reads only those pieces. That is why it works in the browser with layers that would not fit in memory.
- **The sampling method is chosen under** Sampling options **.** Bilinear (recommended) averages the four neighbouring cells weighted by distance, which avoids abrupt jumps between adjacent cells; Nearest neighbour returns the value of the cell as it is. The interpolation applies **only to the continuous variables**: categorical ones, such as Köppen, are always read by nearest neighbour, because averaging climate codes means nothing.

When a point falls in a cell with no data (the sea, or outside the extent of your layers) the variable is left empty. The Drop points with no data in any bioclimatic variable checkbox, ticked by default, leaves them out of the table; if you untick it, they are kept with gaps and each analysis will decide what to do with them.

5.4 Köppen-Geiger and soil type

Two **categorical** variables complete the picture. The **Köppen-Geiger** classification sums up the climate in a label (cold steppe arid, temperate with dry winter…) that is understood at once and is widely used in descriptive ecology. It is just another layer, in a small file, and it is loaded like the rest.

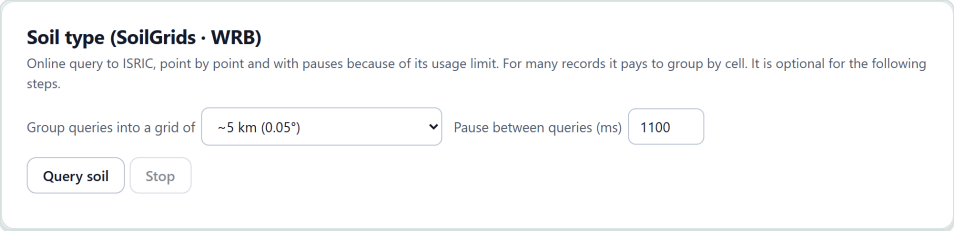
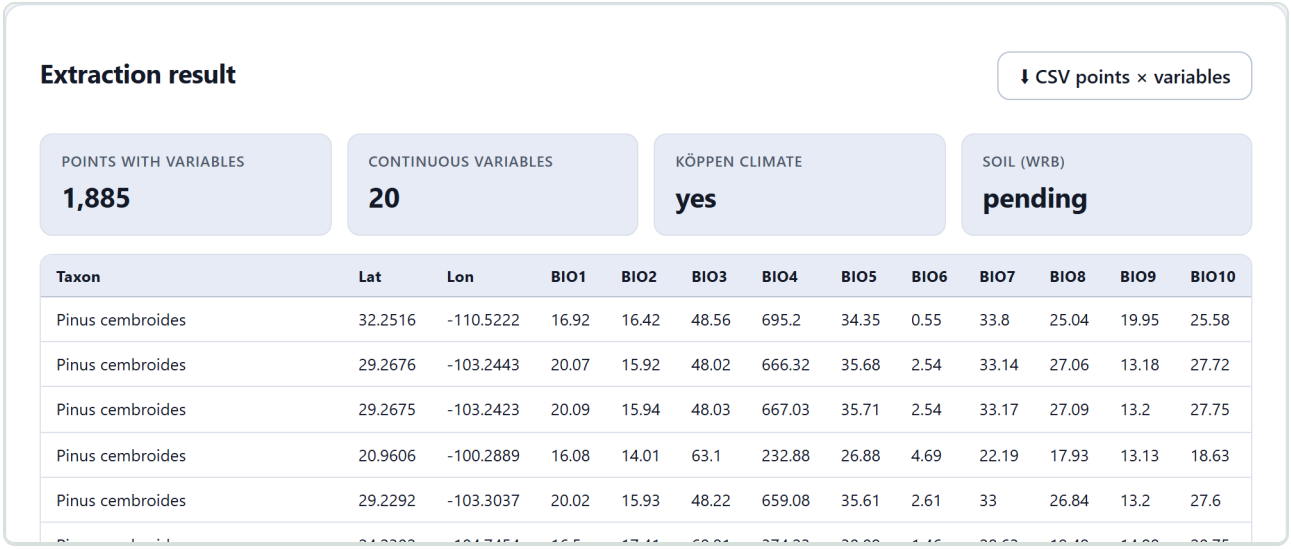


Figure 5.2. The soil type is queried online, point by point and with pauses. It is the only feature of the app that sends coordinates to an external service, which is why it is kept separate and optional.

> **SOIL TAKES TIME**
>
> The public soil service limits how many queries per second it accepts, so the app asks slowly: with hundreds of points the query can take several minutes, and it can be stopped and resumed. For a model with many records, first decide whether soil is really needed: in most climatic niche studies it is not.

5.5 The result

The extraction produces the table that feeds everything else: one row per record, one column per variable.



Taxon	Lat	Lon	BIO1	BIO2	BIO3	BIO4	BIO5	BIO6	BIO7	BIO8	BIO9	BIO10
Pinus cembroides	32.2516	-110.5222	16.92	16.42	48.56	695.2	34.35	0.55	33.8	25.04	19.95	25.58
Pinus cembroides	29.2676	-103.2443	20.07	15.92	48.02	666.32	35.68	2.54	33.14	27.06	13.18	27.72
Pinus cembroides	29.2675	-103.2423	20.09	15.94	48.03	667.03	35.71	2.54	33.17	27.09	13.2	27.75
Pinus cembroides	20.9606	-100.2889	16.08	14.01	63.1	232.88	26.88	4.69	22.19	17.93	13.13	18.63
Pinus cembroides	29.2292	-103.3037	20.02	15.93	48.22	659.08	35.61	2.61	33	26.84	13.2	27.6

Figure 5.3. The example: 1,885 points with 20 continuous variables and the Köppen climate. The download button delivers the complete table as CSV, the file you attach to a paper or open in another program.

5

THE PINYON PINE'S ENVIRONMENT IN THREE NUMBERS

With the 1,885 cleaned records of the example, the central half of the points (from the first to the third quartile) lives like this:

- **Annual mean temperature** between **14.8 and 17.9 °C**, with a median of 16.2 °C (extremes 2.6 and 22.7).
- **Annual precipitation** between **414 and 551 mm**, with a median of 486 mm (extremes 69 and 2,098).
- **Elevation** between **1,473 and 2,210 m**, with a median of 1,978 m (extremes 27 and 3,065).

And by climate type: **1,272** records in cold steppe arid climate (BSk), **227** in temperate with dry winter and warm summer (Cwb), **186** in hot steppe arid (BSh), **95** in hot desert arid (BWh) and **71** in temperate with no dry season (Cfb). The portrait is that of a pine of cool, semi-arid mountains.

5.6 The thematic map

The last card colours the points by the value of the variable you choose. It is the quickest way to see whether a variable has geographic structure, and whether the gradient it shows makes sense.

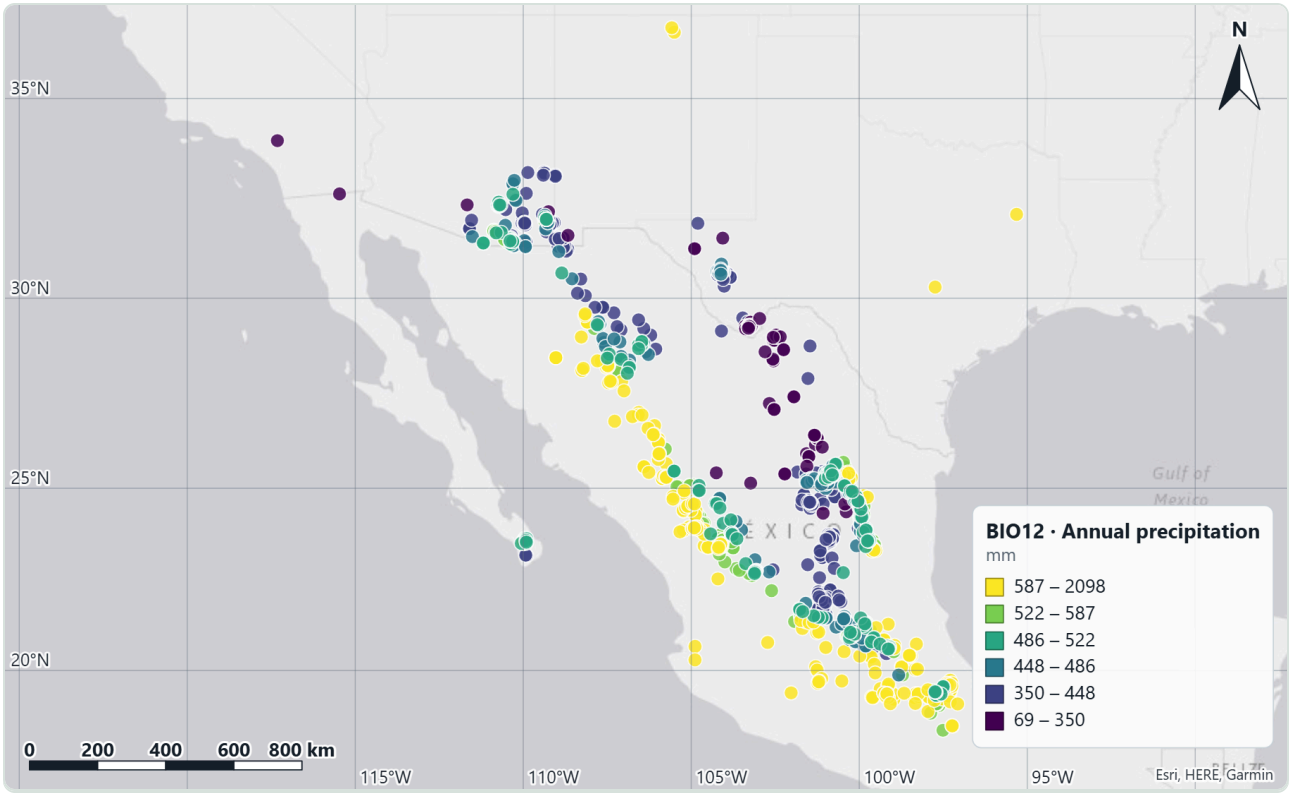


Figure 5.4. The records coloured by **BIO12**, annual precipitation, in six classes. The contrast stands out between the Sierra Madre Occidental, with its intermediate values, and the driest points on the high plateau and in the north. The map is edited and exported with the same studio as in chapter 4.

5.7 Extremes give records away

Before moving on, look at the minimum and maximum values of two or three variables. It is a free quality check: the environment exposes records that geography could not.

THREE EXTREMES IN THE EXAMPLE

- The lowest record is at **27 m elevation**, in the Mexicali Valley, with **69 mm** of annual rainfall. A mountain pine does not live in the desert at sea level: it is a planted tree or a wrong coordinate.
- Two records in Texas, at 143 and 169 m, with 1,096 and 849 mm of rainfall: urban areas in the east of the state, far removed from the conditions of the rest.
- The wettest reaches **2,098 mm** in central Veracruz, at 689 m. Again, a value that calls for review.

None of the three had been flagged by the geographic rules of Block 3, because they lie within the general distribution area. They show up when you look at the environment. They can be removed by going back to Block 2 with a filter, or kept with the decision noted down.

5.8 Which resolution to use

Resolution	Cell	When
30 seconds	≈ 1 km	Local or mountain studies, where relief changes the climate within a few kilometres. The files are huge: it pays to crop them first.
2.5 minutes	≈ 4.6 km	The sensible middle ground for a state-wide or regional study.
5 minutes	≈ 9 km	National studies with records of uneven precision.
10 minutes	≈ 18 km	Continental studies and teaching: it is the one used in the examples in this manual.

The rule of thumb: **the cell should not be finer than the precision of your coordinates**. With records georeferenced to 5 km, a 1 km layer gives a false sense of detail.

5.9 Common problems

What you see	What happened and what to do
"… files read, none recognised as WorldClim or Köppen"	The file names do not follow the expected pattern, or you chose a folder that does not contain the `.tif` files. Rename them or go into the right subfolder.
Many points with no data	Your layers do not cover the whole area of the records (the typical case of a cropped layer) or the points fall in the sea. Compare the extent of the layers with that of your data.
The extraction takes a long time	Normal with 30-second layers and thousands of points: 20 large files are being read. The progress bar advances layer by layer.
The temperature values look multiplied	Some older distributions store temperature in tenths of a degree. If BIO1 comes out as 162 instead of 16.2, your layers are of that kind.
The soil query stops halfway	The service throttled the queries. Press the button again: it carries on where it stopped and keeps what has already been queried.
Köppen comes out empty	The Köppen-Geiger layer is missing (it is a separate file) or it was left out of the file selection.
The button to continue to Block 6 warns you instead of moving on	You have not yet pressed Extract raster values →. The correlation is computed on that table, so it has to be created first; once the extraction is finished, Blocks 7, 9 and 10 become available too (Block 8 opens after the statistics of Block 7).

6

BLOCK 6 · FEWER VARIABLES, A BETTER MODEL

Correlation and variable selection

The nineteen bioclimatic variables often say the same thing. This block shows how much they overlap, computes the VIF live and proposes a handful of variables with little redundancy, without taking away the ones you consider essential.

6.1 What collinearity is and what it breaks

6.2 The correlation matrix

6.3 The variance inflation factor

6.4 The recommended selection

6.5 Pinning the variables that matter

6.6 The pair viewer

6.7 How many variables to use?

6.8 Common problems

6.1 What collinearity is and what it breaks

Two variables are collinear when one can be predicted well from the other. In climate data this is the rule, not the exception: where it is hotter the rainfall tends to differ, and as elevation rises temperature falls. The problem is not that collinearity exists, but what it causes when it is ignored.

What it breaks	How it shows
Variable importance	If BIO5 and BIO10 measure almost the same thing, the model shares the credit between them and neither looks important. With only one of them, that variable comes out as decisive. The ecological conclusion changes although the data do not.
The coefficients	In linear models (GLM, GAM) the coefficients become unstable: they change sign and size when a handful of records is removed. Tree-based models tolerate it better, but their importance measure still suffers.
Projection to another climate	This is the most serious damage and the least visible. A model fitted with two correlated variables learns the relationship between them *as it is today*; if that relationship changes in the future scenario, the model extrapolates over combinations it has never seen.

6.2 The correlation matrix

The matrix shows the correlation between every pair of variables, computed from the values extracted at your records, not from the climate of the whole country: what matters is how they relate **where your species lives.**

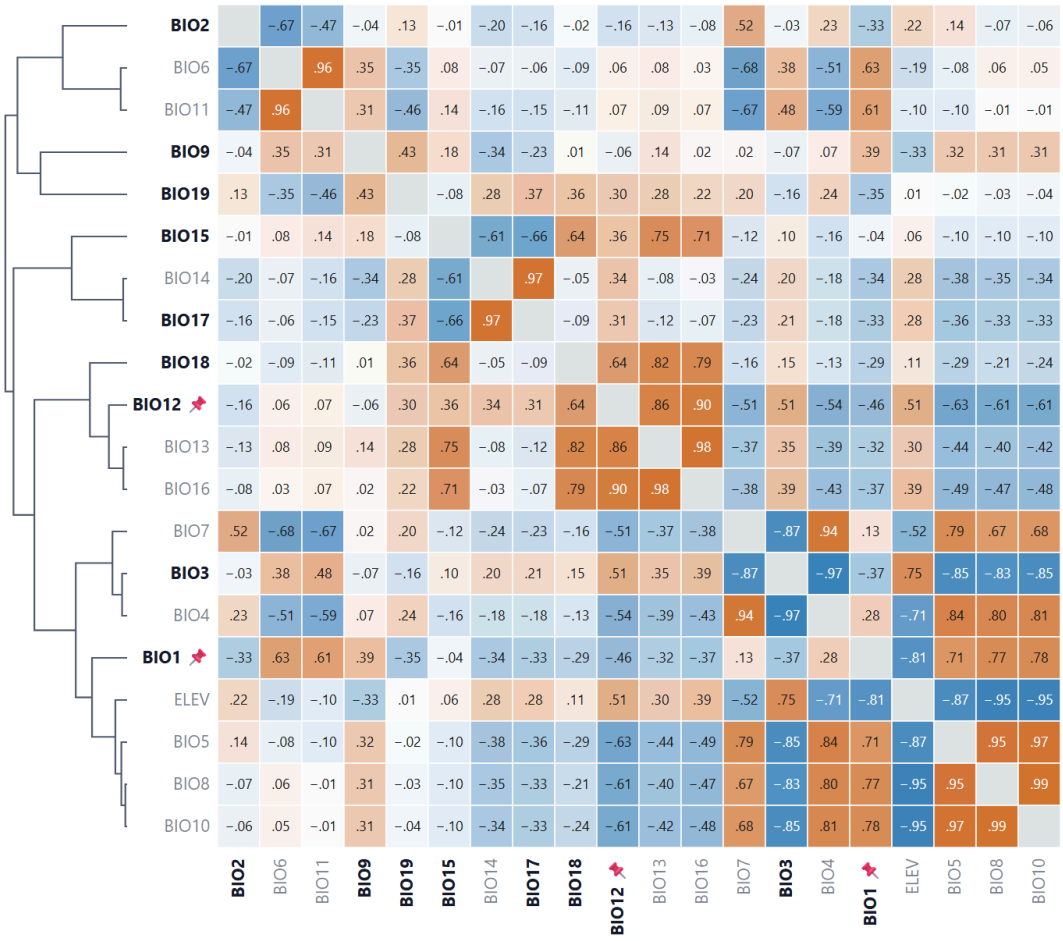


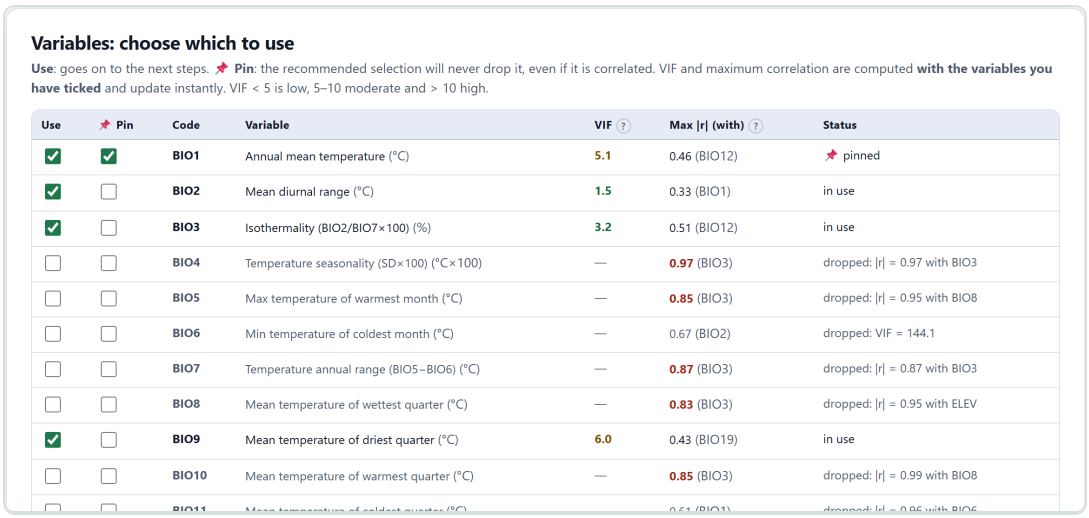
Figure 6.1. The example matrix as drawn on its card; the card's ⬇ PNG button exports it just as it looks. The tree on the left groups the variables that are alike: BIO6 and BIO11 (0.96), BIO14 and BIO17 (0.97), BIO13 and BIO16 (0.98), BIO8 and BIO10 (0.99). In bold, the selected variables; with a 📌 pin, the pinned ones.

- **Pearson or Spearman.** Pearson measures linear association; Spearman measures whether one variable increases when the other does, even if the relationship is curved. With climate variables they usually agree; when they differ a lot, there is a non-linear relationship worth looking at.
- **The order is not alphabetical.** Rows and columns are ordered by hierarchical clustering, so the blocks of colour that appear are real groups of interchangeable variables.
- **Click a cell** and the pair opens: two histograms and a scatter plot to help you decide which one to keep.

6.3 The variance inflation factor

> **WHAT THE VIF IS**
>
> Correlation looks at pairs; the **VIF** looks at sets. Each variable is regressed on **all the others** in the set and VIF = 1 / (1 − R^2) is computed. If the others explain 90% of its variation, the VIF is 10. A high VIF means that the variable is almost a combination of the others, even if it has no high pairwise correlation: that is the case the matrix alone cannot detect.
>
> The thresholds **5** and **10** are widely used reading conventions, not laws: below 5 collinearity is low; between 5 and 10, moderate; above 10, high.

In the variables table, the VIF and the maximum correlation are recomputed **with the variables you have ticked** every time you tick or untick one. You see the effect of each decision at once, and the **Status** column says why each variable was dropped: because of a high pairwise correlation or because of its VIF.



Variables: choose which to use

Use: goes on to the next steps. 📌 **Pin**: the recommended selection will never drop it, even if it is correlated. VIF and maximum correlation are computed **with the variables you have ticked** and update instantly. VIF < 5 is low, 5–10 moderate and > 10 high.

Use	📌 Pin	Code	Variable	VIF	Max |r| (with)	Status
☑	☑	BIO1	Annual mean temperature (°C)	5.1	0.46 (BIO12)	📌 pinned
☑	☐	BIO2	Mean diurnal range (°C)	1.5	0.33 (BIO1)	in use
☑	☐	BIO3	Isothermality (BIO2/BIO7×100) (%)	3.2	0.51 (BIO12)	in use
☐	☐	BIO4	Temperature seasonality (SD×100) (°C×100)	—	**0.97** (BIO3)	dropped: |r| = 0.97 with BIO3
☐	☐	BIO5	Max temperature of warmest month (°C)	—	**0.85** (BIO3)	dropped: |r| = 0.95 with BIO8
☐	☐	BIO6	Min temperature of coldest month (°C)	—	0.67 (BIO2)	dropped: VIF = 144.1
☐	☐	BIO7	Temperature annual range (BIO5−BIO6) (°C)	—	**0.87** (BIO3)	dropped: |r| = 0.87 with BIO3
☐	☐	BIO8	Mean temperature of wettest quarter (°C)	—	**0.83** (BIO3)	dropped: |r| = 0.95 with ELEV
☑	☐	BIO9	Mean temperature of driest quarter (°C)	6.0	0.43 (BIO19)	in use
☐	☐	BIO10	Mean temperature of warmest quarter (°C)	—	**0.85** (BIO3)	dropped: |r| = 0.99 with BIO8
☐	☐	BIO11	Mean temperature of coldest quarter (°C)		0.61 (BIO1)	dropped: |r| = 0.96 with BIO6

Figure 6.2. The variables table. BIO4 was dropped for its correlation of 0.97 with BIO3; BIO6 was dropped for a **VIF of 144.1**, although its highest pairwise correlation was only 0.67: the case that only the VIF uncovers.

6.4 The recommended selection

The ✨ Recommended selection button applies a simple, transparent procedure: first it removes the variables above the correlation threshold, keeping from each pair the one that carries more information about the rest; then, on what is left, it removes one at a time the variables above the VIF threshold, recomputing each time. Pinned variables are never touched.

Both thresholds are yours: **|r|** defaults to 0.8 and the **VIF** to 10, the conventions most used in the niche modelling literature. Lowering them leaves fewer, more independent variables; raising them does the opposite.

6.5 Pinning the variables that matter

An automatic procedure knows no ecology. If your species is limited by frost, BIO6 has to be in the model even if the algorithm prefers another variable; if you are going to discuss mean temperature and annual rainfall, BIO1 and BIO12 cannot be missing. That is what the pin is for: **pinning** a variable makes it untouchable.

▸ WITH AND WITHOUT PINS

With the 1,885 records of the Mexican pinyon and the default thresholds (|r| > 0.8, VIF > 10), the **automatic selection** keeps eight variables: BIO2, BIO3, BIO9, BIO16, BIO17, BIO18, BIO19 and elevation, with a maximum |r| of **0.79** and a maximum VIF of **6.1**. Neither BIO1 nor BIO12 survives: the procedure preferred their relatives.

Pressing 📌 Pin BIO1 and BIO12 and recompute turns the selection into nine variables (BIO1, BIO2, BIO3, BIO9, BIO12, BIO15, BIO17, BIO18 and BIO19) with a maximum |r| of **0.66** and a maximum VIF of **8.7**. In other words, requiring the two classic variables **lowered** the maximum correlation of the set, at the cost of one more variable and a somewhat higher VIF. A better result, and an interpretable one too.

! DECIDE ONCE, AND MAKE IT SHOW

The selection made in this block is the one Blocks 7, 8 and 9 use by default; in Blocks 8 and 9 the Those of step 6 button brings it back. If you change your mind later, come back here and recompute; do not leave two different selections in the same piece of work.

6.6 The pair viewer

The last card lists the pairs above the threshold, from the most to the least correlated, and lets you open any of them to see the point cloud.

Highly correlated pairs

Variable A	Variable B	r	In your selection	Decide
BIO8 Mean temperature of wettest quarter	**BIO10** Mean temperature of warmest quarter	0.99	—	Keep BIO8 Keep BIO10
BIO13 Precipitation of wettest month	**BIO16** Precipitation of wettest quarter	0.98	—	Keep BIO13 Keep BIO16
BIO14 Precipitation of driest month	**BIO17** Precipitation of driest quarter	0.97	✓ one	Keep BIO14 Keep BIO17
BIO5 Max temperature of warmest month	**BIO10** Mean temperature of warmest quarter	0.97	—	Keep BIO5 Keep BIO10
BIO3 Isothermality (BIO2/BIO7×100)	**BIO4** Temperature seasonality (SD×100)	−0.97	✓ one	Keep BIO3 Keep BIO4
BIO6 Min temperature of coldest month	**BIO11** Mean temperature of coldest quarter	0.96	—	Keep BIO6 Keep BIO11

Figure 6.3. The pairs above the threshold. Looking at the cloud prevents two mistakes: dropping a variable because of a high correlation that is really produced by a single group of extreme points, and keeping two variables whose relationship is a perfect straight line.

6.7 How many variables to use?

✓ THREE CRITERIA, IN THIS ORDER

They make sense	Every variable in the model should be justifiable in one sentence of the discussion: why that one, and not another, limits your species.
They do not repeat each other	|r| below 0.8 among all of them and VIF below 10; if you can go down to 0.7 and 5 without losing meaningful variables, so much the better.
There are not too many	With few presences, a model with many variables learns the data by heart. A prudent guide is no more than one variable per ten presences.

Five to ten well-chosen variables are enough for almost any niche study; nineteen are never needed.

6.8 Common problems

What you see	What happened and what to do
The matrix is empty	There is no variables table: go back to Block 5 and extract the raster values.
A huge VIF, in the hundreds	It is normal with all nineteen bioclimatic variables together; it means that the variable is almost an exact combination of others. Apply the recommended selection.
The selection leaves out "my" variable	Pin it and recompute. That is exactly what pinning is for.
Pearson and Spearman give very different results	The relationship between those variables is not linear. Open it in the pair viewer before deciding.
Very few variables are left	Raise the thresholds a little, or pin the ones you need. Three ecologically meaningful variables model better than ten redundant ones.
I changed the selection and the results of another block did not change	Blocks 7, 8 and 9 take the selection when you enter them, or, in 8 and 9, when you press Those of step 6. Go back to that block and reload the selection.

7

BLOCK 7 · GETTING TO KNOW THE DATA

Descriptive statistics and the figure studio

Before modelling you need to know what the data look like: the range in which the species lives, whether the distributions have one mode or two, whether a transformation is needed. Every figure is drawn as vectors and can be edited down to the last detail.

7.1 Describe before you model

7.2 The descriptive table

7.3 Histograms and density

7.4 A closer look at one variable

7.5 The normality tests

7.6 Suggested transformations

7.7 The figure studio

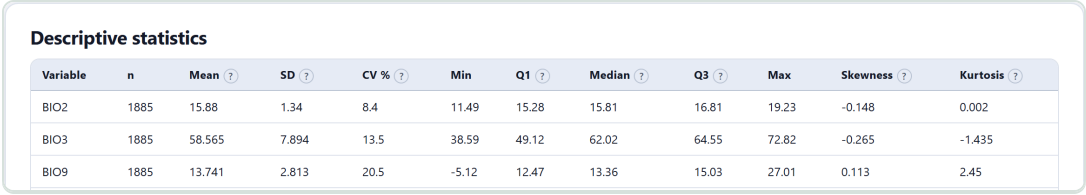
7.8 Common problems

7.1 Describe before you model

This block fits no model: it describes. And yet it is the one that most often changes the course of a study, because it is where you discover that a variable has two modes, that another is dominated by a handful of extreme values, or that the temperature range the species occupies is narrower than was thought.

It is also the first block that uses the **computing engine**: when you press `Run statistical analysis →` the app loads Python inside the browser. The first time takes a while; after that it is immediate.

7.2 The descriptive table



Descriptive statistics

Variable	n	Mean	SD	CV %	Min	Q1	Median	Q3	Max	Skewness	Kurtosis
BIO2	1885	15.88	1.34	8.4	11.49	15.28	15.81	16.81	19.23	-0.148	0.002
BIO3	1885	58.565	7.894	13.5	38.59	49.12	62.02	64.55	72.82	-0.265	-1.435
BIO9	1885	13.741	2.813	20.5	-5.12	12.47	13.36	15.03	27.01	0.113	2.45

Figure 7.1. The eleven measures of each variable. Each heading has its help button with the formula, the reading scale and a warning about the most common mistake.

Measure	How to read it
Mean and median	If they are close, the distribution is symmetric; if the mean is much larger, there is a long tail to the right. To describe the environment of a species, the median is usually the more honest of the two.
Standard deviation	The spread in the units of the variable. It is the one to report alongside the mean.
Coefficient of variation	The relative spread, as a percentage. **Careful:** it only makes sense on scales with an absolute zero, such as rainfall or elevation. For temperatures in degrees Celsius the zero is arbitrary and the CV means nothing: the same data in kelvin give a different number.
Quartiles	Q1 and Q3 enclose the middle half of the data. It is the interval worth quoting when describing the environment of a species.
Skewness	Zero is symmetric; positive, a tail to the right. Values between −0.5 and 0.5 are considered roughly symmetric, and above 1 in absolute value, strongly skewed: this is a reading convention, not a test.
Kurtosis	The weight of the tails relative to a normal distribution (0 = normal). There is no accepted scale of "high" or "low": it is used to compare variables with one another, not against a threshold.

THREE VARIABLES FROM THE EXAMPLE

Of the eight variables that Block 6 selected for the Mexican pinyon: the **diurnal temperature range** (BIO2) has a mean of 15.9 °C and a standard deviation of 1.3, with almost no skewness; **isothermality** (BIO3) has a mean of 58.6% and a kurtosis of **−1.44**, the sign of a flattened distribution with two groups; the **mean temperature of the driest quarter** (BIO9) has a kurtosis of **+2.45**, that is, heavy tails: a few very cold and very warm sites.

7.3 Histograms and density

The histogram grid shows all the variables at a glance, each with its density curve. It is the figure that answers the question no table can: **what shape does the distribution have?**

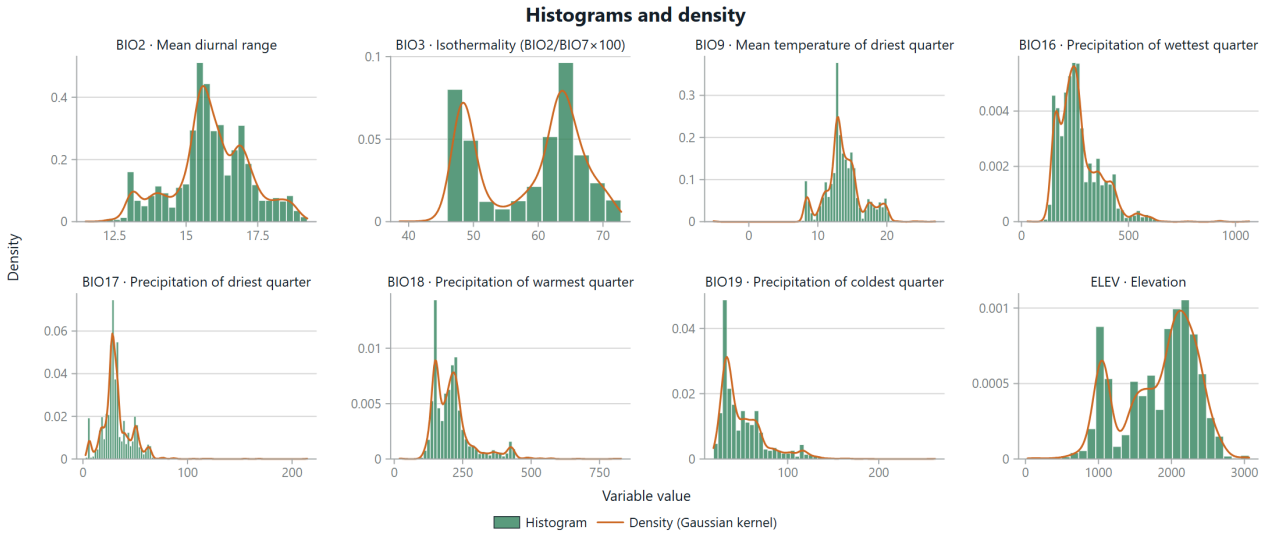


Figure 7.2. The eight selected variables. **Isothermality** (BIO3) and **elevation** are clearly bimodal: two groups of sites with different conditions, which in Block 8 will show up as groups in the ordination. The rainfall variables have the long right tail typical of precipitation.

7.4 A closer look at one variable

When you choose a variable, the app draws five figures of it: a histogram with the fitted normal, a box plot, a violin plot, a Q–Q plot and the empirical cumulative distribution.

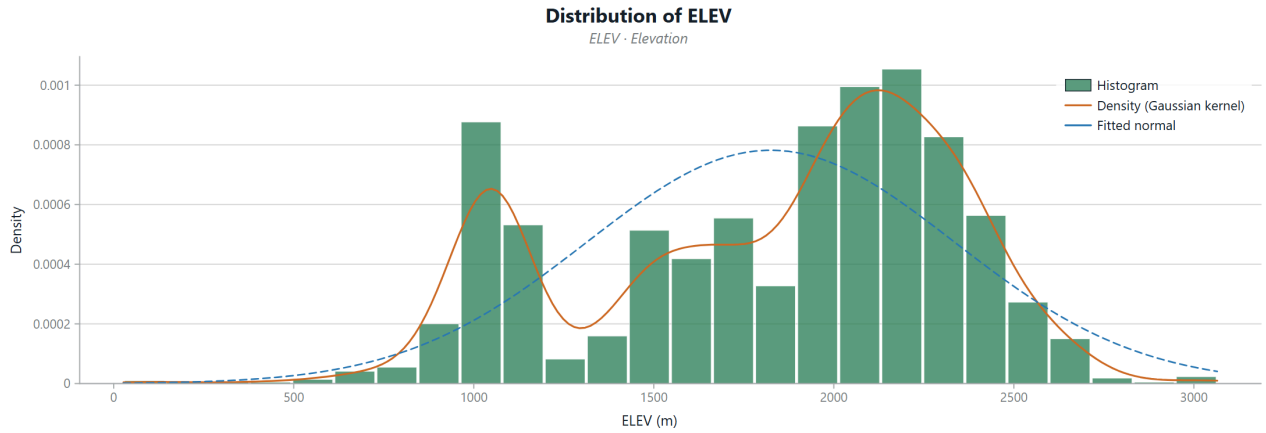


Figure 7.3. The elevation of the 1,885 records. The blue curve is the normal fitted with the same mean and standard deviation: it looks nothing like the data, which have one group near 1,000 m and another, larger one near 2,100 m.

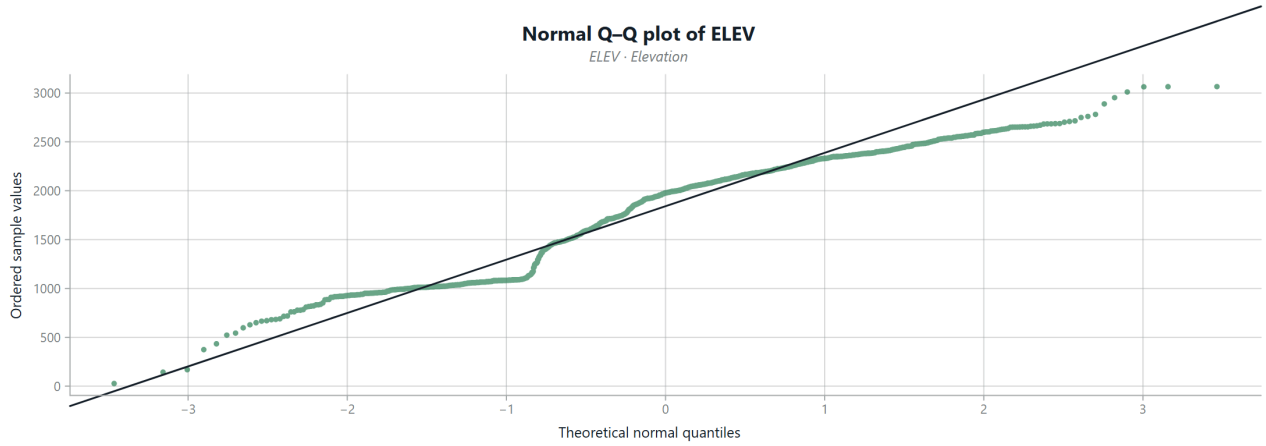


Figure 7.4. The same data in a Q–Q plot. If the variable were normal, the points would follow the line. The **step** in the middle is the signature of a distribution with two modes, and the departures at the ends, that of tails shorter than those of the normal.

> **TWO MODES ARE USUALLY TWO POPULATIONS**
>
> An environmental variable with two modes almost always means that the records come from two different sets of sites: two mountain systems, two subspecies, or one natural population and one cultivated. It is not a flaw in the data: it is information. In the Mexican pinyon, elevation separates the higher populations of the Sierra Madre Occidental and the Altiplano from the lower ones of the north-east and of Baja California Sur.

7.5 The normality tests

The app applies five tests (Shapiro-Wilk, D'Agostino-Pearson, Anderson-Darling, Jarque-Bera and Kolmogorov-Smirnov) and sums them up with a majority verdict.

Normality tests

p > 0.05 → normality is not rejected. The verdict is by majority of tests. Anderson-Darling: normal if A^2 < critical value at 5 %.

Variable	n	Shapiro p	D'Agostino p	A-D A^2	A-D crit. 5%	Jarque-Bera p	KS p	Normal votes	Verdict
BIO2	1885	0	0.0319	11.638	0.785	0.0318	0	0/5	not normal
BIO3	1885	0	0	100.668	0.785	0	0	0/5	not normal
BIO9	1885	0	0	26.142	0.785	0	0	0/5	not normal
BIO16	1885	0	0	43.764	0.785	0	0	0/5	not normal
BIO17	1885	0	0	49.519	0.785	0	0	0/5	not normal
BIO18	1885	0	0	85.7	0.785	0	0	0/5	not normal
BIO19	1885	0	0	88.904	0.785	0	0	0/5	not normal
ELEV	1885	0	0	45.27	0.785	0	0	0/5	not normal

Suggested transformation

Figure 7.5. The five tests for the eight variables of the example: **no variable passes any test** (0 of 5 votes). With 1,885 records, that result was almost inevitable.

! WITH PLENTY OF DATA, EVERYTHING COMES OUT "NOT NORMAL"

Normality tests answer the question "can I rule out that these data come from a normal distribution?". With 50 data points a large departure is needed to rule it out; with 1,885, a tiny departure (and there always is one) is enough for the p-value to be zero. That is why the result in figure 7.5 does not say that the data are odd: it says that the sample is large.

What to do: look at the shape in the histogram and the Q–Q plot, which do tell "slightly skewed" apart from "two modes", and remember that **none of the algorithms of Block 9 requires normality**. Normality matters for classical parametric tests, not for MaxEnt or a random forest.

7.6 Suggested transformations

The last table tries out transformations (log, square root, Box-Cox, Yeo-Johnson) and says whether they improve the symmetry and the p-value. It is useful when the next step is a method sensitive to shape, such as principal component analysis or a linear model.

In the example, the transformations do improve the skewness (BIO2 goes from −0.148 to −0.002 with Box-Cox), but the verdict is still "marginal": with this much data, no transformation will get the tests to accept normality. One more reason not to become obsessed with it.

7.7 The figure studio

Every figure in this block has its own **studio** underneath, with the same philosophy as the one for maps: what you see on screen is what gets exported, and everything can be edited.

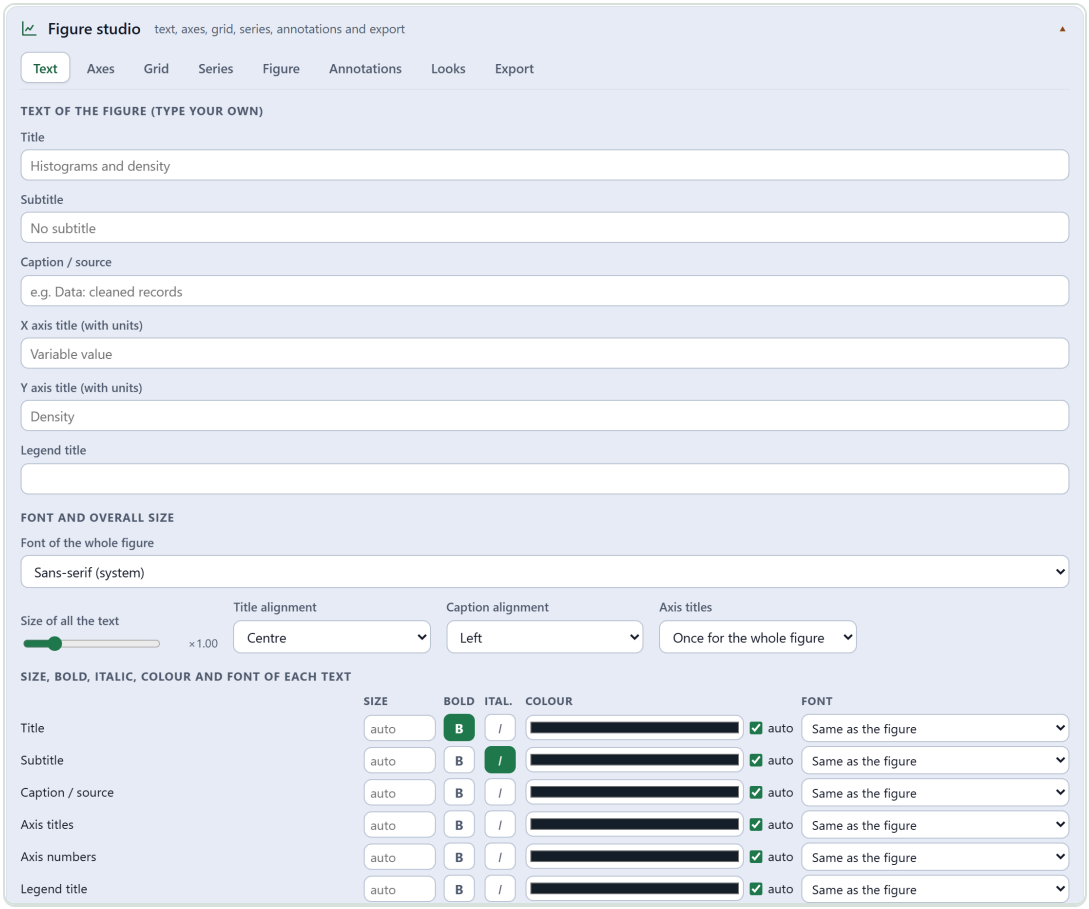


Figure 7.6. The figure studio, with the text tab open: title, subtitle, caption, axis titles and legend title, each with its own size, bold, italic, colour and font.

7

Tab	What it controls
Text	Title, subtitle, axis titles, legend and notes, with full typography.
Axes	Limits, tick marks, number format, logarithmic scale, orientation of the labels.
Grid	Reference lines, their colour and thickness, and the frame of the figure.
Series	Colours, transparency, line width and point size of each series.
Figure	The options specific to each figure: number of histogram bins, violin width, Q–Q points…
Annotations	Text, arrows and lines you want to add on top of the figure.
Looks	Presets, and saving the style as a file to give every figure in the paper the same look.
Export	SVG, PNG, JPEG or WebP; screen size, A4, one or two journal columns, or slide; 150, 300 or 600 dpi.

THE FIGURES IN THIS MANUAL

Figures 7.2, 7.3 and 7.4 are the figures exactly as the app draws them, with no retouching afterwards. Any reader can reproduce them with the data included in the program.

7.8 Common problems

What you see	What happened and what to do
"Preparing the Python engine…" takes a long time	It is the first time: Python and its libraries are being downloaded. On a slow connection it can take a couple of minutes; after that it stays in memory.
Every variable comes out "not normal"	With large samples that is to be expected (section 7.5). It is not a problem with your data and it does not prevent modelling.
A histogram with a single, very tall bar	A few extreme values stretch the axis. Look at them: they are usually the same suspicious records as in Block 5.
The CV comes out huge or negative	You are looking at a variable in degrees Celsius with a mean close to zero. The CV does not apply there.
The variable names come out in the wrong language	Change the language with the ES | EN switch in the top bar: the figures redraw themselves with the translated names.
I want the same figure for every variable	Save the style in the **Looks** tab and load it into the other figures.

8

BLOCK 8 · THE STRUCTURE OF THE DATA

Multivariate analysis, ecology and diversity

Four families of methods on the same table: ordination, clustering, range and niche metrics, and diversity. Everything that can be said about a species before fitting a distribution model.

8.1 What this block does

8.2 Principal component analysis

8.3 Clustering

8.4 Correspondence analysis

8.5 The range

8.6 Niche breadth and overlap

8.7 Diversity

8.8 Common problems

8.1 What this block does

Up to this point each variable has been examined on its own. This block looks at them **together**: which axes of environmental variation exist, whether the records form groups, how large the area they occupy is and how much variety there is within it. Everything is computed on the table from Block 5, with the variables chosen in Block 6, and it starts by pressing Prepare data →, which standardises each variable to mean zero and standard deviation one — essential when millimetres are mixed with degrees and metres.

8.2 Principal component analysis

THE IDEA BEHIND PCA

With eight correlated variables, the records do not fill an eight-dimensional space: they crowd close to a surface of fewer dimensions. PCA looks for the axes of that surface — combinations of the original variables — ordered by the amount of variation they capture. The first summarises as much as a single number can; the second, as much as is left, and so on.

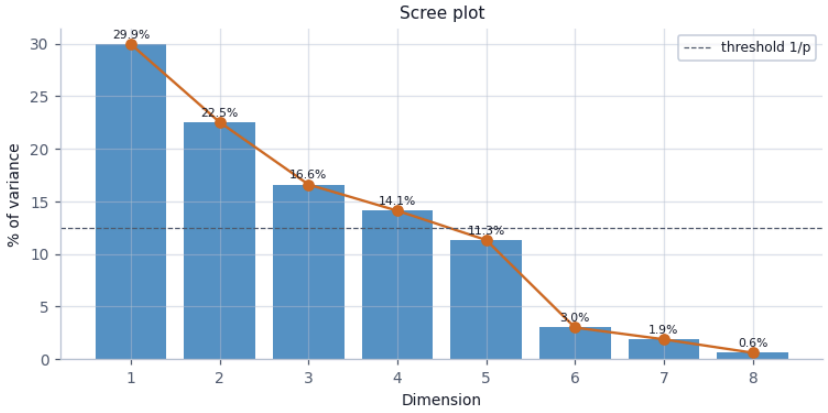


Figure 8.1. The scree plot for the example. The first four components pass the Kaiser criterion (eigenvalue > 1) and together hold 83% of the variation; the first two, 52.5%.

WHAT THE FIRST TWO AXES MEAN

In the example, **dimension 1** (29.9% of the variance) is built on BIO16 — precipitation of the wettest quarter — with a 27.4% contribution, **elevation** with 25.5%, isothermality with 22.9% and BIO18 with 17.4%: it is an axis of *moisture and altitude*. **Dimension 2** (22.5%) is made up of BIO9 (25.1%) and BIO19 (25.0%), temperature and precipitation of the driest quarter: an axis of *dry-season conditions*. Between them they order half of the species' environmental variation.

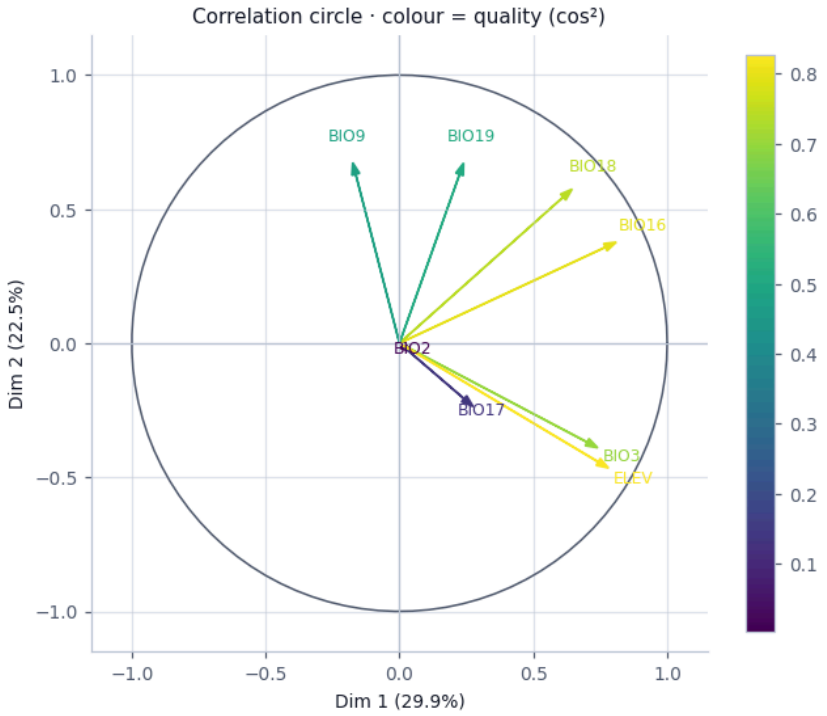


Figure 8.2. The correlation circle. The long arrows — BIO16, elevation, BIO3 and BIO18 — are the variables well represented on the plane; the colour is the quality of representation ($\cos^2$). Two arrows close together indicate variables that say the same thing; opposite arrows, variables that work against each other.

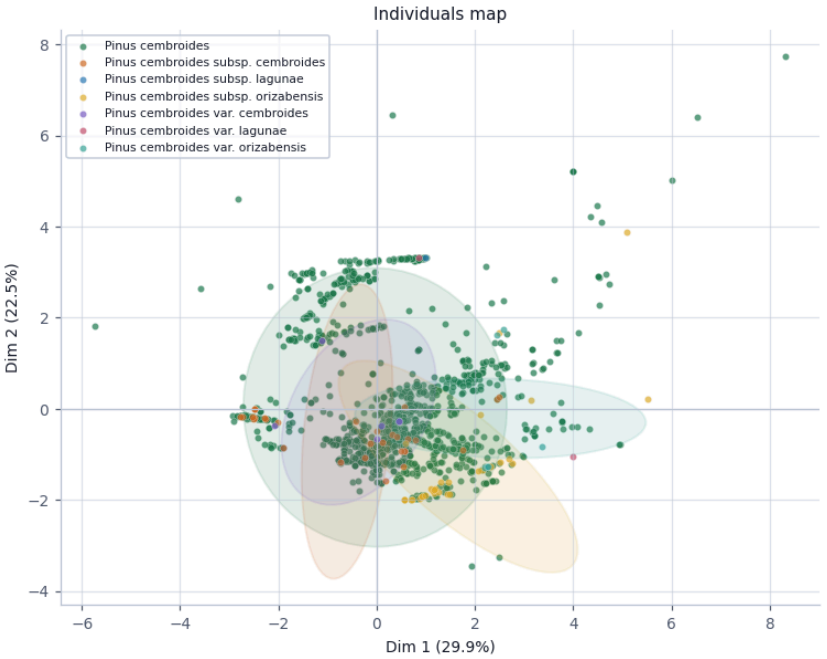


Figure 8.3. The 1,885 records projected onto that plane. Each point is a site; the cloud shows whether there are groups, tails or outlying records — which are usually the same ones that caught the eye in Block 5.

8.3 Clustering

The question clustering asks is whether the records split into sets with different environments. The app answers it in three steps, and they are best taken in order.

1. **Is there a tendency to cluster?** The Hopkins statistic compares the distribution of the data with a random one. By convention, a value above 0.75 indicates a clear tendency to cluster; around 0.5, none at all.
2. **How many clusters?** The elbow method, the mean silhouette and the gap statistic each propose a number. They rarely agree: the choice is a matter of judgement, looking also at whether the clusters make geographic sense.
3. **Run the clustering** with the chosen algorithm — hierarchical with different linkages, k-means, PAM, fuzzy, Gaussian mixture or DBSCAN — and validate it.

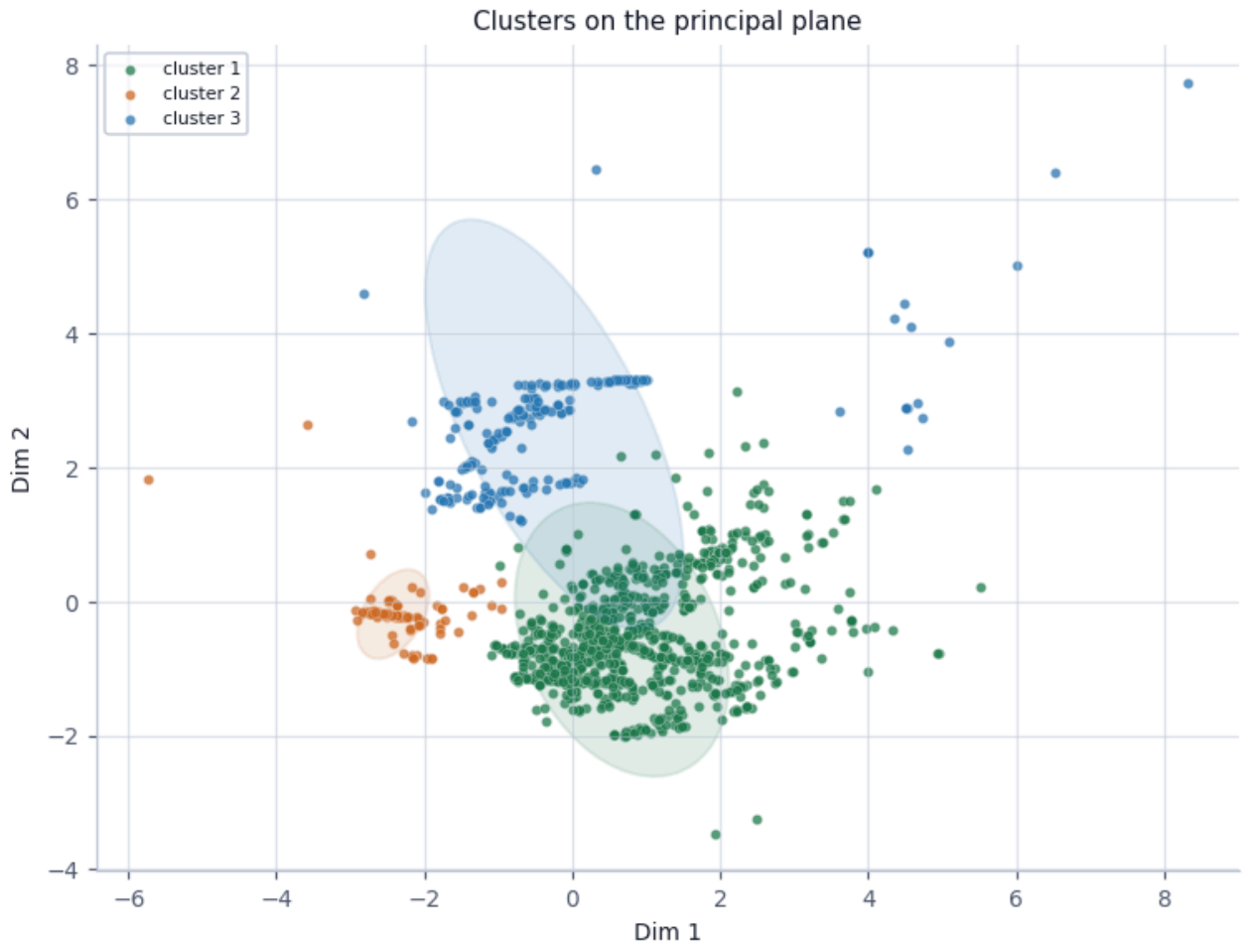


Figure 8.4. The three clusters of the example on the PCA plane. A useful clustering looks like this: clusters occupying different regions of the plane, even if they touch at the edges.

Validation measure	How to read it
Mean silhouette	From −1 to 1. By convention: above 0.7 strong structure, 0.5 to 0.7 reasonable, 0.25 to 0.5 weak, below that no real structure. In the example it is **0.422**: weak but recognisable clusters, which is the usual outcome with continuous environmental data.
Dunn index	The larger, the better separated. It has no accepted scale: it serves to compare partitions of the same data set, not to judge a single one.
Cophenetic correlation	Only for hierarchical clustering: how faithfully the dendrogram represents the original distances. Above 0.8 is considered good; in the example, 0.662.

CLUSTERS ARE NOT SUBSPECIES

An environmental cluster brings together sites with a similar climate, not lineages. It may coincide with a subspecies — and then it is an interesting result — or cut right across it. Before giving a cluster a biological name, look at where it falls on the map and which taxa it contains: the app can colour the map of Block 4 by cluster.

8.4 Correspondence analysis

Whereas PCA works with continuous variables, correspondence analysis works with **tables of counts**: taxon × Köppen climate type, taxon × state, cluster × soil class. It answers questions of the kind "which subspecies is associated with which climate?", and draws them on a map where closeness between a row and a column means association.

With the example, the taxon × Köppen table confirms what Block 5 already suggested: the species in the broad sense is concentrated in the cold arid steppe climate, while the infrataxa of the centre-east appear associated with the temperate climates with a dry winter.

8.5 The range

Three measures describe the area the species occupies, and all three are used in risk assessments.

Metric	What it is and how to read it
Extent of occurrence (EOO)	The area of the minimum convex polygon enclosing all the records. It measures geographic *spread*, not occupied habitat: it includes the cities, seas and deserts that lie between the populations. In the example, **2,797,094 km²**.
Area of occupancy (AOO)	The total area of the cells that contain at least one record. In the example: 1,258 km² with 1 km cells, **4,304 km² with 2 km cells** (the size used by red lists), 13,952 km² with 4 km and 59,200 km² with 10 km.
Nearest neighbour (R)	The Clark and Evans index: 1 means a random distribution, less than 1 clustered, more than 1 regular. In the example it is **0.210** with z = −65.7: strongly clustered.

NEAREST NEIGHBOUR (R)	SPATIAL PATTERN	Z	EOO
0.210	clustered	−65.66	2,797,094 km²

Metric	Value	Red-list reference (area only)
Extent of occurrence (EOO)	2,797,094 km²	no area criterion
Area of occupancy (AOO, 2 km cells)	4,304 km²	no area criterion
Area of occupancy (AOO, 10 km cells)	59,200 km²	—
Number of records	1,885	—

Cell side	Occupied cells	Area of occupancy	Note
1 km	1,258	1,258 km²	—
2 km	1,076	4,304 km²	red-list cell size
4 km	872	13,952 km²	—
10 km	592	59,200 km²	—

Figure 8.5. The range metrics for the example. The app computes the area of occupancy at four cell sizes and marks the one used by red lists, because the figure **depends entirely** on the cell chosen.

! TWO WARNINGS BEFORE QUOTING THESE NUMBERS

The AOO depends on sampling as much as on the species. A well-collected species has more occupied cells than an equally widespread but poorly collected one. The red-list thresholds (AOO < 2,000 km², EOO < 20,000 km² for "vulnerable") were designed for complete inventories, and a threshold on its own **does not assign a category**: the criteria also require fragmentation, decline or fluctuation.

Clustering also measures effort. An R of 0.21 says the points are tightly bunched, but part of that bunching comes from repeated collections along roads and in well-studied mountain ranges. Removing duplicates (Block 3) improves the estimate; removing the bias altogether is impossible with herbarium data.

8.6 Niche breadth and overlap

Levins' **niche breadth** measures, variable by variable, whether the species uses a narrow or a wide range of conditions relative to what is available; its standardised version runs from 0 to 1 and allows variables to be compared with one another. **Overlap** — Schoener's D and Hellinger's I, also from 0 to 1 — compares the niches of two taxa or two clusters: how much of the environmental space they share.

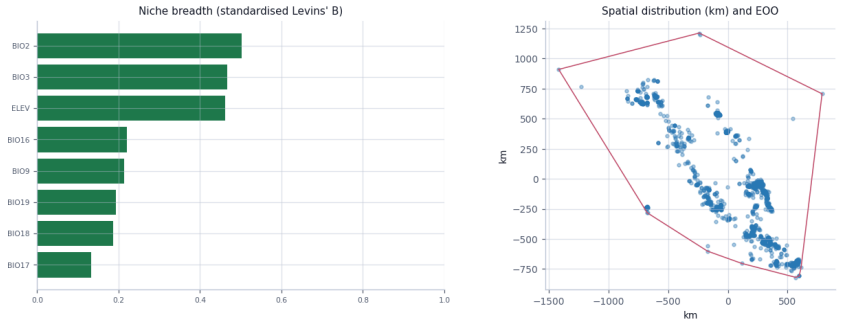


Figure 8.6. Niche breadth and spatial pattern: on the left, the standardised Levins' B of each variable; on the right, the records in kilometres with the convex polygon of the EOO. When there are two or more taxa with enough records, the app also computes the overlap between their environmental clouds.

8.7 Diversity

The last section applies the classic toolkit of community diversity — Shannon, Simpson, Hill numbers, rarefaction, beta diversity — to the table of records, taking as "sites" the cells of a grid, elevation bands, states or the clusters from the clustering.

READ THIS BEFORE INTERPRETING ANYTHING

In a collection of presence records, **the number of records of a taxon measures sampling effort**, not abundance. A species with 800 records is not more abundant than one with 80: it may be better collected, more conspicuous or live beside a road. Diversity indices were designed for communities sampled with comparable effort, and here they are applied to data that do not meet that assumption.

They are useful on two conditions: that you compare **sites with similar effort** — or use rarefaction to even it out — and that you state in the text what you are measuring. With the example data, what is being measured is the diversity of **infrataxa** of the Mexican pinyon, not that of a plant community.

THE EXAMPLE, WITH 31 SITES

With grid cells as sites, the example gives **31 sites × 7 taxa** and 1,867 records. The observed richness of the whole set is **7**, the Chao1 estimate is also **7.0** and the sample coverage is **99.95%**: practically everything there was to collect has been collected, which was to be expected because the "taxa" are the described infrataxa of a single species.

The Shannon index of the whole set is **0.231** and the Simpson index (1 − D) **0.083**: very low values, because one taxon — the species in the strict sense, with 2,281 of the original records — dominates completely. In a real plant community those numbers would point to a problem; here they describe the situation exactly.

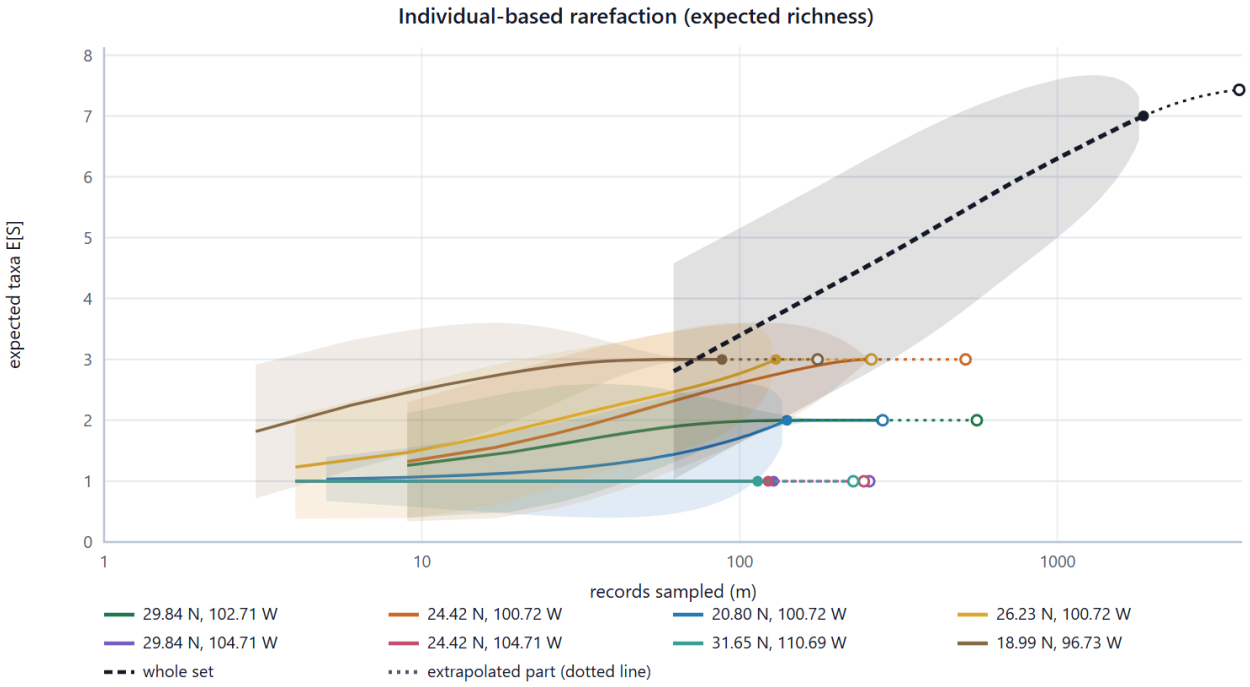


Figure 8.7. Individual-based rarefaction. Each curve tells you how many taxa would have been found with *m* records: this is how sites with unequal effort are compared. The dotted part is extrapolation, and the band, the confidence interval. The black curve is the whole set.

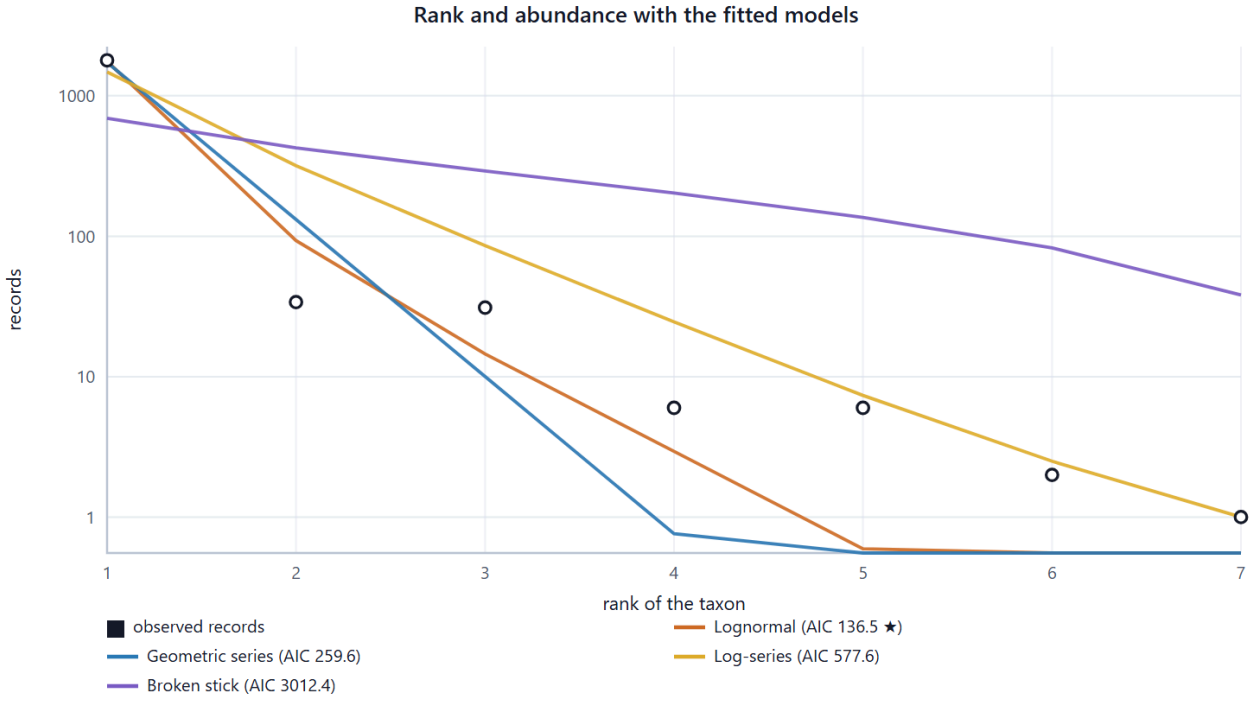


Figure 8.8. The rank–abundance curve. A very steep drop, like this one, is the signature of a set dominated by a single taxon.

The app also computes **beta diversity** between sites — Jaccard, Sørensen, Bray–Curtis and Morisita–Horn — with its partition into turnover and nestedness, and orders the sites by composition with PCoA and NMDS. Every table and figure can be downloaded as CSV and as an image.

8.8 Common problems

What you see	What happened and what to do
"First press "Prepare data" above."	The methods in this block need the standardised matrix. Press Prepare data → at the very top.
The PCA does not separate anything	If the first two axes hold little variance, the variables are not strongly correlated: that is a result, not an error. Look at more dimensions in the eigenvalue table.
The silhouette is low for every *k*	The data are a continuum, not clusters. This is common with climatic variables. Report the silhouette and do not force a partition.
The AOO changes enormously with the cell size	That is expected and must be stated: always report the cell size alongside the number, and use 2 km if you are going to compare with red-list criteria.
Shannon comes out almost zero	One taxon dominates. With records of a single species and its infrataxa, that is normal.
The rarefaction curve of a site is a flat line	That site has a single taxon: there is nothing to rarefy.

9

BLOCK 9 · THE MODEL

Distribution modelling and projection to the future

Ten algorithms and an ensemble, spatial block validation, thresholds, variable importance, response curves, maps, and projection to future climate scenarios with their extrapolation map.

9.1 What this block fits

9.2 Preparing the data

9.3 The ten algorithms

9.4 Validation

9.5 Reading the metrics

9.6 Importance and response curves

9.7 The maps and the threshold

9.8 The ensemble

9.9 Projection to the future

9.10 Common problems

9.1 What this block fits

A distribution model learns to tell the environment of the sites where the species was recorded from the environment **available** in the region, and then applies what it has learnt to every cell of the map. The result is a continuous surface of **suitability**: how similar the climate of each cell is to that of the occupied sites.

The key word is *available*. Since there are no true absences (section 1.1), the contrast is made against a sample of the environment of the whole study area, the **background points**. That is why the area you choose determines what the species is compared against, and why Block 2 mattered so much.

9.2 Preparing the data

The Prepare model data → button does four things and reports on them:

1. **It crops to the chosen extent.** By default this is the country of the records —or the states from Block 2, if you chose them deliberately—. Records outside it do not go in, and the app says how many there are.
2. **It reads the layers onto a grid.** The grid inherits the resolution of your layers; if the number of cells is huge, the app aggregates blocks of cells so that the computation is feasible in the browser.
3. **It thins the presences.** With the Keep a single presence per grid cell checkbox —ticked by default— fifty records of the same tree count once. It is the most effective defence against sampling bias.
4. **It draws the background points** spread across the study area, with a fixed seed so that the result is reproducible.

▸ THE EXAMPLE, PREPARED

With the Mexican pinyon and the WorldClim layers at 10 arc-minutes: extent **Country of the records (MX)**, box from −118.6° to −86.4° and from 14.4° to 32.8°, a grid of **194 × 111** cells of **18.6 km**, 8 predictors. Of the 1,885 cleaned records, **14 fall outside** the extent —those from the United States above the 32.8° parallel— and of the remaining 1,871, 1,468 fall in a cell that is already occupied or has no data: that leaves **403 presences** against **7,751 background points**.

Seeing 1,871 records turn into 403 presences is alarming the first time. No data have been lost: the repetition has been removed. With 18.6 km cells, 403 is the number of *distinct cells* where the species has been recorded, and that is what the model can learn from.

9.3 The ten algorithms

Algorithm	What it does and when it is a good choice
MaxEnt-style	Penalised (L1) logistic regression on linear, quadratic, product and hinge terms, with cloglog output. It is equivalent to maximum entropy modelling and is the sensible starting point in almost any study.
GLM	Generalised linear model with linear and quadratic terms. Transparent, interpretable, and the most sensitive to collinearity.
GAM	Generalised additive model with splines: smooth curves without imposing their shape.
Random forest	Many trees with balanced subsampling. Flexible and robust; its validation uses out-of-bag predictions so as not to inflate the fit.
Boosted trees (BRT)	Trees added one at a time, each correcting the error of the previous ones. Very accurate, and the most prone to overfitting.
Support vector machine	Non-linear boundary with a radial kernel.
Neural network	Five small networks, averaged.
Bioclim	The classic climatic envelope: percentiles of each variable. Simple and useful with very few presences.
Domain	Gower distance to the most similar record.
Mahalanobis	Distance to the environmental centroid, taking the covariance into account.

9.4 Validation

> **WHY SPATIAL BLOCKS AND NOT RANDOM SPLITS**
>
> If the records are split at random, the test points end up a few kilometres from the training points: the model gets them right because they are on the same hill, not because it has learnt anything general. The result is a high and misleading AUC. **Spatial block** validation divides the area into blocks and leaves out whole blocks, so the model has to predict in areas it has not seen. The values drop, and that is as it should be.

Metric	What it measures and how to read it
AUC	The probability that a presence site receives a higher suitability than a background site drawn at random. 0.5 is chance. Usual reading bands: 0.7–0.8 acceptable, 0.8–0.9 good, above 0.9 excellent. They are conventions, and with presence-only data the AUC depends on the study area: values are not compared across different studies.
TSS	Sensitivity + specificity − 1, at the threshold that maximises it. From −1 to 1; by convention, above 0.6 is considered good. It is more stable than kappa and does not depend on prevalence.
Boyce index	Measures whether the higher suitability classes contain proportionally more presences: it is the metric designed for presence-only data. It ranges from −1 to 1; above 0.8 indicates a very consistent model.
Omission (P10)	The fraction of test presences that falls below the 10th-percentile threshold. Ideally close to 0.1.

9.5 Reading the metrics

Model performance ⬇ Metrics CSV

Model	Test AUC	Train AUC	TSS	Sens.	Spec.	Boyce	Omission (P10)	Max-TSS threshold	Details
MaxEnt-style	0.840 ± 0.022	0.892	0.539 ± 0.076	0.81	0.73	0.869 ± 0.051	0.18	0.384	classes lqh; 36 params; A
GLM	0.840 ± 0.019	0.885	0.512 ± 0.067	0.79	0.72	0.901 ± 0.081	0.17	0.530	—
Random forest	0.837 ± 0.016	0.874	0.542 ± 0.082	0.80	0.74	0.958 ± 0.017	0.19	0.361	—
BRT	0.833 ± 0.018	0.928	0.535 ± 0.062	0.77	0.76	0.856 ± 0.086	0.29	0.424	124 iterations

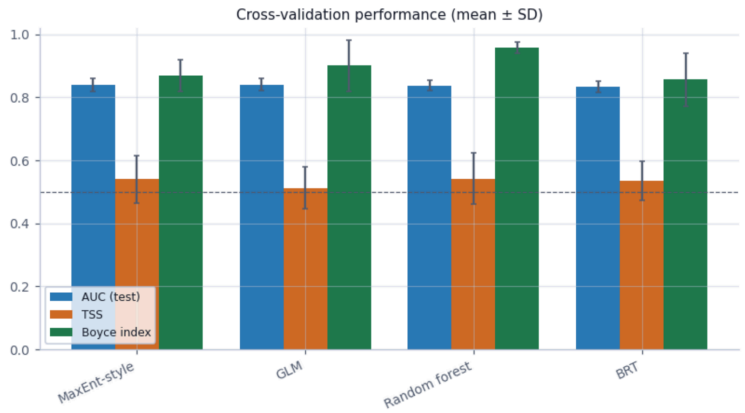


Figure 9.1. The four models of the example, validated with spatial blocks. The **Train AUC** column is not a result: it is there to be compared with the test column.

WHAT THESE NUMBERS SAY

- **The four models are practically equal** in test AUC: MaxEnt-style 0.840, GLM 0.840, random forest 0.837 and BRT 0.833, with standard deviations between ±0.016 and ±0.022. When four different algorithms agree like this, the signal is in the data, not in the method.
- **BRT overfits the most**: its training AUC is 0.928 against 0.833 in testing, a gap of almost 0.10. The random forest (0.874 against 0.837) and the GLM, the most rigid (0.885 against 0.840), have the smallest gaps.
- **The Boyce index separates what the AUC does not**: the random forest reaches 0.958 and BRT stays at 0.856. With almost identical AUCs, the forest ranks the suitability classes better.
- **The TSS lies between 0.51 and 0.54**, below the 0.6 usually asked for: a reminder that with 403 presences and 18.6 km cells the model is useful, not perfect.

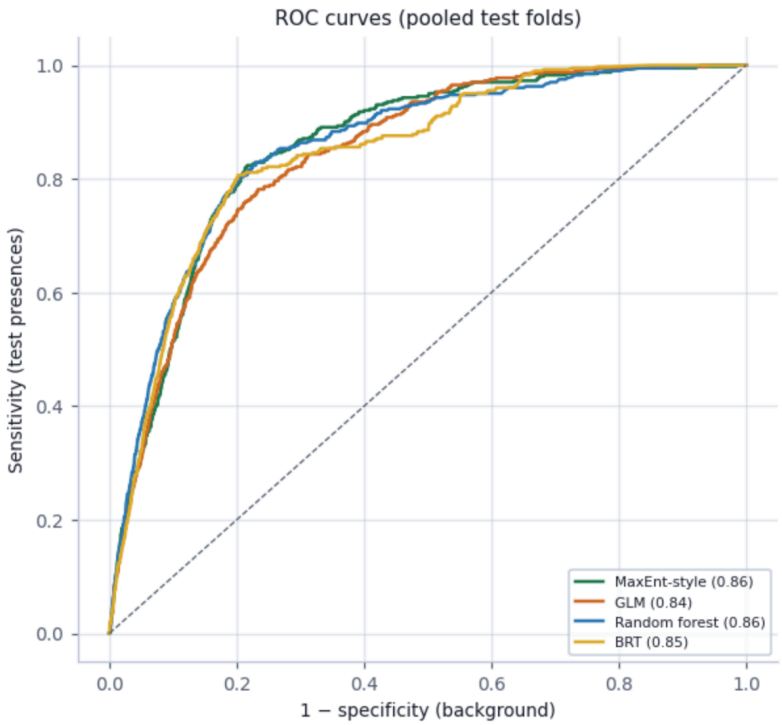


Figure 9.2. The ROC curves. The closer the curve hugs the top left corner, the better the model separates; the diagonal is chance.

9.6 Importance and response curves

Two different questions: **how much** each variable contributes, and **how** the species responds to it.

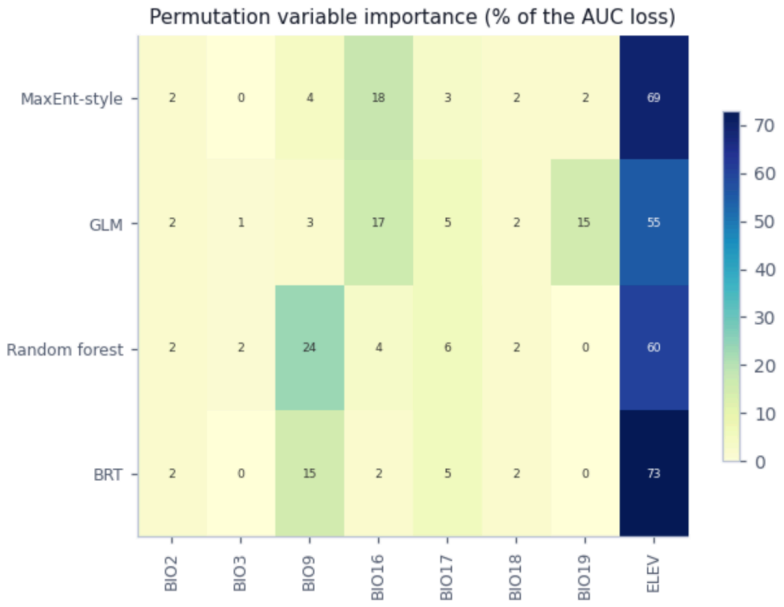


Figure 9.3. Permutation importance, measured **on the test data**: the values of one variable are shuffled and the drop in performance is measured. If a variable contributes nothing, shuffling it changes nothing.

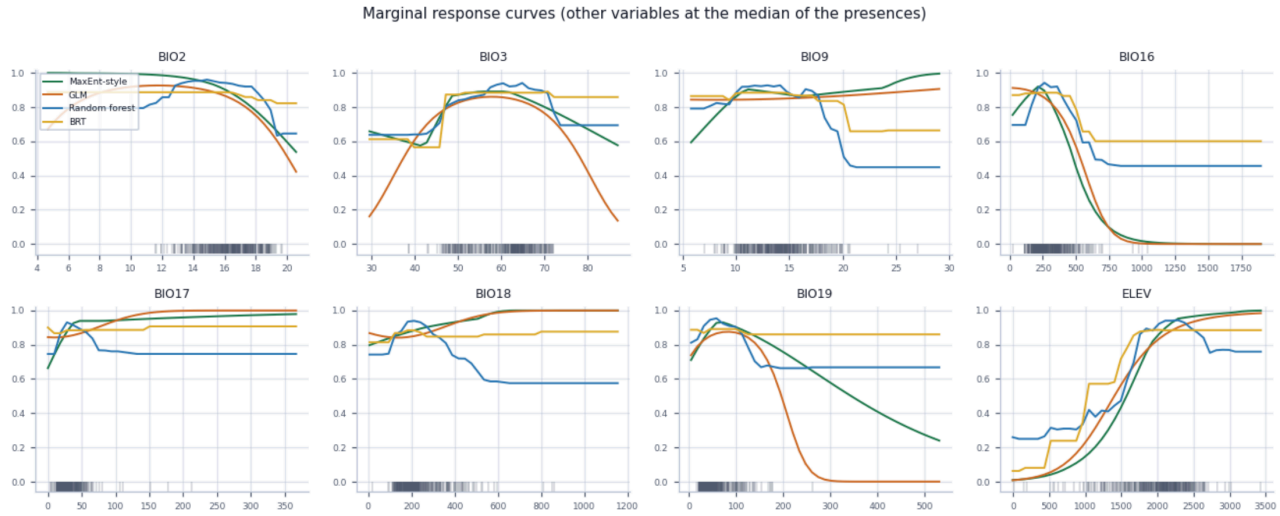


Figure 9.4. The response curves: how suitability changes as one variable moves while the others are held at the median of the presences. A bell-shaped curve indicates an optimum; a monotonic one, a limit.

> **! IMPORTANCE IS NOT CAUSATION**
>
> That elevation comes out as the most important variable in all four models does not mean that elevation itself *causes* the distribution: it may be a variable that travels with the true cause —the cold, the moisture of the mountains, the soil, history—. And if two variables are correlated, the model shares the importance between them arbitrarily; that is why Block 6 comes before this one.

9.7 The maps and the threshold

The app draws three maps per model: the **continuous suitability**, the **presence/absence** map after applying a threshold, and the **uncertainty** between models when there are several.

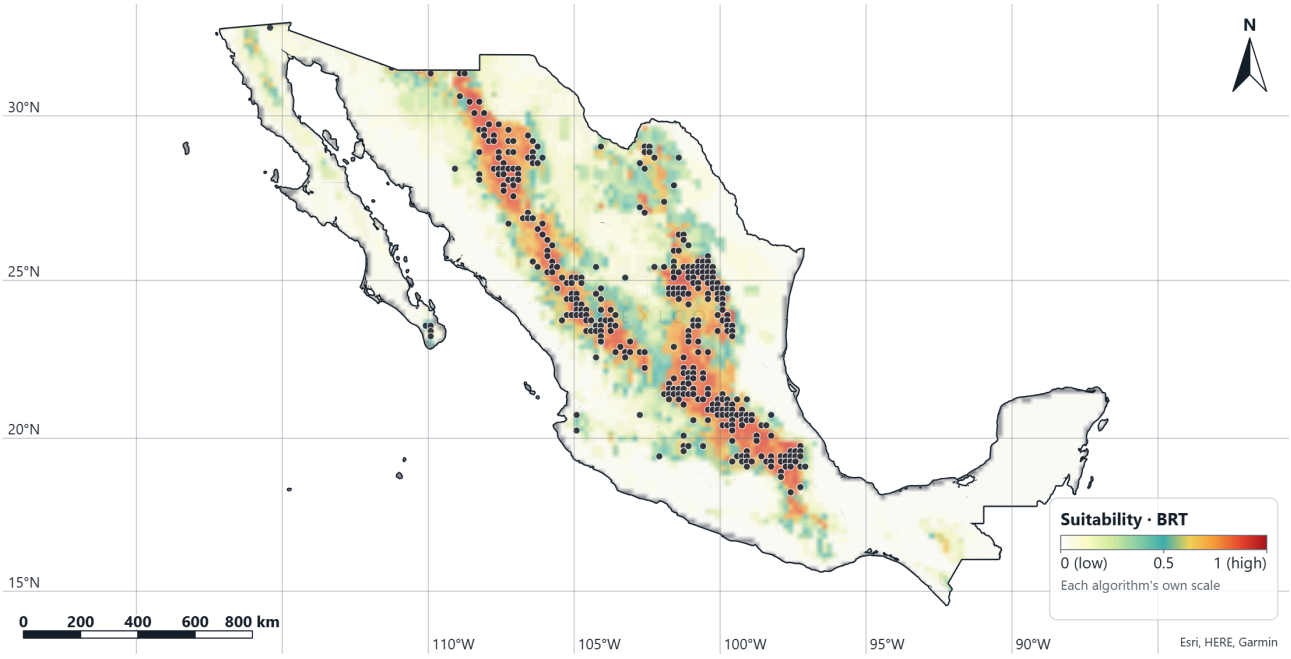


Figure 9.5. The suitability of the boosted trees model over the whole country, clipped to Mexico with the map studio. The dots are the presences used. The two Sierras Madre, the Trans-Mexican Volcanic Belt and the Sierra de La Laguna stand out, and the map points to suitable areas where there are no records: candidates for exploration.

Threshold	What it does
Max. TSS (sens + spec)	The one that maximises sensitivity + specificity. It is the most commonly used and the one the app proposes.
10th percentile of presences	The P10 threshold leaves out the 10% of presences with the lowest suitability. Useful when you suspect some records are poorly georeferenced.
Minimum training presence	Keeps every presence. The most permissive, and the most sensitive to a single error.
Sensitivity = specificity	Balances the two kinds of error.

With the threshold applied, the app computes the **suitable area in square kilometres**: in the example, the MaxEnt-style model with the maximum-TSS threshold (0.384) marks **471,566 km²** of Mexican territory with a climate suitable for the pinyon. That figure changes with the threshold, so you must report which one you used.

9.8 The ensemble

When several algorithms agree, their average usually predicts better than any of them on its own, and their disagreement is an honest measure of uncertainty. The ensemble combines the models by **weighted mean**, **median** or **committee average (votes)**, and weights them by test AUC, TSS, Boyce or equally. Next to the averaged map, the app draws the **agreement map**: where the models say the same thing and where they do not.

9.9 Projection to the future

The last section applies the fitted model to future climate layers. Scenarios are added to a list —each with its climate model, its pathway and its period— and they are all projected in one go.

What it delivers	What it is for
Map per scenario	The future suitability, with the same threshold as the present.
Gain and loss	How much area is gained, how much is lost and how much is kept, in km² and as a percentage.
Shift of the suitable area	How far the centre of the distribution moves, in which direction, and how much it climbs in elevation.
Dispersal assumptions	Unlimited, none, or limited to a distance: three different answers to "will it be able to get there?".
Refugia	The areas that are suitable today and in every future scenario: the first thing to conserve.
Ensemble across scenarios	Mean and agreement between climate models, which is where the greatest uncertainty lies.

> **NEVER PUBLISH A PROJECTION WITHOUT ITS MESS**
>
> The **MESS** map marks the cells where the future climate falls *outside* the range the model saw during training. There the prediction is not an estimate: it is the formula extended blindly. The app hatches those areas on the future maps and lets you export the MESS separately. Without that layer, a suitability map for 2070 promises a precision it does not have.

9.10 Common problems

What you see	What happened and what to do
The map does not cover the whole country	The grid was prepared with a different extent. Change the extent and press Prepare model data → again: the app warns you when the prepared grid no longer matches the chosen extent.
An AUC of 0.99	Be suspicious before celebrating: it usually comes from random validation instead of blocks, from a huge study area, or from duplicated presences.

What you see	What happened and what to do
Very few presences after preparing, or a model that takes too long	The first is the per-cell thinning: if it leaves you below fifteen presences, use higher-resolution layers or enlarge the area. The second is normal for the random forest and the boosted trees, the most expensive: start with MaxEnt and GLM.
There are suitable areas with no records	It may be an interesting prediction —unexplored habitat— or extrapolation: check it against the MESS and against what you know about the species.
The future map is almost all suitable	Check that the future layers have the same resolution and the same variables as those of the present, and look at the MESS.

10

BLOCK 10 · FROM NICHE TO FIELD

Agroclimatic suitability of crops

The same climate data, read with the questions of agronomy: degree days, frost, chilling hours, water balance and length of the growing period, and with them the suitability of a crop, its planting calendar and the factor that limits it in each cell.

10.1 This block does not model your species

10.2 The monthly layers

10.3 The agroclimatic indices

10.4 Suitability of a crop

10.5 The limiting factor

10.6 The crop catalogue

10.7 Comparing crops, and analogues

10.8 Future agroclimate

10.9 Common problems

10.1 This block does not model your species

FIRST THINGS FIRST, TO AVOID A VERY COMMON CONFUSION

The nine previous blocks are about **your species**. This one is not. Block 10 takes the same territory and the same climate and answers a different question: **which crops could grow here?** If you are studying a pine and maize appears, it is not a mistake: maize is simply the first crop in the catalogue. Choose the one you are interested in —there are more than sixty— or enter the requirements of your own.

The two natural uses of this block are to assess the agricultural potential of the region you have already loaded, and to compare the niche of a wild species with the suitability of a related crop.

10.2 The monthly layers

The agroclimatic indices are computed **month by month**: they need minimum temperature, maximum temperature and precipitation for the twelve months —36 WorldClim files, or twelve-band files—. They are loaded as in Block 5: by folder or by files.

IF YOU HAVE NO MONTHLY LAYERS

The 🌡 Use the layers of step 5 button reconstructs an approximate monthly cycle from the bioclimatic layers you have already loaded —mean temperature and extremes, seasonality, annual and quarterly rainfall— and computes a reduced set of indices with it. The app flags those indices as **approximate** and explains the assumption: the thermal peak is placed in July in the northern hemisphere. It is useful for exploring and for learning; for a technical report, load real monthly layers.

10.3 The agroclimatic indices

Once the grid has been read, Compute the indices → produces 21 layers. These are the most widely used:

Index	What it measures and what it is for
Growing degree days (GDD)	The sum of the heat available above a base temperature. It is the currency used to judge whether a crop cycle "fits" at a site. It has no universal scale: each cultivar has its own requirement.
Frost days and months	How often the minimum drops below 0 °C. It determines which crops are impossible without protection.
Frost-free period	The consecutive days without frost: the real planting window in temperate areas.
Chilling hours	Hours accumulated in the cold range of winter. Deciduous fruit trees need them to break bud; the requirement depends on the cultivar, so the figures are indicative.
Heat days	Days above critical thresholds, which in many crops coincide with flowering.

Index	What it measures and what it is for
Reference evapotranspiration	How much water the atmosphere can evaporate. Together with rainfall it gives the water balance.
Water deficit and surplus	How much water is lacking or left over, month by month: the basis of any irrigation plan.
Length of the growing period	The days on which there is enough temperature and moisture at the same time. It is the index that best sums up the general agricultural suitability of an area.
Aridity index and class	The ratio between rainfall and evapotranspiration, with the classes of the United Nations Environment Programme (hyper-arid, arid, semi-arid, dry sub-humid, humid).

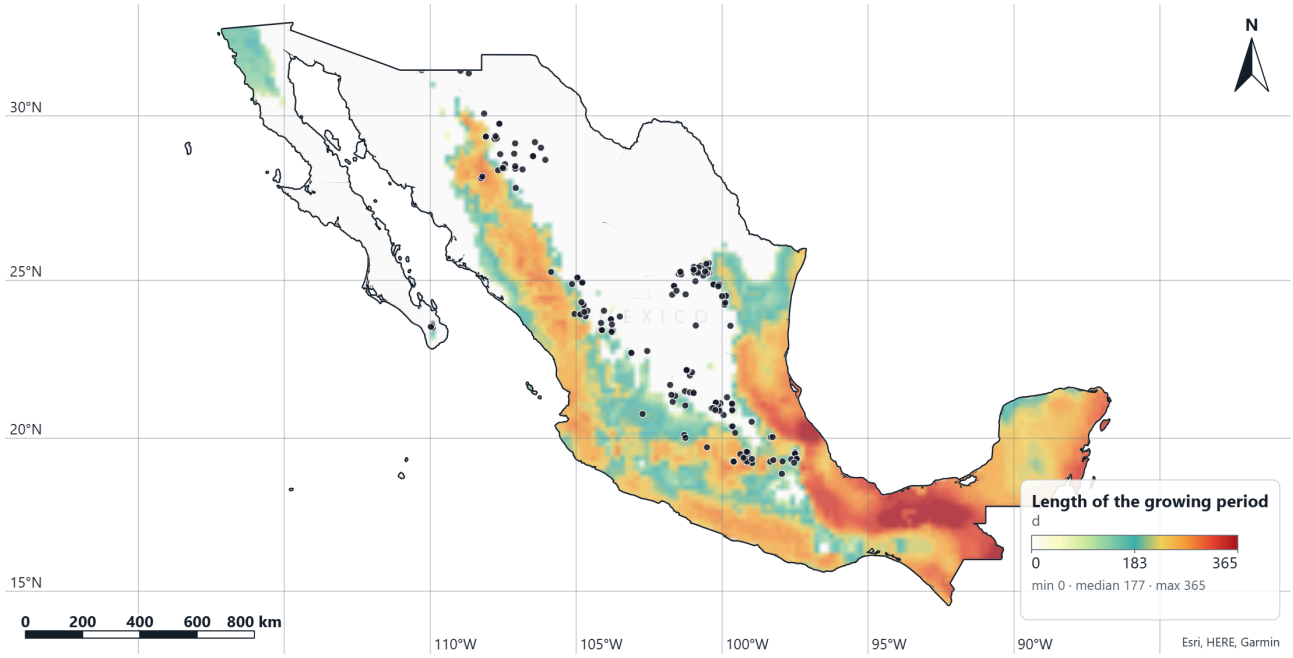


Figure 10.1. The length of the growing period over the whole of Mexico, computed with the example layers. The contrast between the arid north and the humid south-east can be seen at a glance. Each index can be exported as an image, as a georeferenced GeoTIFF or as an ASCII grid.

10.4 Suitability of a crop

The suitability model is an **envelope**: for each cell and for **each possible planting month**, it compares what the crop needs —optimum temperature and lethal limits, water, chilling hours, elevation— with what the site offers over the whole cycle, and keeps the best month.

CROP	CYCLE USED	SUITABLE AREA (≥ 0.5)	% OF THE REGION	MEAN SUITABILITY	BEST PLANTING MONTH
Maize	125 d	1,524,020 km²	48.2 %	0.470	March

MAIN CONSTRAINT
Insufficient water

Figure 10.2. The summary of the suitability of maize over Mexico, with the approximate indices derived from the bioclimatic layers of the example.

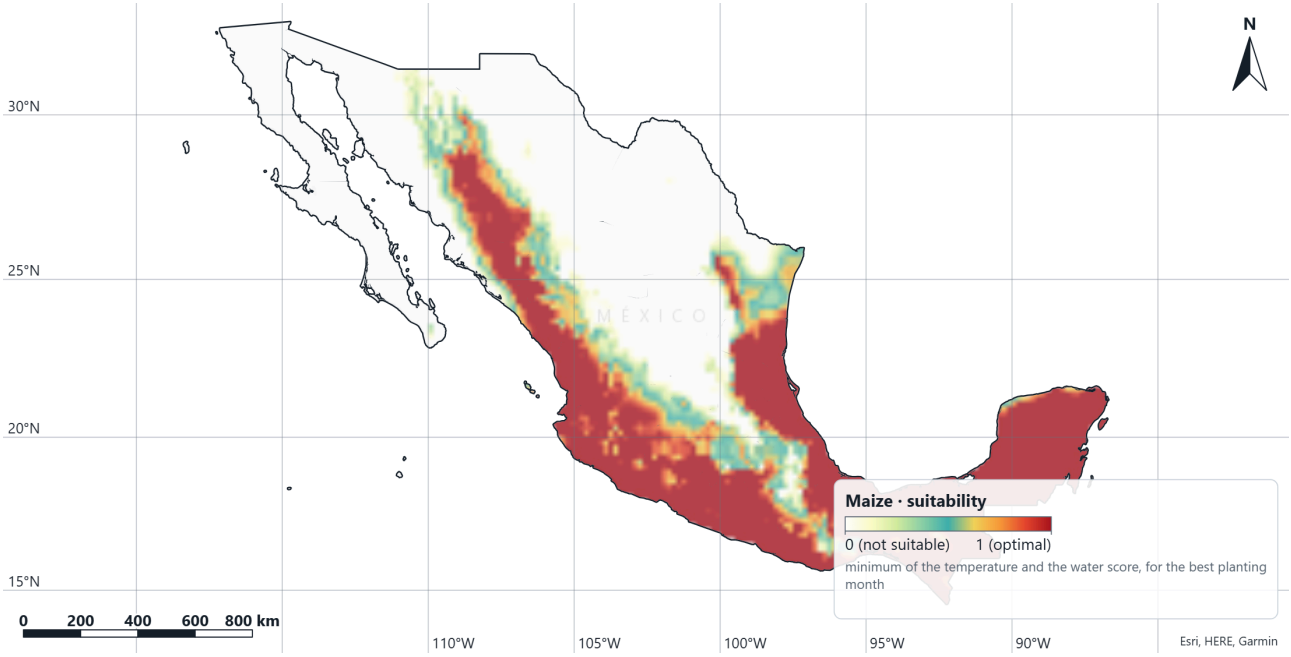


Figure 10.3. The continuous suitability, from 0 to 1. The most suitable areas outline the Pacific and Gulf slopes, the south and the Yucatán Peninsula, while the arid north and the high plateau are left with low values.

MAIZE IN MEXICO, WITH THE EXAMPLE DATA

A 125-day cycle, **1,524,020 km²** with suitability ≥ 0.5 —**48.2%** of the region—, a mean suitability of **0.470**, best planting month **March** and main constraint **insufficient water**. These figures were computed with the reconstructed monthly cycle, so they are good as an exercise, not as a technical recommendation: with real monthly layers they would change, and the optimum month also depends on the cultivar and on management.

10.5 The limiting factor

Knowing that a cell has a suitability of 0.3 does not tell you what to do; knowing *why* does. The limiting factor map paints, in each cell, which of the conditions restricts the crop most in its best planting month.



Figure 10.4. The limiting factor of maize. The ochre of the high plateau and of the north is **insufficient water**; the blue of the high sierras and of northern Chihuahua, **killing frost** or **low temperature**; the green of the Pacific and Gulf slopes, the south and the Yucatán Peninsula, **no constraint**. Each colour suggests a different decision: irrigation, planting date, or a change of cultivar.

10.6 The crop catalogue

The app comes with a catalogue of **more than sixty crops** with their reference ranges: minimum, optimum and maximum temperature, lethal limits, cycle length, water and chilling-hour requirements, and elevation range. Everything can be edited: you can change the values of a crop, save the catalogue as a file and load it again in another session or on another computer.

> **! THE VALUES IN THE CATALOGUE ARE INDICATIVE**
>
> They are published ranges for the species *as a whole*, not for a particular cultivar. A highland landrace of maize and a commercial hybrid for the lowland tropics have different requirements, and the catalogue cannot know which one you use. Before making a real decision, replace the values with those of your own material: that is why the catalogue can be edited, and the app exports the parameters it used together with the results.

10.7 Comparing crops, and analogues

- **Comparison between crops.** It computes the suitability of several at once and draws the map of the **best crop per cell**, plus a table with the suitable area of each one. It is the view that answers "what is worth planting here?".

- **Climatic analogues.** Given a site, it looks for the cells with the most similar climate. It is a way of transferring experience: if a variety does well in a given municipality, where else should it do well?

10.8 Future agroclimate

With monthly layers of a future scenario, the block repeats the whole calculation and delivers the **difference**: how much each index changes, how much suitable area is gained or lost, how the planting calendar shifts and which factor becomes limiting. It is the question that matters most to whoever plans: not "what will the climate be like?" but "what will I have to change?".

The same warnings as in chapter 9 apply: scenarios are projections with uncertainty, it is advisable to use more than one climate model, and you should look at where the future climate leaves the known range.

10.9 Common problems

What you see	What happened and what to do
Maize appears even though you are studying another species	It is the first crop in the catalogue, not a prediction about your species (section 10.1). Choose another one from the list.
The indices are flagged as "approximate"	They were derived from the bioclimatic layers because there are no monthly layers. For a technical report, load the 36 monthly layers.
The map does not cover the whole country	As in Block 9: check the chosen extent and read the climate grid again. The app says which box it used.
The chilling hours come out as zero over the whole map	With the reconstructed monthly cycle, chilling hours are the most fragile index. They need real monthly layers.
The suitable area changes a lot when a parameter is touched	That is to be expected in an envelope model: the result is only as good as the ranges in the catalogue. Document the ones you used.
Soil pH is reported but not applied	The app warns you about it: there is no pH layer, so that requirement is left out of the calculation.

A

APPENDICES · REFERENCE MATERIAL

Appendices

What you look up rather than read: which files the app recognises and how they must be named, the formulas with their sources, the English–Spanish glossary, the error messages with their causes, the bibliography and the third-party components.

A File formats

B Formulas of the indices

C English–Spanish glossary

D Messages and troubleshooting

E References

F Third-party components

A File formats

The app does not convert files or upload them to any server: it reads them on your computer. This is the complete list of what it recognises and of how each thing has to be named.

A.1 · Your own records (Block 1)

Aspect	What it accepts
Format	`.csv`, `.txt` (separated by commas, semicolons or tabs), `.xlsx`, or text pasted from a spreadsheet.
Minimum columns	Species, longitude and latitude. Headers are recognised in Spanish and in English (`especie` / `species`, `longitud` / `lon` / `longitude`, `latitud` / `lat` / `latitude`) and, if they do not match, they are assigned by hand with the three menus.
Optional columns	Country, state, locality, year, month, elevation and uncertainty in metres. They are recognised automatically and used in the filters of Block 2.
Coordinates	Decimal degrees with a point or a comma; degrees-minutes-seconds with symbols or with compass letters. Two checkboxes fix in one go swapped longitude and latitude and a missing minus sign.
Encoding	UTF-8 and the old Windows encoding: accents are repaired on reading.

A.2 · Raster layers (Blocks 5, 9 and 10)

Layer	Expected file name and notes
Bioclimatic	`wc2.1_<res>_bio_1.tif` … `bio_19.tif`, where `<res>` is `30s`, `2.5m`, `5m` or `10m`. GeoTIFF in geographic coordinates (EPSG:4326).
Elevation	`wc2.1_<res>_elev.tif`
Köppen-Geiger	A categorical GeoTIFF with the class codes; the table of names is included in the app.
Monthly	`wc2.1_<res>_tmin_01.tif` … `_12.tif`, and the same for `tmax` and `prec`: 36 files. A twelve-band file per variable is also accepted.
Future scenarios	A 19-band file with the bioclimatic variables of the scenario, or separate `bio_N` layers. They must have the same resolution and extent as those of the present.
Your own regions	GeoJSON of polygons in EPSG:4326, with a recognisable name in the properties of each polygon.

A.3 · What it exports

Type	Formats
Tables	CSV in every block: raw and cleaned records, points × variables, metrics, coefficients, agroclimatic indices, diversity.
Points	Cleaned GeoJSON.
Maps	PNG, JPEG, WebP, TIFF, georeferenced GeoTIFF and SVG; paper sizes at 150, 300 or 600 dpi.
Surfaces	GeoTIFF (Float32, EPSG:4326) and ASCII grid (`.asc`) of the suitability and of each agroclimatic index.
Figures	Vector SVG, PNG, JPEG and WebP from the figure studio.

B Formulas of the indices

The same ones shown in the help cards of the app, with the source they come from. The symbols follow the notation of each original source.

CORRELATION AND VARIABLE SELECTION (BLOCK 6)

Pearson correlation	$r = \Sigma(x-\bar{x})(y-\bar{y}) \,/\, \sqrt{[\Sigma(x-\bar{x})^2 \cdot \Sigma(y-\bar{y})^2]}$
Coefficient of determination	$r^2 = r \times r$
Variance inflation factor	$VIF_j = 1 \,/\, (1 - R^2_j)$

DESCRIPTIVE STATISTICS (BLOCK 7)

Standard deviation	$SD = \sqrt{[\ \Sigma(x - \bar{x})^2 \,/\, (n - 1)\ ]}$
Coefficient of variation	$CV\ \% = 100 \times SD \,/\, \lvert mean \rvert$

MULTIVARIATE ANALYSIS, ECOLOGY AND DIVERSITY (BLOCK 8)

Contribution to the dimension	reference = 100 / (number of variables)
Inertia (correspondence analysis)	total inertia = χ^2 / n
Nearest neighbour	R = mean observed distance / mean distance expected at random
Levins niche breadth	$B = 1 \,/\, \Sigma p^2$; standardised $B = (B - 1) \,/\, (\text{classes} - 1)$
Schoener overlap	$D = 1 - \tfrac{1}{2}\ \Sigma\ \lvert p_{A,i} - p_{B,i} \rvert$
Shannon	$H' = -\ \Sigma\ p_i \cdot \ln(p_i)$
Simpson	$D = \Sigma\ p_i^2$
Pielou evenness	$J' = H' \,/\, \log(S)$
Berger–Parker dominance	$d = N_{max} \,/\, N$
Margalef and Menhinick richness	$D_{Mg} = (S - 1) \,/\, \ln(N)$; $D_{Mn} = S \,/\, \sqrt{N}$
Chao1	$Chao1 = S + f_1^2 \,/\, (2 \cdot f_2)$

MODELLING (BLOCK 9)

TSS	TSS = sensitivity + specificity − 1
Omission	omission = presences below the threshold / test presences

AGROCLIMATE (BLOCK 10)

Growing degree days	$GDD = \Sigma\ \max(0,\ (T_{max} + T_{min})/2 - T_{base})$
Reference evapotranspiration	$ET_o = 0.0023 \cdot R_a \cdot (T_{mean} + 17.8) \cdot \sqrt{(T_{max} - T_{min})}$
Aridity index	AI = annual precipitation / annual ET_o

Besides the formula, each help card in the app gives the reading scale, the most common mistake and the full reference. There are 97 cards spread across Blocks 6 to 10.

C English–Spanish glossary

The terms of the interface and the concepts of the manual, with their Spanish equivalent for anyone who reads the Spanish literature or uses the app in Spanish.

English	Spanish	What it is
Occurrence record	*registro de presencia*	The statement that an organism was at a place at a given time.
Presence-only	*solo presencia*	Data without confirmed absences; the contrast is made against the available environment.
Background points	*puntos de fondo*	A sample of the environment available in the study area.
Accessible area (M)	*área accesible*	The part of the world the species has been able to reach.
Suitability	*idoneidad*	Environmental similarity to the occupied sites, from 0 to 1.
Threshold	*umbral*	The value that turns suitability into presence or absence.
Spatial blocks	*bloques espaciales*	Validation that leaves out whole regions, not single points.
Extrapolation	*extrapolación*	Predicting under conditions the model did not see; it is flagged with the MESS.
Response curve	*curva de respuesta*	How suitability changes as one variable varies.
Permutation importance	*importancia por permutación*	The drop in performance when a variable is shuffled.
Ensemble	*ensamble*	A combination of several models.
Collinearity	*colinealidad*	Variables that predict one another.
Spatial thinning	*adelgazado espacial*	Keeping one presence per cell.
Extent of occurrence (EOO)	*extensión de la presencia*	The area of the convex polygon enclosing the records.
Area of occupancy (AOO)	*área de ocupación*	The summed area of the occupied cells.
Niche breadth	*amplitud de nicho*	How wide the range of conditions used by the species is.
Niche overlap	*solapamiento de nicho*	How much of the environmental space two taxa share.
Rarefaction	*rarefacción*	Expected richness at equal sampling effort.
Beta diversity	*diversidad beta*	The difference in composition between sites.
Growing degree days	*grados-día*	Heat accumulated above a base temperature.
Chilling hours	*horas frío*	The accumulated winter cold that deciduous fruit trees need.
Length of growing period	*periodo de crecimiento*	Days with enough temperature and moisture.

English	Spanish	What it is
Limiting factor	*factor limitante*	The condition that most restricts the crop in each cell.
Climatic analogues	*análogos climáticos*	Sites with a climate similar to that of a reference site.

D Messages and troubleshooting

The warnings the app can show, with their cause and their remedy. Each chapter also has its own table of common problems.

Message	What it means and what to do
"Could not query GBIF"	No connection, or the service is down. Use the bundled example or your own files and try again later.
"the connection to GBIF was lost"	The download broke off after six attempts. The app offers to use the records that did arrive.
"GBIF answered 429 / 503"	Too many requests, or the service is overloaded. The app waits and retries by itself; if it keeps happening, try again in a few minutes.
"Extract the raster values first"	Block 6 needs the points × variables table from Block 5.
"none recognised as WorldClim or Köppen"	The file names do not follow the pattern of appendix A.2.
"The prepared grid uses the extent…"	An informative notice: it says which box the model was prepared with. If you changed the extent afterwards, prepare again.
"N of M records fall outside the extent"	Those records fall outside the chosen box and do not enter the model. It is your decision, not an error.
"Few presences (<15)"	The model will be unstable. Enlarge the area, use higher-resolution layers or switch to an envelope method.
"No monthly layers: the indices will be derived from … bioclimatic layers"	There are no monthly layers: the agroclimatic indices come from a reconstructed cycle and are flagged as approximate.
"The pH range … is reported but not applied"	There is no soil-pH layer, so that requirement of the crop is left out of the calculation.
"Error reading the GeoTIFF files"	A damaged file, or one in a projection other than EPSG:4326.
"Preparing the Python engine…" takes a while	It is the first time in the session: Python is being downloaded into the browser. After that it stays in memory.

E References

The bibliography of the methods the program uses. It is the same list that appears on the home page of the app, where each method links to its reference.

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Besides these, the help cards of the app cite the sources of each diversity index, each normality test and each clustering method: 86 different references in all, each one next to the number it explains.

A

F Third-party components

All the analysis, figures and maps of BioModelling Pro are original code, licensed under the GPL v3 or later. What comes from outside is listed here, each item with its licence; the full breakdown is in `NOTICE.md` and the complete licence texts in `vendor/THIRD-PARTY-NOTICES.txt`.

F.1 · The code

Component	What it is for and under which licence
Leaflet	The interactive maps. 2-clause BSD licence.
geotiff.js	Windowed reading of GeoTIFF files. MIT licence.
html2canvas	Composing the image when a map is exported. MIT licence.
Pyodide	The Python engine inside the browser, with NumPy, SciPy, pandas, scikit-learn, statsmodels and Matplotlib. MPL-2.0 licence and those of each package.
Natural Earth	The boundaries of the 32 states of Mexico (1:10 million scale). Public domain.
Map tiles	Base maps from several providers, used under their terms and with their attribution on the map.
Typefaces	Cormorant, Crimson Pro and Jost, under the SIL Open Font License 1.1 (in this manual only).

F.2 · The data, which also belong to someone

Source	What it is and on what terms
GBIF	The example records (2,400 of *Pinus cembroides*, downloaded on 21 September 2026). Each record keeps the licence of the institution that published it: CC0, CC BY or CC BY-NC. The 70 institutions and the 91 datasets are listed in `datos/GBIF-FUENTES.md`.
WorldClim 2.1	The bioclimatic variables and elevation. Free for academic and non-commercial use, but **they may not be redistributed without prior permission**. Since version 1.1.0 **the program carries no WorldClim layer at all**, not even the clip the example used to bring: they are downloaded once, following the instructions in `datos/LEE-ME.md`.
Köppen-Geiger	The 1 km map of climate classes by Beck et al. (2018). CC BY 4.0: it can be redistributed with a citation, and that is why it does travel with the program.
SoilGrids (ISRIC)	The soil type of Block 5, queried online point by point. CC BY 4.0.

I CITE THE DATA, NOT JUST THE PROGRAM

A result rests on three things, and all three are cited: the program, the **data** you worked with and the **method** you applied. If you download your records from gbif.org instead of from the app, GBIF gives you a download DOI that identifies exactly that dataset: use it. The full breakdown of what is original and what comes from others is in `NOTICE.md`.